

**VIETNAM GENERAL CONFEDERATION OF LABOUR
TON DUC THANG UNIVERSITY
FACULTY OF INFORMATION TECHNOLOGY**



HO THI LINH

MULTI-SOURCE INTEGRATION FOR RECOMMENDATION SYSTEMS

**DOCTORAL DISSERTATION OF
COMPUTER SCIENCE**

HO CHI MINH CITY, 2023

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Ho Chi Minh City, day ... month 08 year 2023

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I hereby declare that this thesis was carried out by myself under the guidance and supervision of Assoc. Prof. Le Anh Cuong; and that the work contained and the results in it are true by author and have not violated research ethics. The data and figures presented in this thesis are for analysis, comments, and evaluations from various resources by my own work and have been duly acknowledged in the reference part.

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Ho Thi Linh

MULTI-SOURCE INTEGRATION FOR RECOMMENDATION SYSTEMS

ABSTRACT

Recommending items that align closely with the user’s interests is crucial for businesses as it helps improve sales revenue. The information collected by recommender systems, including product details, user behavior, and social relationships, etc. comes from diverse and heterogeneous sources. These sources need to be effectively utilized in multimodal recommender systems to enhance the recommendation process. In this research, our focus has been on solving the effective integration of diverse and heterogeneous information sources. The sources we utilize include the Utility Matrix and text sources. In recent studies of multimodal recommender systems, researchers have largely disregarded the combination of these information sources. Moreover, to further enhance the integration of these diverse and heterogeneous sources, we have employed the Transformer model for multimodal recommender systems. In the scope of this thesis, we will focus on two integration approaches, early fusion and late fusion.

We followed the late fusion approach and developed a hybrid recommender system that combines predicted ratings from three individual recommendation models using either voted-based or weighted-based approaches for the prediction task. The voted-based method includes maximum, minimum, and average strategies, while the weighted-based method utilizes a neural network (MLP).

Moreover, according to the early fusion approach, we innovatively combined the Utility Matrix source with textual sources, which has not been extensively explored in the context of multimodal recommender systems. In our work, we extracted and represented features from diverse sources, modalities, and views. We employed various techniques such as CountVectorizer, TF-IDF, and BERT to transform these features into a unified format.

To enhance the integration of multiple sources in recommender systems, we first reduced the dimensions of feature vectors using PCA. Then, we utilized an algorithm that selects the k best features from each source/view/modality and fuses the selected features from different views. The combined output is used as input for a neural network for the task of rating prediction in recommender systems.

To further develop our work from the previous study, we employed a transformer model for multimodal recommender systems. The transformer model's capability to understand intricate connections and interdependencies among the combined data from the Utility Matrix and textual sources enables it to generate predictions with higher precision.

In order to assess the effectiveness of the proposed models, we conducted experiments using the MovieLens and Amazon datasets. The experimental results unequivocally indicate that our proposed Transformer-based Multiview integration for recommender systems has a significantly positive impact on enhancing recommendation accuracy.

TÍCH HỢP ĐA NGUỒN THÔNG TIN CHO HỆ THỐNG KHUYẾN NGHỊ TÓM TẮT

Đề xuất các mặt hàng phù hợp gần với sở thích của người dùng là rất quan trọng đối với doanh nghiệp, vì nó giúp cải thiện doanh thu bán hàng. Thông tin được thu thập bởi các hệ thống khuyến nghị, bao gồm chi tiết sản phẩm, hành vi người dùng và các mối quan hệ xã hội, v.v. đến từ các nguồn đa dạng và không đồng nhất. Những nguồn này cần được sử dụng một cách hiệu quả trong các hệ thống khuyến nghị đa phương thức để tăng cường quá trình khuyến nghị. Trong nghiên cứu này, chúng tôi đã tập trung giải quyết việc tích hợp hiệu quả các nguồn thông tin đa dạng và không đồng

nhất, trong đó, các nguồn mà chúng tôi sử dụng là Utility Matrix và các nguồn văn bản. Trong các nghiên cứu gần đây về các hệ thống khuyến nghị đa thể thức, các nhà nghiên cứu hầu như loại trừ kết hợp các nguồn thông tin này cùng với nhau. Hơn thế nữa, để tăng cường việc tích hợp các nguồn đa dạng và không đồng nhất này, chúng tôi đã khai thác mô hình Transformer cho các hệ thống khuyến nghị đa thể thức. Trong phạm vi nghiên cứu của luận án này, chúng tôi sẽ tập trung vào hai cách tiếp cận tích hợp là early fusion và late fusion.

Theo phương pháp late fusion, chúng tôi đã phát triển một hệ thống khuyến nghị lai để kết hợp xếp hạng dự đoán từ ba mô hình khuyến nghị riêng lẻ bằng cách sử dụng phương pháp tiếp cận dựa trên bình chọn hoặc dựa trên trọng số cho nhiệm vụ dự đoán. Phương pháp dựa trên bình chọn bao gồm các chiến lược tối đa, tối thiểu và trung bình, trong khi phương pháp dựa trên trọng số sử dụng một mạng nơ ron nhân tạo (MLP).

Ngoài ra, theo cách tiếp cận early fusion, chúng tôi đã kết hợp một cách sáng tạo nguồn Utility Matrix với các nguồn văn bản, vốn chưa được khám phá rộng rãi trong ngữ cảnh của các hệ thống khuyến nghị đa phương thức. Trong công việc của mình, chúng tôi đã trích xuất và trình bày các đặc trưng từ nhiều nguồn, thể thức và view khác nhau. Chúng tôi đã sử dụng nhiều kỹ thuật khác nhau như CountVectorizer, TF-IDF và BERT để chuyển đổi các tính năng này thành một định dạng thống nhất.

Để tăng cường cho việc tích hợp nhiều nguồn trong các hệ thống khuyến nghị, trước tiên chúng tôi đã giảm kích thước của các vectơ đặc trưng bằng PCA. Sau đó, chúng tôi sử dụng thuật toán chọn k đặc trưng tốt nhất từ mỗi nguồn/ thể thức / view và kết hợp các đặc trưng đã chọn từ mỗi nguồn/ thể thức / view. Đầu ra của sự kết hợp được sử dụng làm đầu vào cho mạng nơ ron nhân tạo trong nhiệm vụ dự đoán xếp hạng trong các hệ thống khuyến nghị.

Để phát triển hơn nữa công việc của chúng tôi từ nghiên cứu trước, chúng tôi đã

sử dụng một mô hình Transformer cho các hệ thống khuyến nghị đa phương thức. Khả năng của mô hình Transformer là hiểu các mối liên hệ phức tạp và sự phụ thuộc lẫn nhau giữa các dữ liệu được kết hợp từ Utility Matrix và các nguồn văn bản cho phép nó tạo ra các dự đoán với độ chính xác cao hơn.

Để đánh giá hiệu quả của các mô hình được đề xuất, chúng tôi đã tiến hành thử nghiệm bằng cách sử dụng bộ dữ liệu MovieLens và Amazon. Các kết quả thử nghiệm chỉ ra một cách rõ ràng rằng mô hình tích hợp Multiview dựa trên Transformer được đề xuất của chúng tôi cho các hệ thống khuyến nghị có tác động tích cực đáng kể trong việc nâng cao độ chính xác khuyến nghị.

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ABBREVIATIONS

ALS	Alternating Least Squares
BERT	Bidirectional Encoder Representations from Transformers
CB	Content-based
CF	Collaborative Filtering
DNN	Deep Neural Network
FEM	The Feature Extraction Model
GRU	Gated Recurrent Unit
LDA	Latent Dirichlet Allocation
LSA	Latent Semantic Analysis
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MLP	Multi-layer Perceptron
NLP	Natural Language Processing
PCA	Principal Component Analysis
PFM	The Predicting Fusion Model
RMSE	Root Mean Square Error
RNN	Recurrent Neural Network
RS	Recommender system
SVD	Singular Value Decomposition
SVM	Support Vector Machine
TF-IDF	Term Frequency-Inverse Document Frequency

CHAPTER 1. INTRODUCTION

1.1 Introduction the recommender system problem

Nowadays, with the development of technology, people often buy products online more from stores. Businesses need to assure buyers that products are good and they will like them. To create confidence in purchasing products, recommender systems have been built.

Within recommender systems, two primary entities exist: users and items. Users are people who use products/services, while items refer to products/services themselves, such as movies, songs, books, clips, or even other users in the context of friend recommendation problems. The primary objective of recommender systems is to predict a user's preference for a specific item, enabling an appropriate recommendation strategy.

	Jack	Tom	Marry	John	Peter
The Private Life of Sherlock Holmes	5	4	?	?	3
Thor: The Dark World	?	4	3	?	2
Clash of the Titans	?	3	2	?	2
Titanic (1997 film)	4	?	?	1	?
Tom and Jerry	?	3	?	4	2

Figure 1.1 An example of a Utility Matrix for a Movie recommender system

Source: Baltrušaitis et al., 2018

In its essence, a recommender system functions as a tool capable of assisting users in extracting valuable and personalized information from vast reservoirs of data (J. Wang & Tang, 2015). During the mid-1990s, the realm of recommender systems evolved into a distinct domain, prompting researchers to shift their focus towards refining recommendations that incorporated explicit feedback. As a usual scenario, the goal of recommendation is to forecast user preferences using historical data. Fundamentally, by analyzing users' past behavior, recommender systems estimate

users' preferences towards items that haven't existed in their record. A common approach for leveraging historical data involves transforming it into a matrix format denoted as the $User \times Item$ matrix, also referred to as the Utility Matrix. An illustration of this concept is provided in Figure 1.1, which pertains to a Movie recommender system. Within this matrix, ratings are assigned within a scale ranging from 1 to 5. The "?" symbol signifies that a user hasn't provided a rating for a particular item (alternatively, 0 is frequently used to denote unrated items). Further insights into the recommender system problem can be found in Algorithm 1.

Algorithm 1 An algorithm for the recommender system problem

1: Input:

- A set of users U , each user $u \in U$ has features $V_u = v_{u1}, v_{u2}, \dots, v_{ux}$
- A set of items I , each user $i \in I$ has features $V_i = v_{i1}, v_{i2}, \dots, v_{it}$
- A Utility Matrix $R^{n \times m}$, n denotes the number of users, m denotes the number of items.

2: Output: List of items $i_j \in I$ most matched with a user $u_k \in U$.

Do the following tasks:

- 3: Step 1: Build a function F to measure the relevance of user u_k to item i_k .
 - 4: Step 2: Select the top list of items with high relevance (most likely to be selected by users).
-

In a formal context, an abstract representation of recommender systems (depicted in Figure 1.2) can be delineated as follows: Consider a set U comprising n users, a set I encompassing m items derived from the historical data of U , and $R = U \times I$. The recommender system can be defined as:

$$f : R^{n \times m \times d} \longrightarrow \hat{R}^{n \times m} \quad (1.1)$$

Here, d represents the number of dimensional features. The ratings form the foundation of the system, with $d = 1$ signifying a two-dimensional $User \times Item$ space. When supplementary attributes such as time or social interactions are

introduced, the count of additional features becomes $d - 1$. The recommendation algorithm, denoted as f , determines the dimensions that participate in the calculations and their utilization. The outcome of the function f , denoted as $\hat{R}$, takes the form of a prediction matrix.

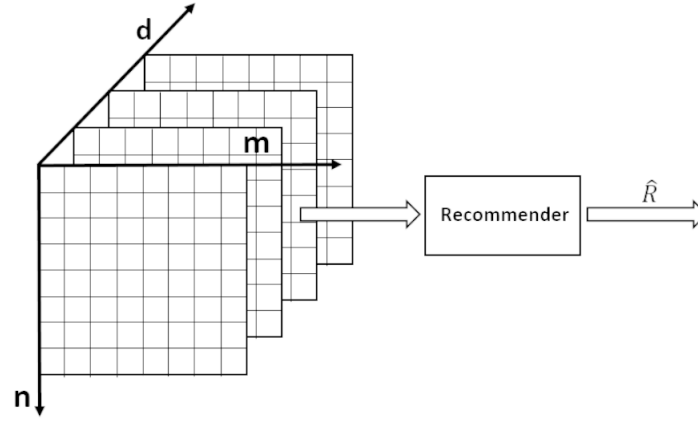


Figure 1.2 An abstract of a recommender system

The recommendation problem refers to the task of suggesting relevant products or services to users based on their preferences, interests, or past behaviors, that is a fundamental challenge in different domains such as e-commerce, content streaming platforms, social media, and personalized advertising.

The objective of RSs is to assist users in discovering new items, enhancing user experience, and increasing user engagement. These systems analyze large amounts of user data, including historical behavior, explicit feedback (examples like reviews or ratings), implicit feedback (examples like purchase history or clicks), demographic information, and contextual data to generate personalized recommendations.

The concept of Modality:

According to (Baltrušaitis et al., 2018), we can see multiple modalities, including the perception of objects, auditory sensations, tactile experiences, olfactory perceptions, and more. In broad terms, a modality pertains to the manner in

which an occurrence unfolds or is perceived. Typically, the term "modality" is linked with sensory modalities, which encompass our fundamental avenues of communication and sensation, like sight or tactile sensations.

Several types of modalities include examples like *natural language*, encompassing both written and spoken forms; *visual signals*, commonly conveyed through images or videos; *vocal signals*, encoding sounds and supplementary cues like prosody and vocal expressions; *auditory signals* (encompassing speech, sounds, and music); the sense of *touch* or *haptics*; perceptions of *smell*, *taste*, and *self-motion*; *physiological signals* like the electrocardiogram (ECG) and *skin conductance*; as well as additional modalities like infrared images, depth images, and fMRI scans, among others.

The concept of Multi-Modality:

Multimodality involves utilizing multiple modes, including linguistic, textual, spatial, and visual modes, among others, in communication practices to generate a specific message or series of messages (Saçak, 2019). In essence, modes are information channels. The information derived from various sources is intricately linked in meaning and occasionally offers supplementary insights. This reveals patterns that remain concealed when analyzing individual modalities separately. These systems integrate heterogeneous and isolated data from diverse sensors, consequently enhancing the accuracy of predictions. For instance, in an emotion detection scenario, to categorize an individual's current mood, one could fuse data collected from an EEG and eye movement indicators. This involves merging two various data sources to accomplish a deep learning task (Shalu et al., 2020).

1.1.1 Multimodal Recommendation Problem

A multimodal recommendation problem refers to a scenario where a system is tasked with providing personalized recommendations to users using data from

multiple modalities or sources of information. Traditionally, recommendation systems have primarily relied on one type of data, such as user behavior history (e.g., past purchases, clicks, or ratings) or item attributes (e.g., genres, tags, or descriptions) to generate recommendations. However, in a multimodal setting, these systems leverage diverse data types to gain a more comprehensive understanding of users' preferences and item characteristics. Figure 1.3 visualizes the multimodal recommendation problem, where features are extracted from the modalities, and then two fusion approaches (early fusion and late fusion) are used to propose models/methods for multimodal recommender systems.

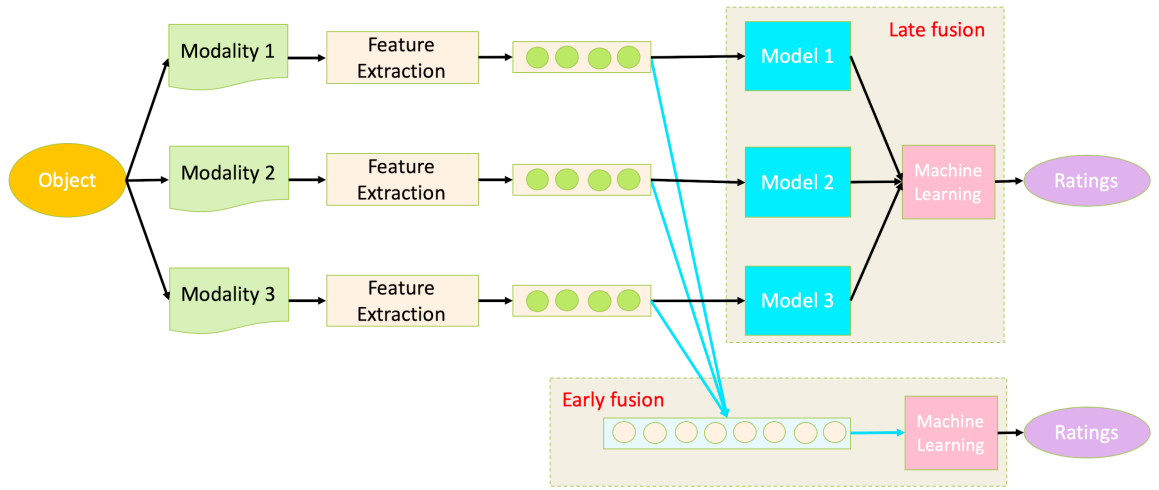


Figure 1.3 Multimodal recommendation problem

(Q. Liu et al., 2023) define Multimodal Recommender Systems as recommender systems designed for items with multimodal features. The main modalities that can be combined in multimodal recommender systems include:

Textual data: This includes textual information such as product descriptions, user reviews, and social media posts. Textual data can provide valuable insights into the sentiment and context surrounding the items and users.

Visual data: Visual information like images or videos associated with items can convey additional details about their appearance and features. For example, in

e-commerce, product images can help capture the style, color, and design of products.

Audio data: In some cases, audio data may be relevant for recommendation, such as in the context of music or podcast recommendations.

Social data: Information from social networks, such as user social graphs, friends' preferences, and interactions, can be used to identify user communities and incorporate social influence in the recommendations.

Contextual data: Additional contextual information like time, location, and device used can be leveraged to provide more relevant and timely recommendations.

The multimodal recommendation problem poses unique challenges in data integration, feature extraction, and model design. Methods like deep learning, transfer learning, and attention mechanisms are often used to effectively combine the different modalities and make accurate recommendations. The integration of multiple modalities helps to overcome the limitations of single-modal recommendation systems and provides a more holistic understanding of user preferences, leading to improved recommendation performance and user satisfaction.

1.1.2 Overview the approaches and methods

1.1.2.1 Approachs for recommender systems

Recommender systems are built to recommend items to users based on predicting users' preference for items by many different factors. Items have the highest level of preferences are items which the users are most likely to purchase and are of interest to. Recommender systems find the matching between users and items, impute the similarities between users and items for recommendation . These systems have benefits for both the users and the service providers. The decision quality and decision-making process has also improved by using these kinds of systems.

There are two categories of recommendations:

- Non-personalized or popularity recommendations, which simply recommend what's popular among all users. Popularity-based recommender systems operate on the principle of favoring popular or trending items. These systems assess which items are currently in vogue or widely favored by users, and provide direct recommendations based on that information. For instance, if a product is frequently purchased by a large number of people, the system recognizes it as highly popular. As a result, when a new user signs up, the system will recommend that popular product to them, increasing the likelihood that the new user will also make a purchase.
- Personalized recommendations, providing recommendations by utilizing data regarding the content or the viewing preferences of the user. Personalization mechanisms can be categorized into three primary types: content-based, collaborative filtering, and hybrid approaches.

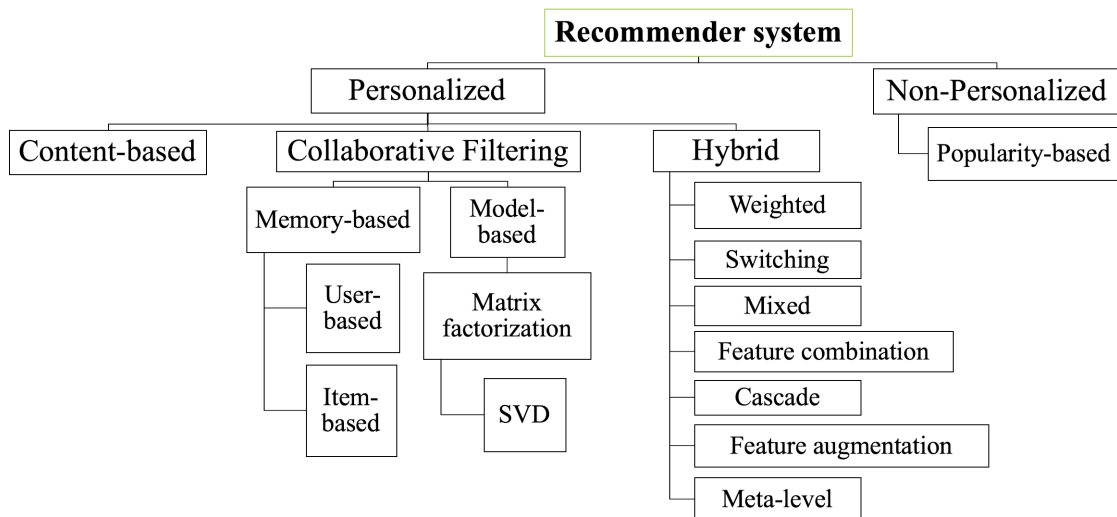


Figure 1.4 Types of recommender systems

Content-based recommender systems:

A content-based recommender system operates based on the principle of finding similar content. For instance, if a user is reading a book, the system will search for

other books with similar content or belonging to the same genre as the book the user is currently reading. To determine similarity, the system utilizes various fundamental attributes.

In order to assess similarity between items, the recommender system calculates distances between them. Different scenarios require different metrics to measure similarity. For numerical data, Euclidean distance is employed, cosine similarity is calculated for textual data, and Jaccard similarity is computed for categorical data.

Cosine Similarity: Cosine Similarity (Adomavicius & Tuzhilin, 2005) is a widely used method in collaborative filtering for recommender systems, which calculates the cosine angle between two vectors to determine their relationship. Equation 1.2 provides the formula for Cosine Similarity, where “ $\cdot$ ” represents the dot product of the rating vectors R_u and R_v for users u and v , respectively.

$$Cos_sim(u, v) = \frac{R_u \cdot R_v}{|R_u| \cdot |R_v|} \quad (1.2)$$

Nonetheless, an important limitation of cosine similarity is that it treats null preferences as negative preferences.

Euclidean distance: The length of a line segment between two users (or items) in Euclidean space is known as the Euclidean distance (Fkih, 2022) and is denoted by d . In practical terms, each user (or item) is depicted by their Cartesian coordinates in relation to the basis of items (or users), and the distance between two users (or items) is computed as the absolute value of the numerical difference of their coordinates. Equation 1.3 provides the formula for calculating the Euclidean distance (d) as an indicator of the correlation between two users, u and v .

$$d(u, v) = \sqrt{\sum_{i \in I_{uv}} (r_{vi} - r_{ui})^2} \quad (1.3)$$

The set of items that are rated by both users u and v is represented by I_{uv} , while r_{ui} and r_{vi} correspond to the ratings provided by user u and v , respectively, for item i . Equation 1.4 presents the Euclidean distance computation for two items, i and j .

$$d(i, j) = \sqrt{\sum_{u \in U_{ij}} (r_{uj} - r_{ui})^2} \quad (1.4)$$

The set of users who rated both items i and j is denoted by U_{ij} , while r_{ui} and r_{uj} indicate the ratings assigned by user u for items i and j , correspondingly. In order to utilize the Euclidean distance as a similarity metric, it needs to be normalized. The Euclidean similarity (ES) measures for users and items are described by equations 1.5 and 1.6, respectively.

$$ES(u, v) = \frac{1}{1 + d(u, v)} \quad (1.5)$$

$$ES(i, j) = \frac{1}{1 + d(i, j)} \quad (1.6)$$

Pearson Correlation Coefficient: Pearson Correlation Coefficient (PCC) (Resnick et al., 1994) is one of the most commonly used methods for finding the linear correlation between two vectors. The PCC result is a value between -1 and +1, where -1 represents a negative correlation, +1 represents high positive correlation, and a value of 0 indicates no correlation, also known as zero-order correlation. Equation 1.7 provides the formula for PCC.

$$PCC_sim(u, v) = \frac{\sum_{i \in I} (r_{u,i} - \bar{r}_{u,I})(r_{v,i} - \bar{r}_{v,I})}{\sqrt{\sum_{i \in I} (r_{u,i} - \bar{r}_{u,I})^2 + \sum_{i \in I} (r_{v,i} - \bar{r}_{v,I})^2}} \quad (1.7)$$

The variables $\bar{r}_{u,I}$ and $\bar{r}_{v,I}$ denote the average ratings of user u and user v , respectively, for the items in the set I that they have co-rated.

Jaccard Similarity: Jaccard Similarity is calculated by dividing the number of users

who have rated both item A and B by the total number of users who have rated either item A or B . This similarity measure is employed to compare the similarity between the items.

Collaborative Filtering recommender systems:

A collaborative filtering-based recommender system operates by leveraging the similarity between users and items, commonly employed in e-commerce and online websites. This system examines the preferences of similar users to make recommendations. The notion of similarity extends beyond user preferences and can also involve similarities between different items. The system is most effective when there is a substantial amount of information available about both users and items, allowing for more accurate and efficient recommendations. There are two types of CF, including memory-based and model-based.

Memory based: Memory-based techniques are utilized for raw data without preprocessing. There are two types, including user based and item based CF. User-based CF identifies users with similar purchasing preferences by analyzing their purchase behavior. The system calculates the similarity between users based on their shared taste in buying items. On the other hand, item-based CF examines the items purchased by a user and identifies similar items. In this approach, the system determines the similarity between different items solely based on the items themselves, rather than considering the users involved in the prediction process.

Model-based: Machine learning algorithms were employed to develop these models. A model is constructed, and recommendations are generated based on it, using only relevant data. This optimization accelerates the system's performance. By adopting this approach, better scalability is achieved, and dimensionality reduction techniques such as MF are frequently applied to enhance efficiency.

Hybrid recommender systems:

Hybrid recommender systems combine multiple recommender system techniques to predict users' preferences effectively. Burke, R. (Burke, 2002) introduced seven approaches of hybrid recommender systems which are detail in Figure 1.5.

Hybridization method	Description
Weighted	A single recommendation is generated by aggregating the scores or votes from multiple recommendation techniques.
Switching	Depending on the current situation, the system alternates between different recommendation techniques.
Mixed	Multiple recommender systems provide simultaneous recommendations.
Feature combination	Various recommendation data sources contribute their features to create a unified recommendation algorithm.
Cascade	One recommender further enhances the suggestions provided by another recommender.
Feature augmentation	The output generated by one technique serves as an input feature for another.
Meta-level	The model trained by one recommender is employed as input for another.

Figure 1.5 Hybridization Methods

Weighted: In the weighted recommender system, the outputs from multiple models are combined using fixed weightings that remain constant throughout the training and testing phases. For instance, a CB model and an item-item CF model can be combined with equal weights of 50% each, contributing to the final prediction. The advantage of employing the weighted hybrid approach lies in its capability to linearly integrate multiple models, thereby enhancing the recommendation process and accommodating the dataset effectively.

Switching: The switching hybrid recommender system dynamically chooses a suitable recommender system according to the specific circumstances. When constructing the model for an item-level sensitive dataset, it is necessary to establish

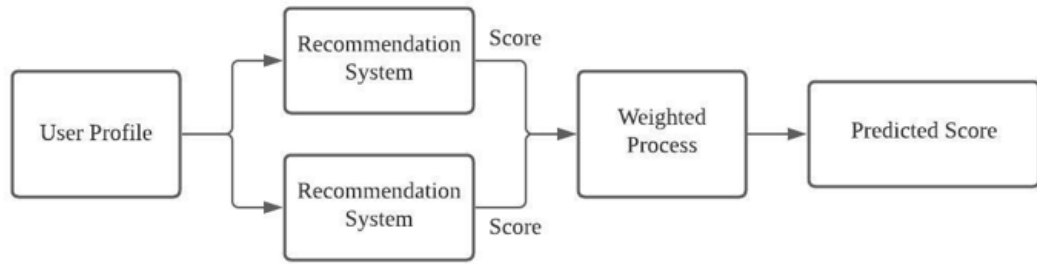


Figure 1.6 A weighted hybrid recommender system

criteria for the recommender selector based on user profiles or other relevant features. The switching hybrid approach adds an extra layer to the recommendation model, enabling the selection of the most appropriate model to utilize. This type of RS is attuned to the strengths and weaknesses of each constituent recommendation model, ensuring optimal performance.

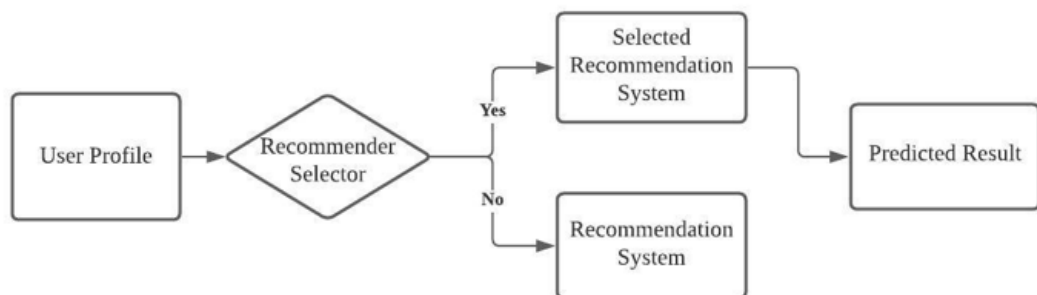


Figure 1.7 Switching Recommendation

Mixed: In the mixed hybrid approach, user profiles and features are initially used to generate distinct sets of candidate datasets. The recommender system then inputs each candidate set to the recommendation model, and combines the predictions to generate the final recommendation results. The mixed hybrid recommender system possesses the ability to make numerous recommendations simultaneously, while effectively fitting partial datasets to the appropriate model for improved performance.

Feature Combination: The feature combination hybrid recommender system

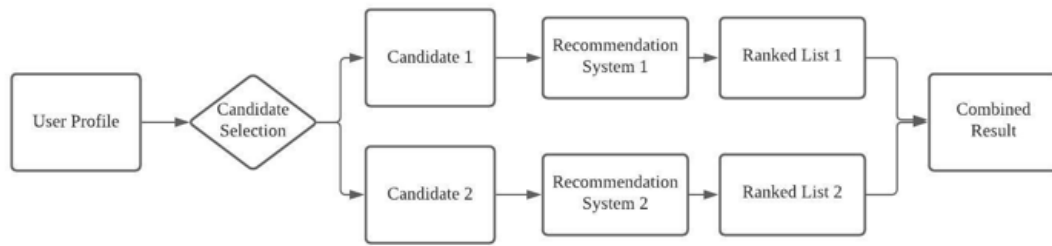


Figure 1.8 A mixed recommender system

incorporates a virtual contributing recommendation model into the existing system, serving as a feature engineering mechanism for the original user profile dataset. For instance, collaborative recommendation model features can be integrated into a CB recommendation model. This hybrid model enables the consideration of collaborative data from the subsystem without relying solely on one specific model.

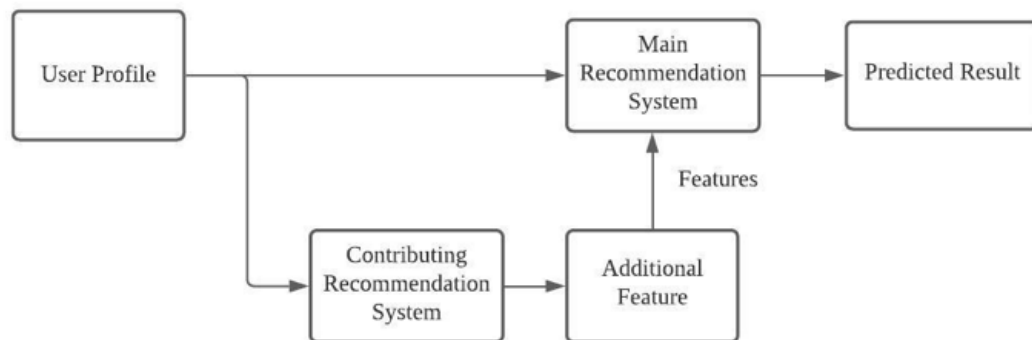


Figure 1.9 A feature combination recommender system

Feature Augmentation: A RS utilizes a contributing recommendation model to generate a rating or classification for the user/item profile. This rating or classification is then incorporated into the main RS to generate the final predicted outcome. The feature augmentation hybrid RS enhances the performance of the main system without altering the primary recommendation model. For instance, the use of association rules allows us to enrich the user profile dataset. By incorporating the augmented dataset, the CB recommendation model's performance is enhanced.

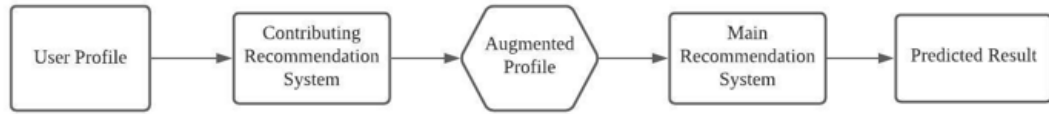


Figure 1.10 Feature Augmentation

Cascade: A cascade hybrid RS establishes a hierarchical structure for recommendation, where the main RS generates the primary outcome, and a secondary model is employed to address minor concerns, such as resolving ties in scoring. In real-world scenarios, datasets often have sparse entries, and the secondary recommendation model proves effective in tackling issues like equal scoring or missing data.

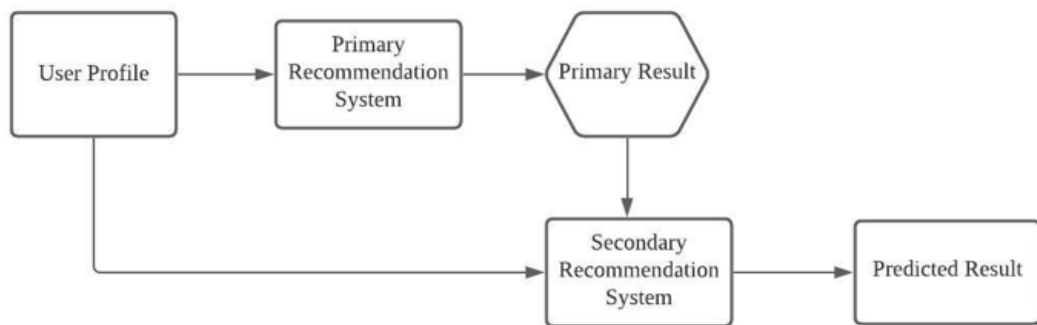


Figure 1.11 A cascade recommender system

Meta-Level: The meta-level hybrid RS and the feature augmentation hybrid RS have similarities in terms of utilizing a contributing model to augment the dataset for the main recommendation model. However, they differ in the approach: the meta-level hybrid system replaces the original dataset with a learned model from the contributing model, whereas the feature augmentation hybrid system incorporates the augmented dataset without replacing the original data.

1.1.2.2 Approachs for multi-modality fusion

Multimodal fusion is the process of integrating multimodal data, which involves combining and fusing information from different modalities to generate recommendations. Deciding on the appropriate fusion technique, such as early fusion (e.g., combining data from different modalities at the input level), late fusion (e.g., combining data from different modalities at the output level), or hybrid fusion (e.g., combining data at both input and output levels), can be challenging and may have a significant impact on the quality and effectiveness of recommendations. Multimodal fusion offers advantages such as resilience, the acquisition of supplementary information, and the sustained functionality of the system even when one or more modalities encounter failure (Pandeya & Lee, 2021).

Early fusion:

Early fusion techniques generate a unified representation by combining input features from various modalities. Subsequently, a solitary model, referred to as h , is trained to capture the connections and interactions among the basic features of each modality. The ultimate prediction can be expressed as stated in (K. Liu et al., 2018).

$$p = h([v_1, v_2, \dots, v_m]) \quad (1.8)$$

where we employ concatenation as a widely observed instance of jointly representing features from different modalities.

Early fusion can be regarded as an initial endeavor in multimodal learning. Since it involves a single model, the training process is straightforward and typically necessitates highly engineered and preprocessed features from different modalities to ensure their alignment or semantic similarity. Additionally, it relies on a single model for predictions, assuming that the model is adequately suited for all the modalities.

Late fusion:

In late fusion, the unimodal decision values are employed and fused using a fusion mechanism denoted as F , which can include techniques like averaging, voting, or a learned model. Assuming that model h_i is applied to modality i (where i ranges from 1 to M), the ultimate prediction is expressed as follows:

$$p = F(h_1(v_1), h_2(v_2), \dots, h_m(v_m)) \quad (1.9)$$

Late fusion (K. Liu et al., 2018) enables the utilization of diverse models on individual modalities, thereby offering increased flexibility. It provides an easier mechanism to handle missing modalities since the predictions are made independently. However, due to late fusion operating on inferences rather than raw inputs, it lacks effectiveness in capturing signal-level interactions between modalities.

Hybrid fusion:

Hybrid fusion is a combination of both feature and decision level fusion for exploiting the advantages of both fusion strategies. Simultaneously, diverse analysis units examine individual features and integrate their decisions through a decision fusion unit. Subsequently, the decisions from previous stages undergo further fusion via a decision fusion mechanism to derive the final decision (Atrey et al., 2010).

1.1.3 Recent studies about multimodal recommender systems

Based on the study conducted in (Singh et al., 2021), it was found that the main focus of the majority of multimodal recommender systems (86%) is on the text-based modality. Nonetheless, researchers have also integrated various other modalities into these systems, including the parameters of temperature (43%), audio (33%), video (29%), images (19%), as well as the parameters of location and society. The inclusion of temporal and demographic parameters is essential for decision-making since preferences can change over time, influenced by cultural

factors and language adaptations based on location. Moreover, physiological parameters are utilized to obtain real-time insights into users' moods, enabling personalized recommendations that cater to their emotional state.

Researchers have employed various modalities in different RSs (Table 1.1). In the domain of music recommendation, text (lyrics, artist, metadata) is combined with audio, as demonstrated by (Domingues et al., 2013; Kuo et al., 2013; Laurier et al., 2008; Oramas et al., 2017). For location/route recommender systems, multimodal information such as location coordinates, mobility options, socio-demographics, text descriptions, photos, and user check-in patterns are utilized, as shown by (Campigotto et al., 2016; X. Wang et al., 2014). In tourist RSs, multimodal data is exploited by (L. Liu, Gan, et al., 2017). Movie RSs leverage history, ratings, tags, poster images, and social relationships, as demonstrated by (Da Costa et al., 2014; Da Costa & Manzato, 2014, 2016; Rafailidis et al., 2017; Z. Zhao et al., 2017). Video RSs utilize user behavior, facial expressions/emotions, and textual descriptions, as seen in the works of (Kaklauskas et al., 2016; Laurier et al., 2008; B. Yang et al., 2007). Finally, e-commerce RSs utilize text information such as feedback, comments, ratings, and product images, as shown by (Peska, 2017).

Based on the relevance of each technique to specific unimodal information, researchers employed a variety of techniques across different modalities. For instance, machine learning techniques such as SVM, MF, KNN, and topic modeling were utilized to assess users' similarity based on factors like rating, word embedding, TF-IDF for content similarity, and audio embedding for audio-based similarity, as illustrated in Figure 1.12. LSA is primarily employed to calculate the similarity between various modalities. Among the techniques applied in multimodal recommender systems, MF and LSA have emerged as the most prominent ones. Table 1.2 provides a summary of the diverse techniques utilized in the construction of multimodal RSs.

Table 1.1 Multimodal recommender systems leverage diverse modalities tailored to each specific application.

Paper	Modality	Application
(Oramas et al., 2017), (Domingues et al., 2013), (Laurier et al., 2008)	Audio, social tags, feedback, usage, text (name, metadata, lyrics, artist), biographies	Music
(Kuo et al., 2013)	Music, video	Recommendation of background music for automated video editing
(Campigotto et al., 2016) , (X. Wang et al., 2014), (L. Liu, Gan, et al., 2017)	Socio-demographics, location, photos, text descriptions, user check-in, patterns, mobility option, venue contex	Recommendation of location/route
(Da Costa & Manzato, 2016), (Da Costa et al., 2014), (Da Costa & Manzato, 2014), (Rafailidis et al., 2017), (Z. Zhao et al., 2017)	User preference with social media and temporal relationship, rating, tags, browsing history, movie-poster image, textual description	Movie
(Kaklauskas et al., 2016), (Arapakis et al., 2009), (Laurier et al., 2008), (B. Yang et al., 2007)	Real-time facial expression, human emotions, mobile data, video content (video query, tags, title), textual data	Video
(Peska, 2017)	Rating, implicit feedback, image of product, comments, location	Ecommerce

1.2 Motivation of the research

Traditional recommendation methods often face challenges such as data sparsity and cold start problems, where certain items or users have limited or no available data. Multimodal recommender systems address these challenges by integrating data from multiple modalities. For instance, even if there are sparse textual reviews for a product, the visual features of the product image can still provide valuable information for generating recommendations. This approach improves recommendation accuracy, especially for new or less popular items and users.

Multimodal approaches have the potential to deliver superior recommendation performance compared to unimodal approaches. By leveraging complementary information from multiple modalities, recommendations become more accurate,

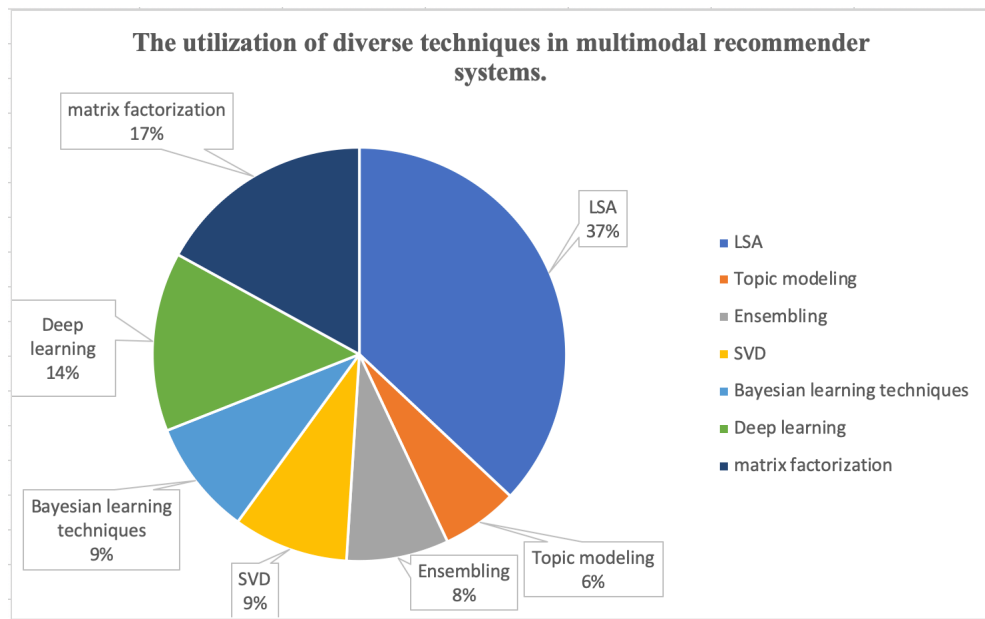


Figure 1.12 The utilization of diverse techniques in multimodal recommender systems.

relevant, and diverse. This leads to increased user satisfaction, engagement, and potentially higher conversion rates across various recommendation scenarios, including e-commerce, multimedia content, and social media.

The advancement of machine learning and deep learning techniques has facilitated the development of more advanced methods to process multimodal data. RNNs, CNNs, deep neural networks, and attention mechanisms have demonstrated great potential in analyzing multimodal data. Leveraging these advancements, researching multimodal recommender systems can unlock opportunities to explore novel approaches that enhance recommendation accuracy and performance.

The aforementioned motivations have inspired us to delve into researching the representation of diverse and heterogeneous information sources, as well as exploring methods and models to integrate them and employ them in the prediction task of recommender systems.

1.3 Objectives and the scope of the study

Table 1.2 Various techniques employed in the creation of multimodal recommender systems tailored to each specific application.

Paper	Application	Technique
(Oramas et al., 2017), (Domingues et al., 2013), (Laurier et al., 2008)	Music	LSA, text and audio embedding, ensemble rankings on MF, SVD++, SVM, deep learning
Kuo et al., 2013	Background music	Multimodal LSA
(Campigotto et al., 2016), (X. Wang et al., 2014), (L. Liu, Gan, et al., 2017)	Recommendation of location/route	Multimodal topic modeling, Bayesian techniques, MF
(Yukawa et al., 2012)	Recommendation for groups	Multi-agent system
(Da Costa & Manzato, 2016), (Da Costa et al., 2014), (Da Costa & Manzato, 2014), (Rafailidis et al., 2017), (Z. Zhao et al., 2017)	Movie	Bayesian personalized ranking (BPR), MF, SVD, SVD++, random walk learning, RNN, CNN
Kaklauskas et al., 2016, (Arapakis et al., 2009), (Jia et al., 2015), (B. Yang et al., 2007)	Video	LSA, SVM, attention fusion function (AFF)
(Peska, 2017)	Ecommerce	Vector space model (VSM)

1.3.1 Objectives of the study

Our primary objective is to propose recommendation models that leverage various approaches to integrate information from diverse and heterogeneous sources. By doing so, we aim to enhance the richness of information available to recommender systems, ultimately leading to improved recommendation accuracy.

Additionally, we are dedicated to addressing the inherent challenges associated with the heterogeneity of information sources in the context of multi-source integration. We strive to develop innovative solutions that overcome the obstacles arising from the disparate nature of these sources. Our goal is to effectively represent diverse and heterogeneous information sources and propose methods and models for integrating information from multiple sources. This will enable the creation of a more comprehensive and effective recommender system.

1.3.2 The scope of the study

1.3.2.1 The Common Problems for Multi-source Integration In Recommender Systems

Integrating multimodal data, such as text, images, audio, or video, requires effective data representation techniques that can capture the inherent characteristics of each modality. Representing different modalities in a unified and meaningful way for recommendation purposes can be challenging due to the heterogeneous data types and the need to extract relevant features from each modality.

Each modality may present its own challenges in data processing and feature extraction. For example, processing text data may require natural language processing (NLP) techniques, while processing image data may require computer vision techniques. Handling different modalities with varying characteristics, data volumes, and data complexities can be challenging and may require specialized techniques for each modality.

Multimodal data may suffer from modality bias, where certain modalities may dominate the recommendation process, while others may be neglected. For example, in a RS that integrates text and image data, the text modality may be given more weight or prominence, leading to biased recommendations. Addressing modality bias and ensuring fair treatment of all modalities in the recommendation process can be a challenge.

Discrepancies in data across different modalities can pose challenges in multimodal integration for recommender systems. Data from different modalities may have inconsistencies, discrepancies, or missing values, which can impact the quality and accuracy of recommendations. Resolving these data discrepancies and ensuring data consistency across modalities may require data preprocessing, data alignment, and data imputation techniques.

Different modalities may introduce noise or irrelevant information in the data,

and handling such noise is an important aspect of multimodal integration. For example, text data may contain typographical errors or irrelevant keywords, while image data may have image distortions or background noise. Proper preprocessing and noise handling techniques are necessary to ensure that noise from different modalities does not negatively impact the accuracy and quality of the recommendations.

Most research on multimodal recommender systems has primarily focused on combining multiple modalities of user and item data. However, there is still very little research on effectively incorporating contextual information, such as location, time, and user behavior, into the recommendation process.

Different modalities or views often possess distinct scales, distributions, or semantics, which complicates their effective combination. While some prior studies have proposed methods to align features across modalities or views, this issue remains unresolved and further research is required.

Data sparsity remains a significant challenge in multimodal recommender systems. Despite employing multiple modalities, some users may not have engaged with any of them, resulting in cold start problems and hindering accurate recommendations.

1.3.2.2 The Scope of the PhD Dissertation

In the PhD dissertation, we will prioritize solving the integration of multi-source approaches in recommender systems as the following:

1. We explore various methods and models to effectively represent diverse and heterogeneous information sources.
2. We propose models that integrate information from diverse and heterogeneous sources (from Utility Matrix and textual sources) to enhance recommendation accuracy by leveraging modern methods and techniques.

3. Although most recent studies on multi-modal recommendation systems exclude the combination of information from a Utility Matrix and textual data, we believe that combining user and item features from a Utility Matrix with textual information will improve accuracy. Therefore, in our study, we will propose different models that can combine these two types of information in recommender systems.

1.4 Methodology

1.4.1 Identifying the research problem

This step involves the initial step of the research process, which is identifying and selecting a specific research topic. It entails exploring different areas of interest, reviewing existing literature, and considering various factors such as novelty, relevance, and feasibility. The goal is to narrow down and define a research topic that aligns with the research objectives.

1.4.2 Surveying recent studies

In this step, the focus is on conducting a thorough literature review to survey recent studies related to the chosen research topic. The purpose is to identify the existing gaps, limitations, or unresolved issues within the field. By reviewing relevant literature, researchers gain insights into the current state of knowledge and identify the specific problem or research question that their study aims to address.

1.4.3 Proposing the model

This step centers around proposing a model or methodology that will be used to investigate the identified research problem. It involves describing the conceptual framework, approach, or theoretical foundation upon which the model will be built. The proposed model is introduced based on existing frameworks or theories, or it can also introduce a new approach.

1.4.4 Evaluating the model through conducting experiments

Under this step, the evaluation of the proposed model is discussed. It entails designing and conducting experiments to assess the performance, validity, and effectiveness of the proposed model in addressing the research problem. This step may involve several substeps:

Choosing the standard datasets: This substep focuses on selecting appropriate datasets for experimentation. Standard datasets that are widely used in the field may be chosen to ensure comparability and benchmarking. The datasets should be relevant to the research problem and provide a representative sample of the data domain.

Implementing: In this substep, the implementation of the proposed model is detailed. It involves translating the proposed model into a concrete software implementation or experimental setup. This may include coding, configuring software tools, or developing algorithms necessary for executing the model.

Comparing results: This step involves evaluating and comparing the outcomes obtained from different approaches, models, or techniques in order to assess their performance or effectiveness. It typically includes measuring various metrics, such as accuracy, precision, recall, or error rates, and analyzing the results to determine which approach or model produces the most favorable outcomes. The purpose of comparing results is to make informed decisions or draw conclusions based on the relative performance of different methods or models.

Analyzing results: This substep centers around analyzing and interpreting the results obtained from the conducted experiments. The analysis may involve statistical techniques, data visualization, or other methods to examine the outcomes of the experiments. The findings are then interpreted and discussed in relation to the research problem, the proposed model, and the existing literature.

These steps provide an overview of the methodology section, covering the steps

from identifying the research topic to evaluating the proposed model through experiments and result analysis. The specific details and depth of each topic will vary depending on the nature of the research and the field of study.

1.5 The contributions

Contribution 1: *Building a hybrid model for recommender systems that integrates the predicted rating outputs from three individual recommendation models using two different methods: voted-based or weighted-based.*

The voted-based method includes max, min, and average approaches, while the weighted-based method utilizes a neural network (MLP). For the weighted-based approach, we combine the predicted ratings from the three individual recommendation models, then the resulting combined ratings are used as input for a neural network to perform the prediction task within the hybrid recommender system. Additionally, we have implemented the KNN algorithm and incorporated it into the individual recommendation models, which can be either user-based or item-based recommender systems. Given the neural network model's capacity to learn from training, our proposed weight-based hybrid recommendation model showcased its superiority over other individual recommendation models.

The first contribution of this work is presented in [P1].

Contribution 2: *Building a multi-modal fusion model for integrating diverse and heterogeneous information sources in recommender systems.*

In our research, we focused on extracting and representing features from user views and item views, derived from the Utility Matrix sources and textual sources. This approach, which combines different types of views, is not commonly explored in the multi-modal approach to recommender systems. Our goal was to transform these features into a unified format using techniques like BERT, TF-IDF, and others. To reduce the dimensionality of the features, we employed dimensionality reduction

methods such as PCA. Additionally, we exploited an algorithm to select the k best features from each source. To fuse the selected features from each source, we proposed a multimodal model that utilizes a neural network for the rating prediction task in recommender systems. By combining the selected features from diverse sources, our model aims to enhance the accuracy of recommendations.

The second contribution of this work is to synthesize the strengths of the methods presented in [P2].

Contribution 3: *Building a Transformer-based Multiview Integration model for Recommender Systems.*

Another significant contribution of our research, which is a further development based on the second contribution, is the utilization of the transformer model for the multimodal fusion model in recommender systems. By incorporating the transformer model, we exploit its ability to learn effective features from the training data, ultimately enhancing the accuracy of predicting user preferences for items. This, in turn, leads to improved recommendation accuracy. The transformer model's capacity to capture complex relationships and dependencies within the combined information from the Utility Matrix and textual sources enables it to generate more accurate predictions. By leveraging the strengths of the transformer model, our proposed transformer-based multimodal integration model provides more precise and reliable recommendations to users.

The third contribution of this work is presented in the work [P3].

1.6 The Dissertation Outline

Chapter 1: Introduction

This chapter introduces the recommender system and the problem and the multimodal recommender systems. It provides an overview of the approaches and

methods, as well as the motivation behind the research. Furthermore, it presents the objectives and scope of the PhD dissertation, along with the methodology used to conduct the study. The chapter also emphasizes the contributions of the dissertation and outlines its structure.

Chapter 2: Background

This chapter presents the fundamental theories of the following: The first, recommender systems, which include the introduction about various recommender systems and performance measures for recommender systems; Finally, machine learning methods, which encompass matrix factorization, BERT, MLP, RNN, Transformer, and Graph Neural Network; The last, feature extraction, which involves TF-IDF and PCA.

Chapter 3: A hybrid model for Recommender Systems

In this study, we have proposed a hybrid model for recommender systems that integrates the predicted ratings generated by three individual recommendation models: content-based, user-based, and item-based. We have employed two distinct approaches: the voted-based method and the weighted-based method. The voted-based method encompasses max, min, and average strategies, while the weighted-based method utilizes a neural network (MLP). The predicted ratings from three individual recommendation models are used as the input for a neural network, doing the prediction task in the hybrid recommender system. Moreover, we have integrated the KNN algorithm into the individual recommendation models, which can be either user-based or item-based CF RSs. In order to evaluate the efficacy of our proposed model, we have conducted comprehensive experiments to measure its performance.

Chapter 4: A multimodal model for fusing diverse information sources in recommender systems

Introducing the issue of heterogeneity in multi-source, multi-modal, and multi-view integration in recommender systems, reviewing studies related to integrating multi-source, multi-modal, and multi-view information in recommender systems, describing the proposed multimodal model for integrating diverse information sources in recommender systems, and presenting experimental results to evaluate the performance of the proposed model.

Chapter 5: A Transformer-based multiview integration model for Recommender Systems

Introducing the problem of multi-source, multi-modal, and multi-view transformation and fusion in recommender systems, reviewing studies related to multi-source, multi-modal, and multi-view transformation and fusion models in recommender systems, describing the proposed Transformer-based multiview integration model for recommender systems, and presenting experimental results to evaluate the performance of the proposed model.

Chapter 6: Conclusions:

This section presents the summary of the thesis, which outlines the results achieved by the PhD thesis, discussions, as well as future research plans.

List of publications: Provide a list of the works that the author has published during the course of the PhD thesis.

References: Include a list of the documents that the PhD thesis has referenced.

CHAPTER 2. BACKGROUND

2.1 Overview about recommender systems

2.1.1 Some types of recommender systems

2.1.1.1 *Basket recommender systems*

In (B. L. Wang & Schelter, 2022), the concept of Next Basket Recommendation (NBR), which seeks to suggest a collection of items to users depending on their past shopping baskets, holds significant appeal within the e-commerce and retail sector.

- A basket constitutes a collection of items, denoted by $b = i_1, i_2, \dots, i_j, \dots, i_{|b|}$, where each i_j belongs to the universal set of items denoted by I .
- In the context of a specific user, we are provided with a sequence of n historical baskets (sequenced chronologically with more recent items towards the end), indicated as $H = [b_1, b_2, \dots, b_i, \dots, b_n]$, where each b_i is a subset of items from I .
- The objective of NBR involves developing a model that takes the input of historical baskets H and predicts the subsequent basket b_{n+1} as a recommendation.

In (Vaishampayan et al., 2022), market basket analysis serves as a data processing approach to unveil connections among various items. Within the retail realm, its core objective is to furnish distributors with insights into customers' purchasing behaviors, empowering them to make optimal decisions based on this information.

2.1.1.2 *Conversational recommender systems*

In (Gao et al., 2021), static recommendation models face challenges in effectively addressing two crucial questions due to inherent limitations: (a) determining the precise preferences of a user, and (b) understanding the reasons behind a user's preference for a particular item. These limitations stem from the approach employed by static models, which learn user preferences without users'

explicit instructions and active feedback. However, the emergence of conversational recommender systems (CRSs) fundamentally transforms this scenario.

Definitions and characteristics:

A CRS is a software system designed to aid users in achieving recommendation-related objectives through engaging in dialogues consisting of multiple turns. The technical architecture of CRS solutions varies based on the system's functionality, such as whether voice input is supported or not. Nevertheless, several typical conceptual components that interact with each other can be identified in these architectures, as depicted in Figure 2.1. A key characteristic of CRS is their task-oriented nature, meaning they support specific recommendation tasks and goals. The primary objective of the system is to offer recommendations to users, aiming to assist their decision-making process or help them discover relevant information. In addition to recommendation tasks, CRS may also involve tasks such as collecting user preferences or providing explanations.

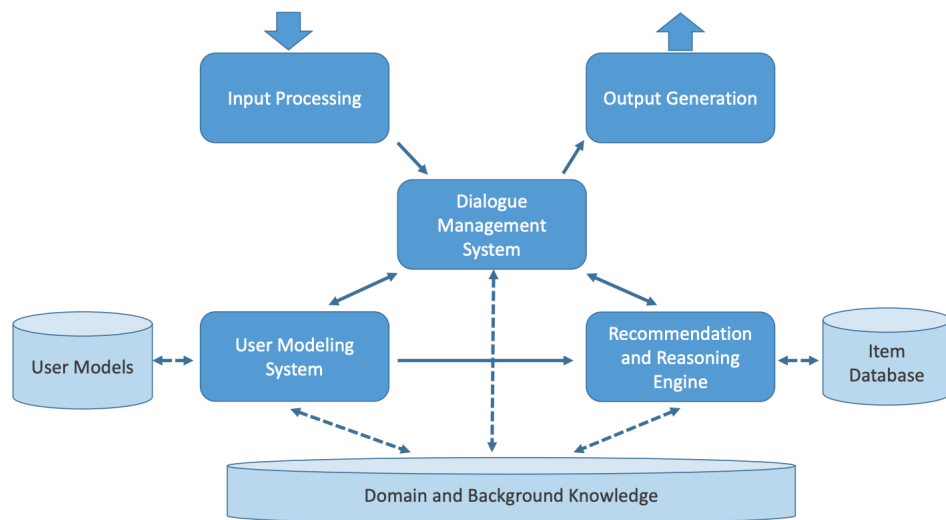


Figure 2.1 A typical architecture for CRSs

Source: [Gao, C., 2021]

A CRS enables multi-turn conversational interactions, setting it apart from

systems that exclusively provide support for question answering (Q&A tools). Commonly found in personal digital assistants, one-shot Q&A-style recommendations allow reliable responses to specific recommendation queries, such as suggesting restaurants. Nevertheless, these systems frequently encounter difficulties in sustaining multi-turn conversations. As a result, a CRS incorporates dialogue state management, either explicitly or implicitly, to keep a record of the conversation history and the current situation.

CRS can be interacted with through voice commands, typed text, or inputs via various means like form fields, buttons, and even gestures. Furthermore, the output isn't constrained and can encompass voice, speech, text, or multimedia content. Moreover, no presumptions are made regarding the driving force behind the dialogue.

There are five key challenges in developing CRSs:

1. **Question-based user preference elicitation:** CRSs offer the capability to explicitly extract user preferences by posing questions. Addressing two crucial inquiries becomes necessary: (1) Which questions should be posed? (2) How should the recommendations be adjusted based on the user's responses? The former revolves around formulating questions that elicit maximum information, while the latter entails utilizing the user's responses to generate more suitable recommendations.
2. **Multi-turn conversational recommendation strategies:** The system is required to engage in repeated interactions with a user and dynamically adapt based on the user's responses across multiple turns.
3. **Dialogue understanding and generation:** A key focus of CRSs is to engage in natural language conversations with humans, emphasizing the need for understanding human intentions and generating responses that are easily comprehensible to humans. However, many CRSs primarily extract essential

information from processed structured data and present the output through rule-based template responses. This approach necessitates significant manual effort to construct rules or templates and results in an over-reliance on preprocessing. Additionally, it hampers the user experience by imposing constraints on the interaction, making it feel unnatural in real-world applications.

4. **Exploitation-exploration trade-offs:** One challenge faced by recommender systems is that each user typically only interacts with a small subset of the available items, leaving a significant number of potentially interesting items unseen. This issue is particularly pronounced for cold-start users. CRSs alleviate this problem by actively exploring unseen items, enhancing the system’s ability to capture user preferences. Consequently, users have the opportunity to express their intentions and receive better-personalized recommendations. However, users have limited time and energy to interact with the system, and unsuccessful exploration can result in wasted time and missed opportunities for accurate recommendations. Furthermore, presenting unrelated items can negatively impact user preferences compared to exploiting previously captured preferences by recommending highly confident items.
5. **Evaluation and user simulation:** Unlike static recommender models that are fine-tuned using offline data, CRSs emphasize enhancing the user experience during real-time interactions. As a result, the evaluation of CRSs entails the consideration of both assessment at the level of individual conversational turns, encompassing recommendation and response generation, as well as an evaluation spanning the entire conversation. However, evaluating CRSs demands a significant volume of online user interactions, which can incur substantial costs. Feasible approaches to tackle this challenge include: (1) making use of off-policy evaluation, which assesses the target policy using logged data from the behavior policy, and (2) integrating user simulators as

stand-ins for real users within the evaluation process.

2.1.1.3 Sequential recommender systems

The presence of sequential dynamics plays a crucial role in numerous contemporary recommender systems, aiming to grasp the contextual aspects of users' recent actions. To capture these patterns, two prominent approaches have emerged: Markov Chains (MCs) and RNNs (Kang & McAuley, 2018).

- MC operate under the assumption that a user's subsequent action can be predicted solely based on their recent actions, or a limited history of actions.
- In contrast, RNNs offer the potential to uncover longer-term semantics and dependencies.

In general, MC-based methods demonstrate superior performance in highly sparse datasets, prioritizing model simplicity, whereas RNNs excel in denser datasets where a higher level of model complexity is feasible.

Sequential recommendation, also known as next-item recommendation, takes into account the sequence of actions and models independent sequential patterns regardless of time. Essentially, sequential models aim to capture the contextual aspects of users' recent activities (Kang & McAuley, 2018).

MC-based Recommendation: Sequential recommendation methods based on MC effectively capture local sequential patterns by assuming that the next item is solely dependent on the previous L items. Existing MC-based sequential recommendation approaches utilize either first-order MCs or higher-order MCs. The first group of methods tends to yield superior results in sparse datasets. However, higher-order MC methods possess two limitations:

1. The MC order L needs to be predetermined before training and cannot be adaptively selected.

2. The performance and efficiency do not scale well with the order L , leading to typically small values for L (e.g., less than 5).

RNN-based Recommendation: An alternative approach explores the utilization of RNNs to capture patterns in user action sequences. RNNs are commonly employed for modeling sequential data, although recent research suggests that CNNs and self-attention can outperform RNNs in certain sequential contexts (Kang & McAuley, 2018).

2.1.2 The performance metrics for recommender systems

2.1.2.1 Predictive Accuracy metrics

Predictive precision or rating estimation metrics deal with the question of how closely a recommender system's predicted ratings align with actual user ratings. This category of measurement is extensively employed to assess ratings that are not binary in nature. It finds optimal applicability in situations where precise rating prediction for all items holds significant importance. The most important measures for this purpose are Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Normalized Mean Absolute Error (NMAE).

Mean Absolute Error (MAE): MAE is employed to calculate the mean of the absolute differences between actual and predicted ratings (Chai & Draxler, 2014). A smaller MAE indicates higher accuracy. Typically, MAE can span from 0 to infinity, wherein infinity signifies the highest error, contingent on the rating scale employed by the application. Computing the MAE is accomplished with the following formula (Herlocker et al., 2004):

$$MAE = \frac{1}{n} \sum_{i=1}^n |d_i - \hat{d}_i| \quad (2.1)$$

where d_i is the actual rating, $\hat{d}_i$ is the predicted rating and n is the amount of ratings.

Root Mean Squared Error (RMSE): RMSE calculates the average of the squared differences between the true and predicted ratings, and subsequently takes the square

root of the result (Chai & Draxler, 2014). As a consequence, large errors may dramatically affect the RMSE rating, rendering the RMSE metric most valuable when significantly large errors are unwanted. The RMSE between the true ratings and predicted ratings is given by (Herlocker et al., 2004):

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (d_i - \hat{d}_i)^2} \quad (2.2)$$

where d_i is the true rating, $\hat{d}_i$ is the predicted rating and n is the amount of ratings.

Normalized Mean Absolute Error (NMAE): This metric is utilized to enable the comparison of MAE across datasets that possess varying scales. Serving as a normalization mean, the evaluation tool for model performance employs the average of the measured data.

$$NMAE(y, \hat{y}) = \frac{MAE(y, \hat{y})}{mean(|y|)} \quad (2.3)$$

2.1.2.2 Classification Evaluation metrics

Precision: The precision is used to know how many items the user really liked considering all recommended items.

$$Precision = \frac{tp}{tp + fp} \quad (2.4)$$

where tp is the number of recommended items the user liked (true positives) and $tp + fp$ is the total number of recommended items (true positives, and false positives). (Júnior et al., 2020)

Recall: The recall is used to calculate the proportion of recommended items that the user likes.

$$Recall = \frac{tp}{tp + fn} \quad (2.5)$$

where tp is the number of recommended items the user likes (true positives) and

$tp + fn$ is the total number of items that user likes (true positives, and false negatives) (Júnior et al., 2020). **F1 Score:** Precision and Recall are used to calculate the F1-score, the formula is given as the following:

$$F1\text{-score} = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \quad (2.6)$$

2.1.2.3 Evaluating recommender systems that provide top-N recommendations

Hit rate: To calculate the hit rate, begin by identifying all items from the user's history in the training data. Then, remove one of these items as part of leave-one-out cross-validation. Utilize the remaining items to generate recommendations, selecting the top 10. If the removed item is among the top 10 recommendations, it is considered a hit. Conversely, if the item does not appear in the top 10, it is not counted as a hit.

Average reciprocal hit rate (ARHR): The ARHR assigns higher credit to recommendations that are ranked closer to the top, as they align with the items that the user has rated.

Cumulative hit rate: In the cumulative hit rate, ratings below a specific threshold, such as those lower than 4, are excluded or considered as rejections.

Rating hit rate: To compute the rating hit rate, individual rating scores are calculated to determine which types of ratings receive more hits. The number of hits for each rating type in the top-N list is summed, and then divided by the total number of items belonging to each rating category within the top-N list.

2.2 The machine learning methods

2.2.1 Matrix Factorization

The Utility Matrix is utilized by the MF approach (Kotu & Deshpande, 2019) to identify latent features that correspond to users and items. If we assume that we have

a rating matrix with a shape of $(n \times m)$ representing the ratings given by n users on m items, the MF method is employed to decompose R into two thin matrices, P and Q , with shapes of $n \times f$ and $m \times f$ respectively. Here, f refers to the number of latent factors that contain critical latent information. Figure 2.2 provides an illustration of how we can decompose the matrix R into matrices P and Q .

To predict each rating r of the pair (user u , item i), we use the following formula:

$$\hat{r}_{u,i} = p_u \cdot q_i^T \quad (2.7)$$

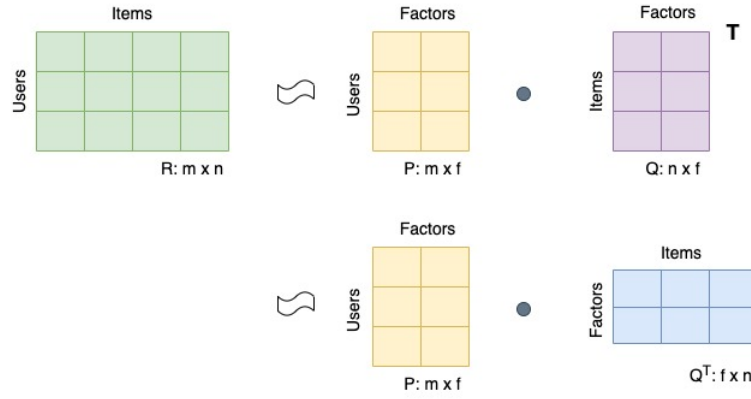


Figure 2.2 The decomposing the Utility Matrix into matrices of latent factors.

where p_u and q_i correspond to the user and item vectors, respectively, both of which have f dimensions. The product of p_u and q_i^T indicates the preference of users for items.

2.2.1.1 Singular Value Decomposition

The SVD algorithm is a useful tool for constructing recommender systems. It involves breaking down a matrix R into a simpler lower-rank approximation, making it easier to manipulate. This is done by factorizing R into two unitary matrices and a diagonal matrix, as shown in the mathematical equation:

$$R = U \Sigma V^T \quad (2.8)$$

Here, R represents the Utility Matrix with dimensions $m \times n$, U represents the singular matrix of the orthogonal left side indicating the relationship between users and latent features with dimensions $m \times r$, Σ represents the diagonal matrix describing the strength of latent features with dimensions $r \times r$, and V represents the singular matrix of diagonal right side indicating the similarity between items and latent features with dimensions $r \times n$. The aim is to extract latent factors by reducing the dimension of R .

Each user is represented by a vector y_u and each item by a vector x_i . The formula in equation 2.9 can be used to calculate the expected rating by a user on an item $\hat{r}_{ui}$:

$$\hat{r}_{ui} = x_i^T \cdot y_u \quad (2.9)$$

2.2.1.2 Alternating Least Square

A commonly employed approach in MF entails employing a predetermined value, often denoted as k (for example, $k \approx 10$), which represents the number of dimensions. With this, we condense the features of each user u , into a vector x_u with k dimensions, while for each item i , a k -dimensional vector y_i is created. These vectors, often referred to as factors, are pivotal in this technique. To forecast the rating assigned by a user u to an item i , the computation $r_{ui} \approx x_u \cdot y_i$ suffices. This equation can be concisely represented in matrix format: Let $x_1, \dots, x_n \in R_k$ correspond to the user factors and $y_1, \dots, y_m \in R_k$ denote the factors related to the items.

Our objective is to approximate the entire ratings matrix R by using the equation $R \approx X_T \cdot Y$. To achieve this, we can frame the problem as an optimization task, where our aim is to minimize an objective function and determine the optimal values for X and Y . Specifically, our goal is to minimize the least squares error of the observed ratings while also applying regularization.

$$\min_{X,Y} \sum_{r_{ui} \text{ observed}} (r_{ui} - x_u^T)^2 + \lambda \left(\sum_u \|x_u\|^2 + \sum_i \|y_i\|^2 \right) \quad (2.10)$$

It is important to observe that the objective function described here is non-convex due to the presence of the term $x_u^T \cdot y_i$. In fact, optimizing this objective is known to be NP-hard. While gradient descent can be utilized as an approximate solution, it tends to be slow and requires numerous iterations. However, it is worth noting that if we consider X and Y as fixed variables, the objective becomes a convex function with respect to the other variable. Consequently, our approach involves fixing Y and optimizing X , then fixing X and optimizing Y , repeating this process until convergence. This iterative technique is commonly referred to as Alternating Least Squares (ALS) (Zadeh et al., 2015).

2.2.2 Bidirectional Encoder Representations from Transformers

BERT, as introduced by (Devlin et al., 2018), stands as a pre-trained language model that finds widespread application across diverse NLP tasks. It takes its foundation from the Transformer architecture, which harnesses a multi-head attention mechanism for the efficient encoding of textual data. The initial representation of input text takes the form of a token sequence, subsequently converted into one-hot vectors. These vectors then navigate through multiple Transformer layers, where inter-word attention computations facilitate the creation of word representations. Notably, BERT also integrates positional information into its framework to bolster performance. As part of its training process, the model conceals certain words during input, predicting them subsequently at the output layer.

BERT has garnered extensive adoption as a preprocessing utility for various NLP tasks, showcasing its prowess through the attainment of cutting-edge outcomes. In the context of this research, we will employ BERT_BASE to produce word vectors. This particular variant of BERT encompasses 12 layers, each containing 768 hidden units, culminating in a grand total of 110 million parameters. Furthermore, an alternative version, BERT_LARGE, incorporates 24 layers, 1024 hidden units, and an impressive

340 million parameters.

2.2.3 Multi-Layer Perceptron

The MLP (Abinaya & Devi, 2022), which is a type of feed-forward neural network, consists of three layers: an input layer, a hidden layer, and an output layer. The input layer receives the input signal, while the output layer performs classification and prediction functions. The MLP's connected mechanism is a series of hidden layers located between the input and output layers, which can be infinite. In a feed-forward network like the MLP, data is processed in a forward path from the input to the output layer. To train all nodes in the MLP, the backpropagation learning technique is utilized. The MLP is capable of addressing non-linearly separable issues and can approximate any continuous function through its structure.

The computations executed by each neuron in the hidden and output layers are defined by Equations 2.11 and 2.12, which involve bias vectors c_1 and c_2 , weight matrices V_1 and V_2 , and activation functions ρ and Φ . The set of parameters to be learned consists of $\theta = \{V_1, V_2, c_1, c_2\}$. Two options for activation functions are the tanh function, given by $\tanh(x) = e^x - e^{-x} / e^x + e^{-x}$, and the logistic sigmoid function, which is $\text{sigmoid}(x) = 1 / (1 + e^{-x})$.

$$hr = \Phi r = \rho V_1 gr + c_1 \quad (2.11)$$

$$gr = \varphi(V_2 h(r) + c_2) \quad (2.12)$$

2.2.4 Recurrent Neural Network

2.2.4.1 Long Short-Term Memory

The LSTM Network, discussed in the work by (Saxena, 2021), represents a more advanced variant of a RNN, a type of sequential network designed to enable the preservation of information. Notably, it possesses the ability to effectively address the issue of the vanishing gradient, which is a challenge encountered by traditional RNNs and is particularly relevant for tasks requiring sustained memory

retention. Figure 2.3 is the intuitive diagram of the LSTM network.

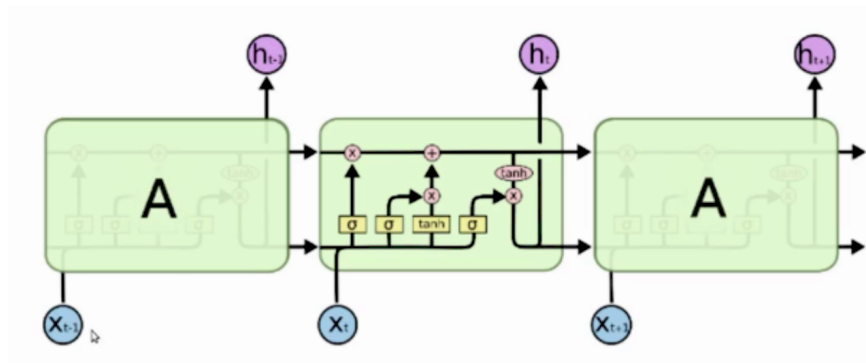


Figure 2.3 The intuitive diagram of a LSTM network.

Source: [Shipra Saxena, 2021]

The architecture of LSTM is comprised of three components: The initial component is referred to as the Forget gate, followed by the second component known as the Input gate, and ultimately, the third component designated as the Output gate.

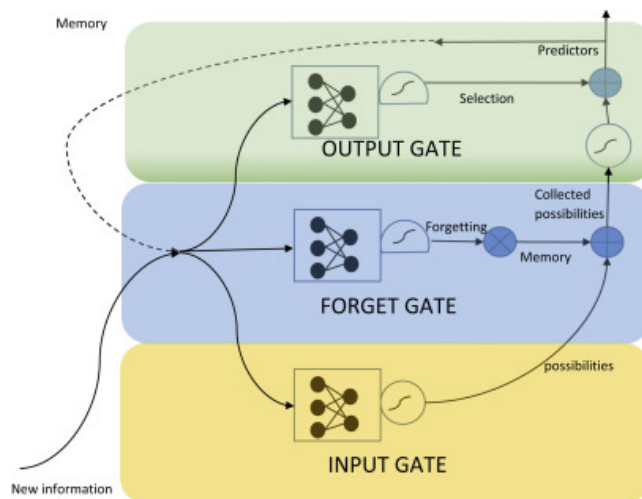


Figure 2.4 A typical LSTM cell structure showing its main three components.

Source: [Greff et al., 2017]

Forget Gate: Within an LSTM network cell, the initial action involves determining whether to retain or discard information from the preceding timestamp. The formula

for the forget gate is provided as follows:

$$f_t = \sigma(x_t \times U_f + H_{t-1} \times W_f) \quad (2.13)$$

where x_t is input to the current timestamp, U_f is weight associated with the input, H_{t-1} is the hidden state of the previous timestamp, W_f is the weight matrix associated with hidden state.

Later, a sigmoid function is applied over it. That will make f_t a number between 0 and 1. If f_t is 0 then the network will forget everything and if the value of f_t is 1 it will forget nothing.

Input Gate: Input gate is used to measure the importance of the new information carried by the input. The equation of the input gate is given below:

$$i_t = \sigma(x_t \times U_i + H_{t-1} \times W_i) \quad (2.14)$$

where x_t is input at the current timestamp t , U_i is the weight matrix of input, H_{t-1} is a hidden state at the previous timestamp, W_i is the weight matrix of input associated with hidden state. Later applying sigmoid function over it, so the value of I at timestamp t will be between 0 and 1.

Output Gate: Below is the equation of the Output gate, which is pretty similar to the two previous gates.

$$o_t = \sigma(x_t \times U_o + H_{t-1} \times W_o) \quad (2.15)$$

Due to the sigmoid function, the value it produces will also fall within the range of 0 to 1. To compute the current hidden state, we employ both O_t and the tanh of the updated cell state, as demonstrated below.

$$H_t = o_t \times \tanh(C_t) \quad (2.16)$$

Interestingly, the hidden state is reliant on both the Long term memory (C_t) and the present output. To obtain the output for the current timestamp, you can simply apply the SoftMax activation to the hidden state H_t .

$$Output = Softmax(H_t) \quad (2.17)$$

2.2.4.2 Gated Recurrent Unit

GRU networks (Shen et al., 2018) are a type of RNNs that address the issues of vanishing and exploding gradients in traditional RNNs when dealing with long-term dependencies. As a gated RNN, GRU comprises an input layer with multiple neurons, an output layer whose size corresponds to the output space, and hidden layers with memory cells that perform the network's primary functions. These memory cells are updated using two gates: the reset gate and the update gate, which filter out irrelevant information from the previous time step and current input. The reset gate determines the importance of the previous information, while the update gate decides whether to ignore the current input and form a "short-circuit connection" from the previous memory cell to the current one.

At each time step, the equations presented below illustrate how the memory cells in each hidden layer of the GRU networks are updated.

The equation for the reset gate is: $r_t = \sigma(W_r x_t + U_r h_{t-1})$.

The update gate is calculated using the equation: $u_t = \sigma(W_u x_t + U_u h_{t-1})$.

The output of the GRU network is obtained using Equations 2.18 and 2.19, respectively.

$$h_t = (1 - u_t)h_{t-1} + u_t \tilde{h}_t \quad (2.18)$$

$$\tilde{h}_t = \tanh((W_h x_t) + r_t \odot (U_h h_{t-1})) \quad (2.19)$$

The GRU network processes the inputs in chronological order, updating the memory cells at each time step using the equations described above. The network outputs the final result only after processing the last element of the input sequence.

2.2.5 Transformer

The Transformer encoder architecture, which is illustrated in Figure 2.5 and was first introduced in (Vaswani et al., 2017), has gained widespread use in natural language processing (NLP) for tasks involving sequence-to-sequence, such as machine translation and language modeling. One of the strengths of this architecture is its ability to effectively capture long-range dependencies, as well as its parallelizability.

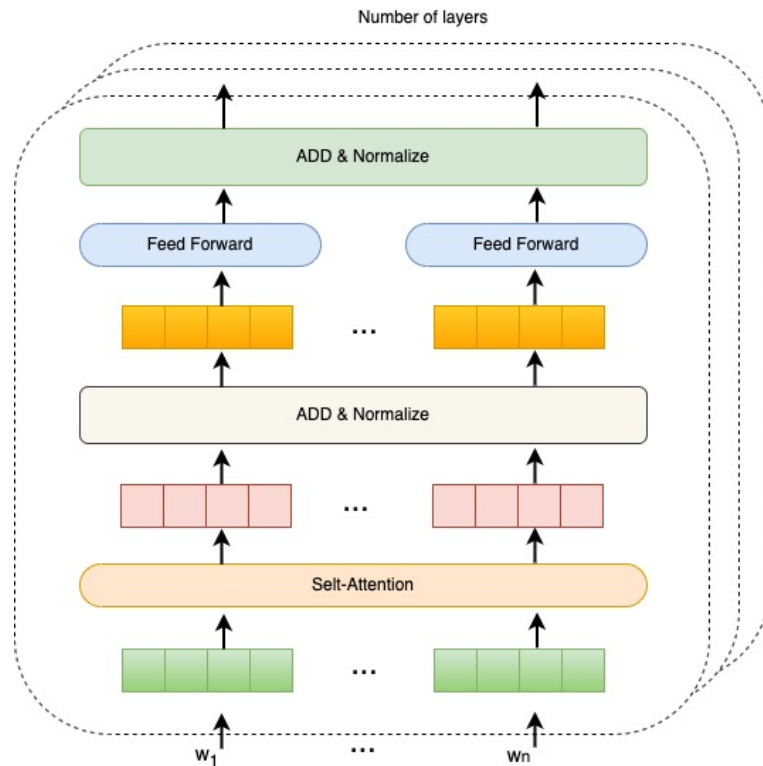


Figure 2.5 An example for the Transformer encoder architecture.

The fundamental components of the Transformer encoder design encompass input embeddings, positional encoding, multi-head self-attention, layer normalization, position-wise feed-forward networks, residual connections, and layer

stacking.

Throughout the training process, input embeddings grasp the significance of input tokens, whereas positional encoding is introduced to tackle the lack of built-in positional context in the Transformer architecture. This addition permits the model to grasp the sequential arrangement of tokens within the input sequence.

At the heart of the Transformer encoder lies the multi-head self-attention mechanism, which empowers the model to evaluate the importance of diverse tokens and effectively grasp long-range relationships. Following self-attention, layer normalization is implemented to normalize the results, contributing to stable training. Position-wise feed-forward networks seize intricate interplays among tokens, while residual connections aid in maintaining gradient flow and mitigating the challenge of vanishing gradients.

Typically, the Transformer encoder structure encompasses multiple identical layers stacked sequentially. The output of each layer serves as the input for the subsequent one, enabling the model to capture hierarchical representations of the input sequence.

In summary, the distinguishing features of the Transformer encoder architecture encompass its self-attention mechanism, layer normalization, and residual connections, which collaboratively establish a potent framework for handling sequence-to-sequence tasks in NLP.

2.2.6 Graph Neural Network

The Graph definition: The representation of a graph (Wu et al., 2020), denoted as $G = (V, E)$, consists of a set of vertices or nodes, denoted as V , and a set of edges, denoted as E .

To indicate a node, v_i , belongs to V , and an edge, $e_{ij} = (v_i, v_j)$, points from v_j to v_i . The neighborhood of a node v is defined as $N(v) = \{u \in V | (v, u) \in E\}$. An adjacency

matrix, denoted as A , is an $n \times n$ matrix where $A_{ij} = 1$ if $e_{ij} \in E$, and $A_{ij} = 0$ if $e_{ij} \notin E$.

A graph may have node attributes, represented by X , where $X \in R^{n \times d}$ is a node feature matrix and $x_v \in R^d$ represents the feature vector of a node v .

Additionally, a graph may possess edge attributes, denoted as X^e , where $X^e \in R^{m \times c}$ is an edge feature matrix and $x_{v,u}^e \in R^c$ represents the feature vector of an edge (v, u) .

The Directed Graph definition: A graph is categorized as a directed graph (Wu et al., 2020) when all its edges have a specific direction from one node to another. In contrast, an undirected graph is viewed as a particular scenario within directed graphs, where a pair of edges with opposite directions exists to connect two nodes. The classification of a graph as undirected is solely dependent on the symmetry of its adjacency matrix.

The Spatial-Temporal Graph definition: A spatial-temporal graph (Wu et al., 2020) is a graph with attributes associated with its nodes that dynamically change over time. The spatial-temporal graph is represented as $G^{(t)} = (V, E, X^{(t)})$, where $X^{(t)} \in R^{n \times d}$ denotes the attribute matrix of the graph at time t .

2.3 Feature Extraction

2.3.1 Term Frequency-Inverse Document Frequency (TF-IDF)

TF-IDF is a technique which is used to evaluate the significance of terms regarding document classification or document content. This method, which is a variation of the Bag of Words model (Eklund, 2018), relies on calculating both term frequency (TF) and inverse document frequency (IDF). The formula for TF may be expressed as:

$$tf_{t,d} = \frac{n_{t,d}}{\sum_k n_{k,d}} \quad (2.20)$$

In the formula for TF-IDF, $n_{t,d}$ represents the frequency of term t in document

d , while $n_{k,d}$ denotes the frequency of all terms in document d . The IDF, on the other hand, is calculated using the following formula:

$$idf_t = \log\left(\frac{|D|}{|D_t|}\right) \quad (2.21)$$

In the context of calculating IDF for the TF-IDF formula, $|D|$ represents the total number of documents in a corpus, whereas $|D_t|$ represents the number of documents in which the term t appears.

2.3.2 Principal Component Analysis

PCA is a technique which is commonly utilized in data analysis and machine learning for dimensionality reduction. By reducing the number of variables while maintaining as much information as possible, PCA can identify patterns in high-dimensional data. It is advantageous for reducing computational complexity, pinpointing significant features or variables, and eliminating noise and irrelevant information from high-dimensional data.

The transformation performed by PCA converts correlated variables into a set of nonlinearly correlated variables, known as principal components (PCs). This transformation ensures that the principal components capture the most variation in the data. The process of PCA involves several steps, as outlined in (Shlens, 2014):

Calculate the covariance matrix based on the sample data. Determine the eigenvalues and eigenvectors of the covariance matrix. Generate the PCs using the following formula:

$$p_i = b_{i1}X_1 + b_{i2}X_2 + \cdots + b_{ik}X_k \quad (2.22)$$

Here, p_i represents the i^{th} PC, b_{ik} represents the weight, and X_k represents the variable.

CHAPTER 3. A HYBRID MODEL FOR RECOMMENDER SYSTEMS

3.1 Introduction

Recommender systems play an increasingly crucial role in various industries such as entertainment, services, and e-commerce. They assist businesses in boosting their sales revenue by suggesting items that closely align with a user's preferences and requirements.

Recent research has primarily focused on similarity-based recommender systems, which encompass two approaches: content-based filtering and collaborative filtering. Content-based filtering systems recommend items that share similar characteristics with those that a user has previously shown interest in, without considering the similarities among users. On the other hand, collaborative filtering systems employ a memory-based model and offer two approaches: user-based and item-based. In the user-based approach, ratings from users who possess similar preferences to the target user are utilized to suggest items, disregarding item similarities. Conversely, the item-based approach identifies a set of items that exhibit the highest level of similarity to the target item, and the ratings of those items specified by the target user are employed to predict their preference for the target item, without considering user similarity.

To leverage the metrics derived from the aforementioned three models, we propose a hybrid model that combines them using minimum, maximum, and average operations on ratings from three individual models. Additionally, we suggest a weighted combination approach using a neural network.

This study presents two main contributions. Firstly, we propose a hybrid model to enhance RSs. Secondly, we conduct experiments on the ML-GroupLens 20M dataset, demonstrating that our weighted-based hybrid model outperforms other

models in terms of Precision and RMSE metrics. Furthermore, we experiment on the Amazon sub-dataset and illustrate that our weighted-based hybrid model, employing a multi-strategy combination, surpasses other individual models in terms of MAE, RMSE, and Precision metrics.

3.2 Related Works

Content-based RSs suggest items to users by assessing their similarity to previously preferred or expressed interest items. These systems rely on input data such as item and user descriptions or content. (Lehinevych et al., 2014) investigated the relationship between features like audio, visualization, and text and the similarity of movie content. They computed three similarity matrices based on Gaussian Mixture Models and Latent Semantic Indexing. (McFee et al., 2012) aimed to enhance content-based similarity by optimizing the content-based similarity metric. (Kant & Bharadwaj, 2012) improved content-based filtering by integrating collaborative diverse predictions with an item representation scheme and fuzzy similarity measures to enhance recommendation diversity. (Peng et al., 2010) constructed a content-based RS utilizing a weighted unidirectional graph to depict item similarity based on semantic relations. (Ghauth & Abdullah, 2010) proposed a framework that recommends learning materials with similar content, utilizing ratings from proficient learners to indicate material quality. (Cantador et al., 2010) introduced and evaluated different content-based recommendation models that employ weighted lists of social tags to define users' and items' profiles. (Zenebe & Norcio, 2009) proposed a fuzzy set theoretic approach that incorporates a representation method for item features and user feedback using fuzzy sets, along with a content-based algorithm based on various fuzzy set theoretic similarity measures. (Tintarev & Masthoff, 2006) compared Lin's taxonomy-based measure and the WASP measure with human judgments of similarity for a news RS.

In (Aggarwal et al., 2016), collaborative filtering RSs rely on the concept of

similarity and offer two approaches: model-based (e.g., Bayesian networks, MF, Restricted Boltzmann machines) and memory-based (e.g., user-based or item-based). (Agarwal & Chauhan, 2017) demonstrated that the prediction accuracy depends on the similarity measures employed to identify similar users, and better similarity measures result in enhanced predictions. (Mahara et al., 2016) introduced a novel similarity approach that considers users' rating habits using the Mean Measure of Divergence. (Alhijawi & Kilani, 2016) presented a genetic algorithms-based recommender system, SimGen, which computes similarity values between users without relying on well-known similarity measures like Pearson correlation and vector cosine-based similarity. (Patra et al., 2015) proposed a similarity measure that employs Bhattacharyya similarity to determine the importance of each pair of rated items. (H. Liu et al., 2014) incorporated a sigmoid function to penalize and reward similarity, and modified the PSS measure by combining it with the modified Jaccard to account for the small proportion of common ratings. (Mao et al., 2013) introduced a similarity impact factor to enhance similarity measures in CF recommendation algorithms. (Kim et al., 2011) developed collaborative error-reflected models that utilized the similarity of like-minded users and the similarity of items rated by the same user to predict ratings. (Ahn, 2008) improved recommendation performance in cold-start scenarios using a novel heuristic similarity measure.

Other research related to the hybrid-based approach, which is also known as a main approach of RSs, can be mentioned as in (S. Yang et al., 2017), they adapted the latest Statistical Relational Learning approaches to construct a practical hybrid job RS. Additionally, their approach allows for a principled trade-off between system precision and recall to be tuned, satisfying the typical requirement in RSs that false positives are more undesirable and should be penalized more severely than false negatives. A hybrid recommendation model was proposed by (Jiang et al., 2020), which utilizes deep emotion analysis and multi-source view fusion to provide personalized recommendations based on user-post interaction ratings, implicit

feedback, and auxiliary information in a hybrid RS. The model employs the HITS algorithm to process the data set, filter out highly influential users and posts, and design calculation methods for measuring post similarity and calculating K nearest neighbors, while also utilizing cooperative training strategies for the fusion of two recommended views. A movie RS has been proposed by (Geetha et al., 2018) that aims to suggest movies to users based on their interests and the popularity of movies. The system can recommend movies to both new and existing users by mining movie databases to gather important information such as popularity and attractiveness. To provide more precise recommendations, the system uses a combination of CB and CF techniques, known as hybrid filtering. (Sharma et al., 2022) suggested a hybrid RS that combines CF and CB, consisting of three phases: identifying similar users based on profile matching, selecting candidate items for each similar user using vector representations of user profiles and item contents, and using the Resnick prediction equation to calculate prediction values and suggest items to the target user in the final phase. In their study, (Thakkar et al., 2019) investigated various methods for combining predictions from user-based and item-based CF (UbCF and IbCF) to reduce overall prediction error. Specifically, they used multiple linear regression (MLR) and support vector regression (SVR) to combine predictions from UbCF and IbCF. (Walek & Fojtik, 2020) introduced Predictory, a monolithic hybrid RS for suggesting suitable movies. It combines a CF module, a CB system, and a fuzzy expert system that evaluates various parameters to determine the final list of recommended movies. (Kiran et al., 2020) developed a hybrid RS that utilizes deep learning and embeddings to tackle the cold-start problem. By integrating side information about users and items into a very deep neural network, the proposed solution learns non-linear latent factors, and improves accuracy through decreasing learning rate and increasing weight decay values cyclically varied across epochs. (Gatzioura et al., 2019) proposed HybA, a hybrid RS that addresses playlist continuation based on similar concepts rather than similar users, using LDA and

Case-Based Reasoning. HybA identifies semantic characteristics of a started playlist, reuses the most similar historical ones, and provides support for beyond accuracy dimensions such as increased coherence or diverse items' discovery, using a graph model to overcome the semantic gap between music descriptions and user preferences. (Biswas & Liu, 2022) developed a hybrid RS that uses collaborative filtering with ALS and deep learning to improve recommendation performance and address the "cold-start" problem. Their approach involves utilizing ALS outputs to influence recommendations from a DNN that incorporates various types of information within a big data processing framework. In their paper, (Vall et al., 2019) presented two hybrid RSs that combine music playlist data with any type of song feature vector representation to enhance collaborative filtering. These systems leverage curated music playlists to incorporate collaborative information into the recommendation process. (Chalkiadakis et al., 2023) created a hybrid RS for the tourism industry that includes a Bayesian preferences elicitation component and a novel CB recommendations component. The CB component incorporates two semantic similarity measures, a hierarchy-based algorithm and a non-hierarchy-based algorithm, including the Weighted Extended Jaccard Similarity (WEJS). The user model is built based on ratings of generic images corresponding to types of points of interest. (Vahidi Farashah et al., 2021) proposed a four-phase hybrid method that employs link prediction and combines a CF with a CB. The method utilizes Density-based spatial clustering of applications with noise algorithm (DBScan) and Deep Neural Network (DNN) algorithm for user clustering and classification, age, gender, and occupation-based similarity criteria, and the improved Friendlink algorithm to calculate user similarity and suggest higher-rated movie services to new users. (Logesh & Subramaniaswamy, 2019) identified research problems in e-Tourism applications and proposed a solution for improving personalized recommendations. They developed the Personalized Context-Aware Hybrid Travel RS (PCAHTRS) by incorporating contextual information from users.

(Walek & Fajmon, 2023) proposed the Eshop recommender, a hybrid RS that uses CF, CB approaches, and a fuzzy expert system to recommend products based on user preferences and activity. The fuzzy expert system considers similarity level, coefficient of purchased product, and average rating. (Ngaffo et al., 2020) suggested a hybrid model-based recommendation approach that combines a user-based approach and an item-based approach. In order to estimate the probability of a user rating an item, they utilize Bayesian inference, showcasing the simplicity of memory-based methods and the effectiveness of model-based methods. In their paper, (Khan et al., 2020) introduced a hybrid RS model called Deep Semantic based Topic driven Hybrid RS (DST-HRS). This model leverages the semantic information of item descriptions, which is influenced by its topics, to extract embeddings. The embeddings are then integrated into Probabilistic Matrix Factorization (PMF) to overcome the sparsity issue in RS. By incorporating textual information such as reviews, synopsis, comments, plots, etc., DST-HRS efficiently exploits the semantics of the items. The research conducted by (Do et al., 2020) was centered on weighted hybridization. Instead of utilizing fixed weights for combining CF and CB, their aim was to suggest a straightforward approach to dynamically weighting the combination. Collaborative filtering faces two main challenges: (1) no explicit product ratings from customers for recommendations, and (2) difficulty recommending products to new customers with no preferences. To overcome the lack of ratings, (Oyebode & Orji, 2020) developed an algorithm to infer customer preferences from transaction data. For the cold-start issue, they proposed a hybrid RS combining item-based CF with demographic-based approach. (D. Yang et al., 2021) proposed TP-HR, a personalized hybrid RS that combines a time-aware CF algorithm with a temporal association rule-based recommendation algorithm to address concept and user interest drifting. The time-aware CF algorithm tracks preference dynamics with time-aware neighbor selection and item rating prediction functions, while the temporal association rule-based recommendation algorithm

mines effective rules while considering the time context of users' historical behaviors. (Y. Yang et al., 2020) proposed a hybrid RS called U2CMS, which combines CF and CB similarity models with Markov chains to improve sequential recommendation. U2CMS utilizes a higher-order Markov chain to capture sequential patterns and text information of content for accurate item relationship modeling.

In the provided research about hybrid RSs above, several authors proposed different approaches to combining CF and CB RSs to provide users with more accurate and personalized recommendations. The studies have explored various techniques, including deep emotion analysis, latent factor modeling, hierarchical similarity measures, and fuzzy expert systems. Additionally, the research focused on addressing the cold-start problem by incorporating user and item side information into deep learning models. Several authors have proposed hybrid RSs for different domains, including movie recommendation, music playlist continuation, and e-tourism. Many studies employ machine learning algorithms such as clustering, classification, and regression to develop the hybrid systems.

Our research incorporates both CB and memory-based CF RSs. Previous studies on CF models have primarily concentrated on individual approaches, such as content similarity, similarity among users with similar preferences, or similarity between items rated by the same user, to generate recommendations. In contrast, our approach introduces a hybrid model that combines predicted ratings from three individual recommendation models using various multi-strategy combinations. The combined result is then utilized as input for a neural network to predict user preferences for items in RSs.

3.3 The Proposed Model

Our first step is to demonstrate how to estimate the ratings from three individual recommendation models: CB, user-based CF, and item-based CF. We then introduce

a hybrid recommendation model using multi-strategy combination, that leverages the strengths of each individual model. Lastly, we present the mathematical formalization of our proposed hybrid model.

3.3.1 Basic models

The basic models of RSs mentioned in our work include content-based, user-based, and item-based collaborative filtering.

3.3.1.1 Content-based recommender systems

Each user has a unique level of preference for each item, which can be quantified as a value for each user-item pair if it is known in advance. This value can be assigned based on the users' ratings for items. The collection of all ratings, including unknown ratings that require prediction, constitutes a matrix known as a Utility Matrix. Building a Utility Matrix is crucial in recommender systems because it is nearly impossible to make recommendations to users without it, except by constantly suggesting the most popular items. Therefore, the creation of a Utility Matrix is of utmost importance in recommender systems. Algorithm 2 details the steps to build a CB recommender system.

3.3.1.2 Collaborative filtering recommender systems

CF is a popular technique in RSs that utilizes the behavior and preferences of similar users to make recommendations. One of the most widely used algorithms in CF is the K-Nearest Neighbors (KNN) algorithm. There are different variations of the KNN algorithm, including user-based and item-based.

- ***The user-based KNN algorithm*** (Herlocker et al., 2004) in CF RSs involves finding the K users who are most similar to a target user based on their historical behaviors or ratings, and then recommending items that those similar users have liked or rated highly. The similarity between users can be measured using various distance metrics, such as cosine similarity or Pearson correlation

Algorithm 2 An algorithm for building a CB recommender system

- 1: **Training stage:**
 - 2: Input: Given a dataset containing items' properties and a Utility Matrix $R^{n \times m}$ represents the level of preference of the i^{th} user for the j^{th} item, where n denotes the number of users and m represents the number of items.
 - 3: Output: A model for the rating prediction
 - 4: **Step 1:** Building the Feature Extraction Model (named as FEM)
 - Doing the task: Feature extraction for each item
 - Return vector of features for each item using CountVectorize and TF-IDF. Suppose that we obtain a feature vector denoted by x_i for a given item i^{th} .
 - 5: **Step 2:** Using the Utility Matrix for returning a list of ratings rated by a given user u .
 - 6: **Step 3:** Building the Prediction Model (named as PM)
 - Using feature vectors from Step 1 and ratings from Step 2, the feed them all into a Regression model to generate ratings (i.e. prediction)
 - Training the model for the prediction
 - 7: **Inference stage:**
 - 8: Input: Given a pair (user u , item i)
 - 9: Using FEM to return a feature vector item i , denoted by x_i
 - 10: Using the Utility Matrix for returning a list of ratings rated by a user u , denoted by *list_ratings_by_u*
 - 11: *list_ratings_by_u* and vector x_i as input for the PM to generate the output (i.e. the rating).
-

coefficient.

- ***The item-based KNN algorithm*** (Sarwar et al., 2001) is a variation of the KNN algorithm used in RSs. It involves finding similar items to the items a target user has interacted with or rated, and recommending the most similar items to the user. The formula for the item-based KNN algorithm involves computing the similarity between each pair of items in the dataset, finding the K most similar items for each item the target user has interacted with, and aggregating the ratings or preferences of the target user for those items.

CF models rely on the ratings provided by other users for recommending items. The ratings that are not specified are referred to as "unobserved" or "missing", while the ones that are specified are referred to as observed ratings. CF methods are based on the idea that unspecified ratings can be estimated due to the high correlation between observed ratings across various users and items. Memory-based methods are also known as neighborhood-based CF algorithms. These algorithms make predictions on user-item ratings based on their neighborhoods. There are two ways to define neighborhoods in such methods as user-based and item-based.

We consider $U = u_1, u_2, u_3, \dots, u_N$ and $I = i_1, i_2, i_3, \dots, i_M$ as the sets of users and items, respectively. The user-item rating matrix is denoted by $R = (r_{ij})_{N \times M}$, where $i = 1, 2, 3, \dots, N$, and $j = 1, 2, 3, \dots, M$. The UserKNN and ItemKNN methods for predicting the rating of user u on item i are represented by pseudocode as shown in Algorithm 3 and Algorithm 4, respectively.

After obtaining the predicted ratings from the three single models, we employed them as inputs for hybrid recommendation models that employed minimum, maximum, average, and weighted ratings.

3.3.2 The hybrid recommendation models

Algorithm 3 An algorithm for building a user-based recommender system

```

1: procedure UserKNN()
2:   for u=1 to N do
3:     Caculate  $Sim_{u_{u'}}$ 
4:   end for
5:   Sort  $Sim_{u_{u'}}$  //Sort in descending order of similarity
6:   for k=1 to K do
7:      $K_u \leftarrow k$  //k nearest users of a user  $u$ 
8:   end for
9:   for i=1 to M do
10:     $bottom\_result = \sum |Sim_{u_{u'}}|$ 
11:    if  $bottom\_result \neq 0$  then
12:       $\hat{r}_{ui} = \bar{r}_u + \frac{(r_{ui} - \bar{r}_u) \cdot Sim_{u_{u'}}}{bottom\_result}$ 
13:    else
14:       $\hat{r}_{ui} = 0$ 
15:    end if
16:  end for
17: end procedure

```

Algorithm 4 An algorithm for building a item-based recommender system

```

1: procedure ItemKNN()
2:   for i=1 to M do
3:     Caculate  $Sim_{i_{i'}}$ 
4:   end for
5:   Sort  $Sim_{i_{i'}}$  //Sort in descending order of similarity
6:   for k=1 to K do
7:      $K_i \leftarrow k$  //k nearest items of an item  $i$ 
8:   end for
9:   for u=1 to N do
10:     $bottom\_result = \sum |Sim_{i_{i'}}|$ 
11:    if  $bottom\_result \neq 0$  then
12:       $\hat{r}_{ui} = \frac{r_u \cdot Sim_{i_{i'}}}{bottom\_result}$ 
13:    else
14:       $\hat{r}_{ui} = 0$ 
15:    end if
16:  end for
17: end procedure

```

To leverage the strengths of singular models and improve the accuracy of recommendations, we have developed a hybrid model (shown in Figure 3.1) that combines predicted ratings from three different filtering methods: content-based (M1), user-based (M2), and item-based (M3). To predict rating scores for users on items, we utilized various methods that incorporated rating scores from three individual models (M1, M2, M3). These methods included the voted-based approach, which encompassed averaging-based, maximum-based, and minimum-based techniques. Additionally, we employed a weighted-based approach using a neural network (MLP) to estimate the rating scores. The predicted ratings from these three models are denoted as follows:

- X_1 is a set of the rating outputs from model M_1 .
- X_2 is a set of the rating outputs from model M_2 .
- X_3 is a set of the rating outputs from model M_3 .

The algorithm 5 for the proposed hybrid recommender system using multi-strategy combination follows the scheme below.

Algorithm 5 An algorithm for the proposed hybrid recommendation model.

- 1: **Training stage:**
 - 2: Input: X_1, X_2, X_3
 - 3: Output: A model for the prediction task
 - 4: **Step 1:** Choosing a voted-based approach or a weighted-based approach for fusion.
 - 5: **Step 2:** Building the Prediction Model
 - 6: **2.1. For the voted-based approach:**
 - 7: **if** voted-based approach is the maximum value **then**
 - 8: **for** each pair(user u , item i) **do**
 - 9: $\hat{r}_{ui} = \max(x_{1i}, x_{2i}, x_{3i})$
 - 10: **end for**
 - 11: **else if** voted-based approach is the minimum value **then**
 - 12: **for** each pair(user u , item i) **do**
 - 13: $\hat{r}_{ui} = \min(x_{1i}, x_{2i}, x_{3i})$
 - 14: **end for**
 - 15: **else if** voted-based approach is the average value **then**
 - 16: **for** each pair(user u , item i) **do**
 - 17: $\hat{r}_{ui} = \text{mean}(x_{1i}, x_{2i}, x_{3i})$
 - 18: **end for**
 - 19: **end if**
 - 20: **2.2. For the weighted-based approach:**
 - The rating outputs from X_1, X_2, X_3 are given as input to a neural network to perform the prediction task for the hybrid recommender system.
 - Utilizing a basic feedforward network model denoted as $y = f(x)$, where $x = [x_1, x_2, x_3]$, and w_0, w_1, w_2 , and w_3 represent the weights of the neural network.
 - Define $f(x) = \sum_{i=0}^d w_i x_i$, where d represents the dimensionality of the data, which in this context is set at $d = 3$.
 - 21: **2.3. Training prediction model**
 - 22: **Inference stage:**
 - 23: Input: Given a triangle (
 - x_{1i} is the rating output from X_1 .
 - x_{2i} is the rating output from X_2 .
 - x_{3i} is the rating output from X_3 .)
 - 24: Using the Prediction Model.
-

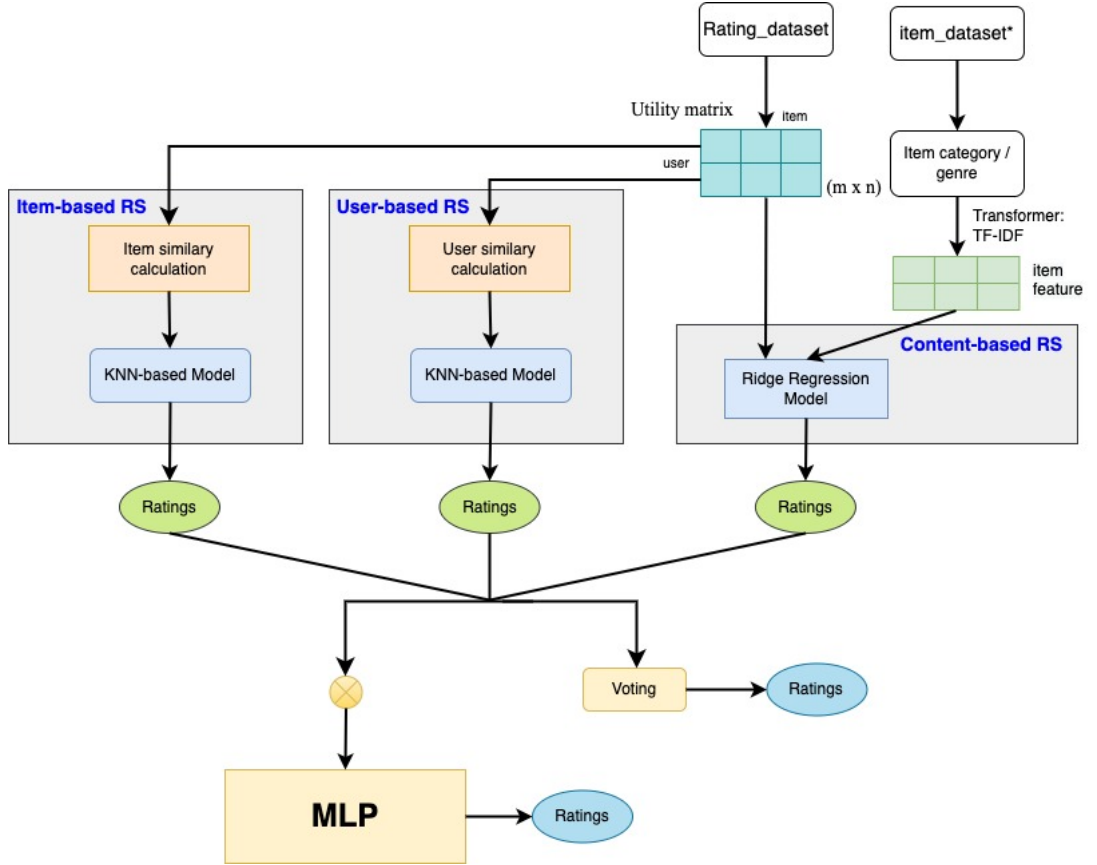


Figure 3.1 A proposed hybrid model for recommender systems.

3.4 Experiments and Results

3.4.1 The dataset summary

Preprocessing datasets:

To conduct our experiments, we employed a reduced-size version of the ML-GroupLens 20M dataset. The rating data in this mini dataset consists of 6,822 ratings contributed by 4,991 users for a set of 1,840 movies.

We have also conducted on two sub-datasets from Amazon, namely Electronics and Toys and Games. For enhancing the accuracy of our analysis, the data underwent preprocessing, involving the exclusion of users who seldom provided ratings for items, as well as the removal of items with limited community ratings. Furthermore, in instances where a user allocated multiple ratings to a single item, only the most

Table 3.1 The MovieLens dataset overview

Data	Property	Users	Movies
Movies	movieId, title, genres		27278
Rating	userId, movieId, rating, timestamp	4991	1840

recent rating was retained. The Electronic sub-dataset consists of 49,701 ratings from 1,042 users and 4,841 items, while the Toys and Games sub-dataset contains 63,440 ratings from 3,797 users and 2,901 items.

Describing datasets:

Each rating entry of the MovieLens dataset is linked to a user identifier number, a movie identifier number, a rating score, and a timestamp. Additionally, the movie data contains 27,278 titles, each of which is associated with a movie-id number, a title, and a genre that belongs to one of the 21 predefined categories. We have provided a summary of the dataset we used in Table 3.1. We chose this dataset because it is widely adopted by researchers and developers who are working on recommender systems. A data sample of the MovieLens dataset is shown in Figure 3.2.

	userId	movieId	rating	timestamp	title	genres
0	43712	2141	4.0	1488502592	American Tail, An (1986)	Adventure Animation Children Comedy

Figure 3.2 A sample data of the MovieLens dataset.

These details about the Amazon datasets can be found in Table 3.2. The information we use with the Amazon dataset includes rating scores, user identifiers, item identifiers, review texts, item categories, and item descriptions. A data sample of the Amazon dataset is presented in Figure 3.3.

	0
overall	5.0
reviewerID	AIL4HD8BRUHV9
asin	0963469150
reviewText	Monopoly is fun, but it takes FOREVER to play,...
category	[Toys & Games, Games, Card Games]
description	[Quiddler, winner of over 20 Best Game Awards,...]

Figure 3.3 A sample data of the Amazon dataset.

Table 3.2 The Amazon sub-datasets overview

Dataset	Ratings	Users	Items
Electronic	49701	1042	4841
Toys and Games	63440	3797	2901

3.4.2 The evaluation metrics

To evaluate our model, we employ indicators MAE, RMSE, and Precision. Equation 3.1 outlines the formula for calculating the precision index, with tp representing the number of correctly recommended items and fp indicating the number of incorrectly recommended items.

$$Precision = \frac{tp}{tp + fp} \quad (3.1)$$

3.4.3 The experimental results

3.4.3.1 The experimental setups for datasets and models

The datasets are splitted into 80% for training and validating, 20% for testing. Table 3.3 presents the details of how the number of rating scores is divided for the training, testing, and validation tasks.

In the content-based RS, we utilize the properties of each item to create an item feature vector based on its item category. For the Electronic sub-dataset, the item

Table 3.3 Splitting the datasets for training, testing, validating tasks

Dataset	Train	Test	Validation
Electronic	39761	7952	1988
Toys and Games	50752	10151	2537

feature vector has a dimension of 594. Similarly, for the Toys and Games sub-dataset, the item feature vector has a dimension of 304. We also utilize the Ridge Regression model, which is a variant of Linear Regression with regularization for the prediction task.

In our work, we conducted experiments using a neural network (MLP) for the rating prediction task in RSs. The neural network was configured with specific parameters, including 10 epochs, a batch size of 100, the Adam optimizer, a first hidden layer consisting of 120 nodes, a second hidden layer with 64 nodes, and an output layer containing 1 node.

3.4.3.2 Experimental results

For the MovieLens dataset, we conducted a performance comparison of several recommender systems using the RMSE and Precision metrics for individual models, namely, user-based, item-based, and CB, as well as our hybrid model based on minimum, maximum, average, and weighted ratings. The results of this evaluation are presented in Table 3.4, where we have defined Δ_1 as the absolute difference between the actual rating (y_j) and the predicted rating ($\hat{y}_j$) being less than or equal to 0.5, and Δ_2 as the absolute difference being less than or equal to 1.0. As presented in Table 3.4, the user-based CF is the individual model that achieved the highest precision of 61.72%. On the other hand, the weighted-based hybrid model achieved the highest precision of 66.67% and the best RMSE of 1.073, which is significantly better than the best-performing single model.

For the Electronic dataset, the experimental results can be found in Table 3.5.

Table 3.4 The comparison of the Precision and RMSE metrics among recommendation models on the MovieLens dataset

Model	Precision	Precision	RMSE
	Δ_1	Δ_2	
Content-based	2.59 %	7.44 %	3.265
User-based CF	32.18 %	61.72 %	1.222
Item-based CF	1.21 %	6.31 %	3.538
Maximum voted-based hybrid model	32.18 %	61.72 %	1.222
Minimum voted-based hybrid model	1.14 %	6.26 %	3.548
Average voted-based hybrid model	7.08 %	16.09 %	2.373
Weighted-based hybrid model with MLP	36.62 %	66.67 %	1.073

The hybrid recommendation models combined the output results from the individual recommendation models using the combination strategy and used them as inputs for a neural networks (MLP). The experimental results indicate that the hybrid model with MLP, achieved the highest recommendation accuracy with an MAE index of 0.656, an RMSE index of 0.897, and a Precision index of 83.36%.

Table 3.5 The results of the strategy combination for fusing in a neural network on the dataset Electronic

Model	MAE	RMSE	Precision
Content-based	4.329	4.437	2.43%
Item-based	4.329	4.431	2.98%
User-based	0.783	1.270	82.13%
Maximum voted-based hybrid model	0.776	1.244	82.13%
Minimum voted-based hybrid model	4.404	4.505	2.43%
Average voted-based hybrid model	2.967	3.101	5.73%
Weighted-based hybrid model with MLP	0.656	0.897	83.36%

The experimental results for the Toys and Games dataset can be found in Table 3.6. The results indicate that the hybrid model with MLP, achieved the highest recommendation accuracy with an MAE index of 0.599 and an RMSE index of 0.856, a best Precision index of 81.40%.

Table 3.6 The results of the strategy combination for fusing in a neural network on the dataset Toys and Games

Model	MAE	RMSE	Precision
Content-based	4.294	4.392	1.37%
Item-based	4.269	4.368	1.52%
User-based	1.327	2.101	68.14%
Maximum voted-based	1.305	2.052	68.14%
Minimum voted-based	4.355	4.448	1.35%
Average voted-based	3.123	3.308	4.47%
Weighted-based hybrid model with MLP	0.599	0.856	81.40%

3.5 The Chapter Summary

In our study, we constructed hybrid recommendation models that estimate users' ratings by combining the outputs of three individual RSs using various methods such as max, min, average, and weighted-based approaches. Our findings on the MovieLens dataset demonstrate that the weighted rating-based hybrid model outperformed both the fusion-based hybrid models (max, average, min) and other individual models in terms of prediction precision.

Additionally, we conducted experiments on two sub-datasets from the Amazon to evaluate a proposed deep learning-based hybrid recommendation model, which employs a combination strategy for concating the outputs of the three individual RSs. The results reveal that the deep learning-based hybrid recommendation model outperforms the individual recommendation models.

CHAPTER 4. A MULTI-MODAL MODEL FOR FUSING DIVERSE INFORMATION SOURCES IN RECOMMENDER SYSTEMS

4.1 Introduction

RSs are created with the purpose of discerning user preferences for specific products or services, with the ultimate aim of facilitating product exposure to potential customers, making them a crucial aspect of commerce. As a result, the development of accurate recommendation methods has become a significant and intriguing research problem. Similar to previous studies, we refer to products, services, or other items of user interest as "items". The objective is to determine the level of interest for user-item pairs. Most research approaches the RS problem as a machine learning challenge, wherein a generalized model is constructed based on known user and item information in order to generate interest levels for unknown user-item pairs.

In the conventional approach, the recommendation system problem is typically depicted as a matrix that represents user preferences for different items. The rows of the matrix represent users, while the columns represent items. The values in the matrix, often integers ranging from 1 to 5, denote the ratings given by users to items. This matrix, also known as the Utility Matrix, serves as an example of how the RS task involves constructing a model that can predict ratings for the empty cells based on the known values in the Utility Matrix.

	Item1	Item2	Item3	Item4	Item5	...
User1	4	5	3	1		
User2	4					
User3		3	5	2	4	
User4		5			1	
User5	5			2		
...						

Figure 4.1 An example of a Utility Matrix.

There are two primary techniques for creating RSs: CF and CB. CF predicts the rating (r) of an item (i) by a user (u) by examining the ratings of other users similar to a user u who have previously rated an item i . This method can be implemented using two well-known techniques: Neighborhood-based CF (Feng et al., 2020; Geetha et al., 2018) and MF (Hoyer, 2004; Z. Wang et al., 2021; Xue et al., 2017; E. Yang et al., 2021; H. Zhang et al., 2021), which rely only on the information contained in the Utility Matrix. On the other hand, CB filtering, as seen in some studies (Geetha et al., 2018; Ghauth & Abdullah, 2010; Reddy et al., 2019b; Singla et al., 2020b; Wahyudi et al., 2020), assumes that if a person enjoys item i , they are likely to enjoy similar products. Similarity between items can be computed using the Utility Matrix or item profiles from other sources. However, CB only predicts ratings for similar items, whereas CF overcomes this limitation by utilizing a collaborative mechanism that detects a user's preference for an item based on similar users who have rated that item. Nonetheless, CB can perform well in predicting ratings for new items that are similar to past items, particularly when using more detailed product descriptions. Recent research (Geetha et al., 2018; Ghazanfar & Prugel-Bennett, 2010; Mulay et al., 2022; Nayak et al., 2018; Okaka, 2018; Otegbade, 2016; Walek & Fajmon, 2023) has combined both these approaches using ensemble learning, which integrates multiple individual models to create a stronger combined model.

Beyond the Utility Matrix, there are numerous other sources of information that can be leveraged for item recommendations, such as the user's profile, item descriptions, and item reviews. These additional sources of information offer opportunities for utilizing various data representation techniques and machine learning models to achieve the objective of item recommendation. In general, there are multiple modalities available to assess the level of user interest in an item for a given user-item pair (u, i). Recent studies have explored a multimodality approach to the RS problem, as demonstrated in (Conceiç ao et al., 2019; K. Liu et al., 2018; Ren et al., 2022; J. Wang et al., 2022).

Our research aims to address this challenge by utilizing a variety of information sources, which we refer to as multimodal data, to develop a unified model for item recommendations. Our goal is to make the most of different sources of information, including the Utility Matrix (which contains ratings between user-item pairs), user profiles, item characteristics, and item reviews. To achieve this, we propose a novel combined model that solves two key issues: first, how to build models based on these information sources that effectively extract useful information and represent it in a unified manner; and second, how to combine this information in a way that complements and integrates it to predict ratings.

Our proposed approach involves utilizing MF to determine latent feature vectors of users and items. We then apply a range of transformation techniques to these vectors, as well as the feature vectors of users and items, to convert them into a common form. For other modalities, such as item reviews, descriptions, and categories, we propose representations based on natural language processing techniques. We also apply a feature selection method to identify the best feature subset and remove noisy features. Finally, we combine these vectors using neural networks to create a multi-modal feature descriptor that can be utilized by RSs. The aim of this approach is to enrich the information provided to RSs and improve their performance.

We have made significant contributions to the field, which include:

1. Introducing a multi-modal transformation and fusion model that utilizes a deep neural network to integrate diverse features from multiple sources. This includes:
 - Obtaining latent features of users and items using MF and then reducing their dimensions through PCA.
 - Transforming item features from the item dataset and user features through various techniques (such as TF-IDF and BERT) and then reducing their

dimensions through PCA.

2. Conducting experiments to demonstrate how our multi-modality transformation and fusion model enhances the performance of RSs.

4.2 Related Works

RSs have been the focus of numerous studies, which have explored different ways of providing users with helpful recommendations. Two common approaches are CB and CF. Specifically, some researchers have proposed various CB RSs across different domains. These systems utilize different approaches and techniques to recommend movies, hotels, learning materials, journals/conferences, books, products, courses, clothing, restaurants, and food items. The RSs consider user preferences, genre correlation, movie features, social network interactions, academic background, and domain-specific knowledge to generate personalized recommendations. They employ methods such as CB, topic modeling, deep neural networks, and machine learning algorithms to provide accurate and relevant recommendations to users. (Reddy et al., 2019b) employed genre correlation to create a CB RS that suggests movies based on user preferences, while (Wahyudi et al., 2020) combined hotel categories and city features to recommend the highest rated hotels to users. (Singla et al., 2020b) introduced a movie recommendation framework that determines similarities between movies using publicly available features such as plot, rating, country of production, and release year. (Ghauth & Abdullah, 2010) proposed a framework that recommends learning materials based on content similarity and user ratings. Meanwhile, (Y. Zhao & Shen, 2016) conducted an empirical study of movie ratings to develop a preference model that eliminates impractical predictions when predicting user preferences. (Model, 2020) proposed a system architecture to overcome the cold start problem of RSs by using product item documents processed through TF-IDF algorithm and Vector Space Model, allowing for queries to find similar items to those seen by the user. (Deldjoo

et al., 2016) introduced a novel CB RS that employs an automated analysis of video content to extract representative stylistic features (such as lighting, color, and motion), based on established approaches of Applied Media Theory. An intelligent CB RS was proposed by (Haw et al., 2022) to provide users with access to product recommendations as well as dashboard systems containing descriptive analytics modeled using ontology. The dashboard offers insight into past data and comprises five primary features: the Highlight Dashboard, Customer Dashboard, Advanced Search, Pivot Table and Pivot Chart, and Report. In (Van Dat et al., 2022), a new CB recommendation approach based on the Gaussian Mixture Model was proposed to enhance accuracy and provide more sensitive results for probabilistic recommendation problems. (Sridhar et al., 2023) introduced a CB movie recommendation model that utilizes a hybrid model combining Monarch Butterfly Optimization (MBO) and Deep Belief Network (DBN) to recommend movies based on a user's Facebook profile. MBO is used for feature selection, while DBN is utilized for classification in the RS. A latent factor model was suggested for recommendation by (Van den Oord et al., 2013), where the latent factors are predicted from music audio in cases where they cannot be obtained from usage data. The study compared a conventional method that uses a bag-of-words representation of audio signals with deep CNNs. (D. Wang et al., 2018) introduced a computer science CB Journals & Conferences RS with a web service at <http://www.keaml.cn/prs/> that prioritizes suitable journals or conferences based on manuscript abstracts. The system uses a web crawler to update the learning model and employs a hybrid model with chi-square feature selection and softmax regression for fast online response. In (Mooney & Roy, 2000), exploring content-based book recommending involves automated text-categorization of semi-structured text from the web using LIBRA, a prototype system that uses Amazon.com's book database and textual meta-data. User ratings of a selected set of training books are used to learn a user profile through a Bayesian learning algorithm, generating a ranked list of

recommended titles from the catalog. (Prando et al., 2017) introduced a RS that recommends specific products to new e-commerce users based on their social network interactions. The RS uses a CB approach to analyze user data from social networks, including direct posts, content likes, and page likes, to improve the user experience by suggesting items in their preferred category. In (Rohani et al., 2014), the proposed algorithm enhances CB RSs by utilizing social networking features, taking into account the interests and preferences of users' friends and faculty mates and utilizing a hierarchy tree structure to exploit all interests and preferences. The algorithm was incorporated into MyExpert, an Academic Social Network (ASN), along with three other recommender algorithms to propose relevant academic items to members of the online community. (Afoudi et al., 2019) presented the Restaurant CB RS, which prioritizes restaurants according to their numerous features. This system uses a CB approach, and the article explores the impact of feature selection on its performance while using an alternative method for calculating similarity between items. A new CB RS for literature recommendations has been proposed by (Bagul & Barve, 2021), which utilizes LDA and Jensen-Shannon distance as its basis. A movie RS developed by (Reddy et al., 2019a) is based on the user's potential genre preferences, achieved through CB filtering utilizing genre correlation. A movie recommendation framework called FLEX was introduced by (Singla et al., 2020a), which employs a CB filtering method and combines Doc2Vec and TF-IDF approaches. FLEX utilizes publicly available movie features, such as plots, ratings, production countries, and release year, to generate a recommendation list based on movie similarities. (Morsomme & Alferez, 2019) introduced a course RS for the Liberal Arts bachelor program at University College Maastricht that complements academic advising by recommending courses based on student interests and academic background. The system uses a topic model to provide course recommendations and a sparse predictive model for warnings and preparatory course suggestions based on past academic performance and expertise level. A CB clothing

RS was proposed by (Gharaei et al., 2021) using a DNN to eliminate the need for manual feature extraction by obtaining the cloth category and producing the necessary features in a large and useful volume for predicting unobserved item ratings. (Shahbazi & Byun, 2019) used the XGBoost algorithm for recommending items by analyzing user click information from the Jeju online shopping mall dataset. A unified food recommendation framework was proposed by (Vairale & Shukla, 2021) to identify food items using CB features and domain knowledge. The recommender model utilizes food items rating patterns and similarity scores to generate diet recommendation lists for thyroid patients.

The CF approach has been used in several research studies, such as a research-paper RS was proposed by (Murali et al., 2019) that utilizes CF approach to suggest the most suitable research papers to a user based on their queries and similarities with other users' queries, thereby saving time and effort. By using a CF technique and cosine similarity with k-nearest neighbor, (Gupta et al., 2020) overcame the limitations of CB. Although Euclidean distance is preferred, cosine similarity was used because it provides almost the same accuracy and equidistance of movies. Obtaining supplementary features like audio information and text content can be difficult for most RSs, as some models rely on them to improve performance. However, (Mohammed & Manjaiah, 2020) have developed an efficient deep learning method for collaborative recommendation, called DLCRS, that does not require the use of any additional information beyond the interaction between users and items. (Valdiviezo-Diaz et al., 2019) introduced a Bayesian model that provides recommendations comparable to those of MF models, while also being explainable. The model employs both user-based and item-based CF, recommending items based on similar users and items, respectively. (Madani et al., 2019) proposed a recommendation approach for suggesting relevant courses to learners based on social and CF, considering the learner's profile and social content. (Nilashi et al., 2019) developed a soft computing method using dimensionality reduction and prediction

machine learning techniques to enhance the scalability of prediction from large numbers of user ratings. The method was evaluated on a substantial dataset obtained from the TripAdvisor platform. (Bobadilla et al., 2020) incorporated prediction errors (reliabilities) into deep learning layers to improve prediction and recommendation quality by recommending highly predicted items that are also found to be reliable. The proposed deep learning architecture consists of three stages that correspond to three levels of abstraction: (a) real prediction errors, (b) predicted errors (reliabilities), and (c) predicted ratings (predictions), each of which requires a learning process using MF, Multilayer Neural Network, and hidden factors. To address the cold-start problem, (F. Wang et al., 2021) proposed a Trust-based CF (TbCF) algorithm for basic rating prediction, and a novel approach called UIT_hybrid that integrates user trust into existing CF-based methods to supplement rating information. UIT_hybrid offers multiple perspectives to improve the robustness, accuracy, and diversity of recommendations. In their work, (Hu et al., 2019) developed the Transfer Meeting Hybrid (TMH) model for cross-domain recommendation utilizing unstructured text. The TMH model incorporates a memory network to extract relevant content from unstructured text and selectively transfers knowledge from a source domain via a transfer network. A new bio-inspired clustering ensemble approach was proposed by (Logesh et al., 2020) to enhance user-based CF. The proposed method combines swarm intelligence and fuzzy clustering models for clustering aggregation. (W. Wang et al., 2019) suggested incorporating geographic and temporal influences into POI recommendation through a fusion model. To achieve this, they proposed a unified collaborative recommendation framework that utilizes network representation learning to calculate trust-enhanced user similarity in user-based CF and fuses it with the geographic and temporal influences to suggest potential POIs. (Fernández-Tobías et al., 2019) evaluated MF models that utilize metadata as a bridge between items in different domains for cross-domain CF. By leveraging the underlying knowledge graph

connecting items from different domains, they demonstrated that their models can offer superior recommendations to new users while balancing recommendation accuracy and diversity. (Sachdeva et al., 2019) proposed a recurrent version of VAE that passes the consumption sequence through a RNN to handle temporal dependencies. By modeling the probability distribution of future preferences at each time-step, the model showed improved accuracy in capturing temporal information and outperformed the current state-of-the-art. (Zou et al., 2020) suggested a neural network-based exploration policy to quickly adapt to user preferences by learning from feedback data. Their approach encodes the exploration policy in the weights of multi-channel stacked self-attention neural networks and trains it with efficient Q-learning to maximize users' overall satisfaction in the RS. The approach considers satisfied recommendations resulting from the exploration bonus as the delayed reward that improves the quality of the user profile. (Alhijawi & Kilani, 2020) proposed BLIGA, a genetic-based RS that evaluates possible recommendation lists instead of individual items. BLIGA uses a hierarchical evaluation approach with three fitness functions: semantic similarity between items, similarity in satisfaction level between users, and predicted ratings to select the best recommendation list. *The provided research studies focus on various aspects of CF RSs, including addressing the cold-start problem, enhancing scalability, utilizing additional features, improving diversity and accuracy of recommendations, handling temporal dependencies, and incorporating trust and social content. These studies propose different approaches, such as deep learning, Bayesian models, transfer learning, bio-inspired clustering, and neural network-based exploration policies. Many of these studies also evaluate their proposed methods on large datasets and compare their performance with existing state-of-the-art methods. However, some limitations have been identified, such as the difficulty in obtaining supplementary features like audio information and text content, the cold-start problem, and the reliance on user ratings and historical data, which may not always reflect users' current preferences.*

Additionally, some approaches require extensive computation and training, limiting their practicality in real-world applications. Nonetheless, the proposed methods have shown promising results in enhancing recommendation performance and overcoming some of the challenges associated with traditional RSs.

In addition, other deep learning-based methods have been investigated. (Ahmed & Ghabayen, 2022) presented a Bi-GRU model architecture for predicting review rating and polarity. (Gezici et al., 2019) used RNNs for analyzing reviews and predicting user ratings. (E. Yang et al., 2021) developed a federated collective MF algorithm that safeguards user privacy while accurately predicting user preferences. (Z. Wang et al., 2021) introduced a Visual Recurrent Convolutional MF (VRConvMF) model that utilizes descriptive texts and posters to extract textual and visual features. (H. Zhang et al., 2021) proposed a FeatureMF model that incorporates item features into the MF framework to enrich item representation. Lastly, (Xue et al., 2017) proposed a new approach based on deep MF that utilizes both explicit and implicit information.

Recently, there has been an increase in multimodality-based approaches in the field of recommendation research. (Conceição et al., 2019) utilized bag-of-words and TF-IDF models along with descriptors from visual data using a deep convolutional network to create a low-dimensional sparse representation of each video document through multimodal fusion. (Feng et al., 2020) integrated users' local relations and global ratings to enhance prediction accuracy and robustness in sparse data. (J. Wang et al., 2022) proposed a Fine-grained Multimodal Fusion Network (FMFN) that combines textual and visual factors for detecting fake news. (Ren et al., 2022) proposed a deep course recommendation model for multimodal fusion that employed multimodal features using LSTM with an attention mechanism. They used course video, audio, title, and introduction to extract features, and constructed a user preference graph using historical interactions to represent user

preferences through collaborative signals. (Lei et al., 2023) introduced Learning the User's Deeper Preferences (LUDP), a novel module that aggregates item ID embeddings and multi-modal features to create an item-item similarity graph and user preference graph. (Mu & Wu, 2023) proposed a personalized multimodal movie RS using deep learning algorithms to mine hidden features of movies and users. They used these features to predict movie scores based on input information. (Ren et al., 2022) proposed a course recommendation approach that uses a deep model combining multimodal feature extraction through LSTM networks and attention mechanism. The model includes video, audio, title, and introduction data from courses and user demographic information, as well as explicit and implicit feedback data to create a comprehensive learner profile.

Traditionally, CF models have relied on either item content similarity or similarities among users with similar preferences or items rated by the same user to suggest items to users. MF-based RSs use a formula to estimate users' preference levels for items from users' and items' latent feature vectors extracted from a Utility Matrix. Recently, neural networks have been used to fuse item feature vectors or/and user rating vectors, and the results applied to RSs. Multimodal RS studies have refrained from integrating information from both the Utility Matrix and textual sources due to potential issues of interference and reduced accuracy.

In contrast, our research adopts a multimodal approach, leveraging various techniques to extract and represent information effectively from multiple modalities. We use reduced-dimension user and item latent features and feature vectors obtained through PCA instead of the original feature vectors of users and items. This approach reduces the dimensionality of the data and computational costs. The best k feature selection algorithm is applied to filter out noise for the feature vectors before integrating them and feeding them as input to a Feedforward Neural Network model for rating prediction. Our proposed model has demonstrated better recommendation

performance when compared to both unimodal recommendation models and SVD RSs.

4.3 The Proposed Model

4.3.1 Our Method for Multimodal Fusion

Multimodal fusion involves the process of integrating information from various sources or modalities, including text, images, audio, video, and other data forms. The aim of multimodal fusion is to generate a comprehensive and accurate representation of a particular event or situation by merging information from multiple sources. Multimodal fusion techniques are used in the context of machine learning and artificial intelligence to enhance the performance of models that operate on intricate and heterogeneous data. By merging information from diverse modalities, these models can gain a better understanding of the underlying patterns and relationships in the data, and thus make more precise predictions or decisions.

In our proposed model, we utilize an early fusion approach. We present the architecture of our proposed multimodal model for the recommendation problem in Figure 4.2. The model consists of two main components: the first module is designed for extracting and representing features from various information sources of users and items, while the second module is used for integrating these features for the prediction task. Our multimodality-based recommendation algorithm follows the following general scheme:

Algorithm 6 The Method for Multimodal Fusion

- 1: **Training stage:**
 - 2: Input: Given a dataset containing multi-modality of a set of users and items.
 - 3: Output: A model for predicting ratings
 - 4: **Step 1:** Building the Feature Extraction Model (named as FEM)
 - 5: **for** each modality **do**
 - 6: Run FEM()
 - 7: **if** user modality **then**
 - 8: $V_u = v_{u1}, v_{uk}$ //A set of k feature vectors for a given user.
 - 9: **end if**
 - 10: **if** item modality **then**
 - 11: $V_i = v_{i1}, v_{ij}$ //A set of j feature vectors for a given item.
 - 12: **end if**
 - 13: **end for**
 - 14: **Step 2:** Building the Predicting Fusion Model (named as PFM)
 - Combining feature vectors from Step 1: $\leftarrow V_{concat}$
 - Feed V_{concat} into a Regression model to generate ratings (i.e. prediction)
 - Training prediction model
 - 15: **Inference stage:**
 - 16: Input: Given a pair (user u , item i)
 - 17: Using FEM to return feature vectors for user u and feature vectors for item $i \leftarrow V_u = v_{u1}, v_{uk}$ and $V_i = v_{i1}, v_{ij}$
 - 18: Combining V_u and V_i then use the result as input features for the PFM for generating the output (i.e. the rating).
-

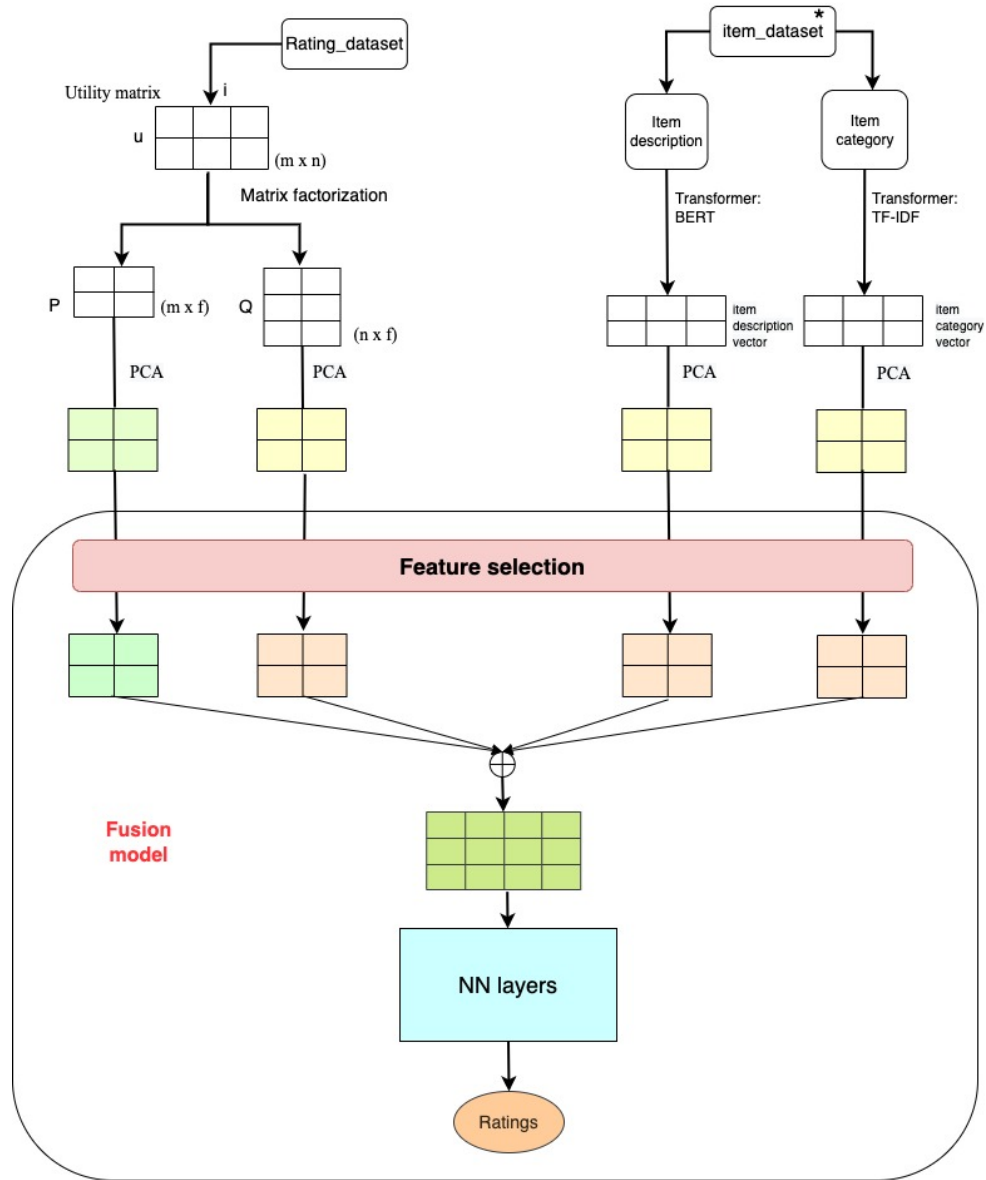


Figure 4.2 The proposed transformation and fusion model for recommender systems.

4.3.2 The Feature Extraction Model (FEM)

In our proposed model, the component responsible for extracting and representing features is of utmost importance. We generate various features to depict users and items by sourcing information from four different sources, namely the Utility Matrix, user reviews, item descriptions, and item categories. The process is outlined below:

Feature Extraction from Utility Matrix: We employ MF technique to decompose

the given Utility Matrix into two lower rank matrices, which capture latent features of users and items. Specifically, from the $(m \times n)$ Utility Matrix representing m users and n items, we obtain:

- A $(m \times f)$ matrix that represents user features, where each user is represented by a vector of f features. Each row in this matrix corresponds to a specific user and is considered as a modality of that user.
- A $(n \times f)$ matrix that represents item features, where each item is represented by a vector of f features. Each row in this matrix corresponds to a specific item and is considered as a modality of that item.

Review-Feature: For a given pair (user u , item i) with a review written by user u for item i , we utilize a pre-trained BERT model to generate a vector representation of the review text. This vector representation is then considered as a modality of a user u .

Description-Feature: Similarly, for a description of item i provided in text format, we use a pre-trained BERT model to generate a vector representation of the description. This vector representation is considered as a modality of an item i .

Category-Feature: The category of item i is comprised of a list of categories to which the item belongs. To convert these category features into a vector representation, we use a TF-IDF measure instead of BERT. This vector representation is also considered as a modality of an item i .

To summarize, our approach involves generating two feature vectors for a user u (considered as two modalities of a user u) and three feature vectors for an item i (considered as three modalities of an item i) when given a pair of (user u , item i). These feature vectors are then processed through the PCA algorithm to obtain the final feature representations. We apply the PCA technique to reduce the dimensionality of the data and extract important representative features, followed by an algorithm to select the

best k features. These feature vectors are then input into the fusion component for predicting the rating of (u, i) .

In essence, the process of extracting and transforming feature vectors for a given user u and item i involves the following steps:

Algorithm 7 The Procedure Feature Extraction Module

- 1: **Procedure FEM:**
 - 2: **Step 1:** Extracting feature kinds including:
 - $Rat(u)$ //Rating-Feature for user u
 - $Rat(i)$ //Rating-Feature for item i
 - $Rev(u)$ //Review-Feature for user u
 - $Des(i)$ //Description-Feature for item i
 - $Cat(i)$ //Category-Feature for item i
 - 3: **Step 2:** Using PCA to transform feature vectors
 - 4: **for** each user u **do**
 - 5: $f_{u1} \leftarrow Rat(u)$
 - 6: $f_{u2} \leftarrow Rev(u)$
 - 7: **end for**
 - 8: **for** each item i **do**
 - 9: $f_{i1} \leftarrow Rat(i)$
 - 10: $f_{i2} \leftarrow Des(i)$
 - 11: $f_{i3} \leftarrow Cat(i)$
 - 12: **end for**
 - 13: **Step 3:** Using the SelectkBest algorithm from the sklearn library to select a subset with k best features from transformed feature vectors obtained after Step 2.
For implementing this method, do the following tasks:
 - Instantiate the SelectkBest class, define the scoring function (the mutual information scoring function) and value of k .
 - Train the algorithm on our dataset.
 - Transform it to obtain the selected features.
-

4.3.3 The Predicting Fusion Model (PFM)

We will use a neural network to integrate the extracted features from the different modalities and generate a rating indicating a user u 's interest in an item i . The process

involves identifying and extracting the modalities' features, followed by the algorithm 8 outlined below to accomplish the task.

4.4 Experiments and Results

4.4.1 The dataset summary

Preprocessing datasets:

We performed experiments on the MovieLens and Amazon sub-category datasets to showcase the efficacy of our proposed model. Prior to the experiments, we applied preprocessing to both datasets to identify users and items that met our predefined filter criteria. For the MovieLens dataset, we only included ratings of items that were rated by at least 100 users and filtered out users who rated 200 or more items.

Similarly, for the Amazon - Electronics subdataset, we selected ratings of items that were rated by 100 or more users and removed the ratings of users who rated 58 or more items. In the case of the Amazon Toys and Games sub-dataset, we considered only ratings of items rated by a minimum of 30 users and filtered out users who rated 15 or more items. Lastly, for the Amazon Video Games sub-dataset, we only included ratings of users who rated at least 25 items. You can find additional information about the datasets below.

Describing datasets:

Our experiment utilized the Amazon dataset, which can be downloaded from <https://jmcauley.ucsd.edu/data/amazon/>. Specifically, we focused on sub-datasets such as Electronic, Video Games, and Toys and Games. The experiment relied on two distinct files, namely the rating file and the metadata file. The rating file contained various attributes, including overall rating scores, verified reviews, reviewTime, reviewerID (identifying reviewers), asin (identifying items), reviewerName, reviewText, summary (content summary of reviews),

Algorithm 8 Procedure PFM

- 1: **Step 1: Procedure FEM:**
 - 2: **for** each pair (user u , item i) **do**
 - 3: FEM(pair (user u , item i))
 - 4: $f_{u1}, f_{u2} \leftarrow$ user u //user feature vectors
 - 5: $f_{i1}, f_{i2}, f_{i3} \leftarrow$ item i //item feature vectors
 - 6: $f_s \leftarrow$ set of features //obtain a subset of selected features
 - 7: **end for**
 - 8: **Step 2:** Transforming data
 Once a subset of selected features has been identified in Step 1, fit the algorithm to the data and obtain the chosen features through the following transformations:
 - 9: $f_{u1_fs} \leftarrow f_{u1}$ //transform user feature vectors
 - 10: $f_{u2_fs} \leftarrow f_{u2}$ //transform user feature vectors
 - 11: $f_{i1_fs} \leftarrow f_{i1}$ //transform item feature vectors
 - 12: $f_{i2_fs} \leftarrow f_{i2}$ //transform item feature vectors
 - 13: $f_{i3_fs} \leftarrow f_{i3}$ //transform item feature vectors
 - 14: **Step 3:** Combining features
 Concatenate f_{u1_fs} , f_{u2_fs} , f_{i1_fs} , f_{i2_fs} , f_{i3_fs} and we obtain the final representation of the input:

$$F = \text{Concat}(f_{u1_fs}, f_{u2_fs}, f_{i1_fs}, f_{i2_fs}, f_{i3_fs})$$
 - 15: **Step 4:** Building the prediction model
 - Designing a Multi-layers Perceptron model for generating ratings.
 - For the training data set containing pairs (user u , item i), using Step 1, Step 2 and Step 3 to convert them to corresponding feature vectors.
 - Training the prediction model
 - 16: **Step 5:** Inferencing
 - 17: **Input:** Given a pair (user u , item i)
 - 18: $F \leftarrow$ Using the Step 1, Step 2 and Step 3.
 - 19: **Input** the feature vector F to the prediction model for generating the rating for the pair (user u , item i)
-

unixReviewTime, vote, image, and style. Similarly, the metadata file contained details such as category, tech1, description, fit, title, also_buy, tech2, brand, feature (listing the item features), rank, also_view (related items also viewed), details, main_cat (main item categories), similar_item, date, price, asin, imageURL (links to image files), and imageURLHighRes.

Furthermore, we also employed the MovieLens dataset, which is available for download from <https://grouplens.org/datasets/movielens>. The experiment relied on two primary files: the rating file, which included identification attributes such as userId (identifying users), movieId (identifying movies), rating, and timestamp; and the metadata file of movies, which contained details such as movieId, title, and genres.

Table 4.1 displays the specifics of the MovieLens and Amazon datasets that have been chosen.

- The MovieLens dataset contains 77764 ratings from 27042 users for 8203 movies. It comprises the identification of users, items, rating scores, timestamp, title, and genres.
 - On the other hand, the Amazon sub-datasets rating data includes overall, verified, reviewTime, reviewer identification, item identification, reviewer name, reviewText, summary, unixReviewTime, vote, image, style, item categories, tech1, description, fit, title, also_buy, tech2, brand, feature, rank, also_view, details, the item main categories, similar_item, date, imageURLHighRe, the item price, and imageURL.
1. The Electronic dataset comprises 85065 ratings from 1042 users for 21200 items.
 2. The Video and Games dataset consists of 98769 ratings from 2291 users for 24708 items.

3. Lastly, the Toys and Games dataset contains 71780 ratings from 5462 users for 3028 items.

Table 4.1 The overview of datasets

Dataset	Dataset sub-category	Ratings	Users	Items
MovieLens		77764	27042	8203
Amazon	Electronic	85065	1042	21200
	Toys and Games	71780	5462	3028
	Video Games	98769	2291	24708

4.4.2 The experiments and analyzing results

4.4.2.1 Evaluation metrics

We assess the effectiveness of the proposed model by employing two metrics: RMSE and Precision.

4.4.2.2 Experimental setups

During the model development, we split the data into training and testing sets, with 80% of the data allocated for training and the remaining 20% for testing. For validation purposes, we utilized 20% of the training data during the training process.

To extract features of item description and user reviews from the Amazon dataset, we used a "bert-base-uncased" pretrained model with a tokenizer setup of maximum length 100. For the matrix factorization step, we defined 50 factors, 500 epochs, a learning rate of 0.0003, and an L2 norm of vectors of 0.04. We also set 30 components and 15 best features for each modality in PCA.

In the MovieLens dataset, we used the same specifications as the Amazon dataset for matrix factorization, i.e., 50 factors, 500 epochs, a learning rate of 0.0003, and an L2 norm of vectors of 0.04. For PCA, we selected 20 components and 15 best features for each modality.

We employed a deep neural network with three layers to fuse item feature vectors and latent feature vectors of users and items in both datasets. The first hidden layer

contained 64 nodes, the second hidden layer contained 32 nodes, and the output layer contained 1 node. Furthermore, we set the batch size to 100, epochs to 50, and used the Adam optimizer with a learning rate of 0.0002 and the "relu" activation function.

4.4.2.3 Experimental results

To evaluate the effectiveness of our proposed model, we compared its performance with that of the SVD recommendation model, which utilized the same datasets for training and testing. On the MoviesLen dataset, Table 4.2 displayed the experimental results, highlighting the superiority of our multi-modal recommendation model that combined item genre information and rating scores. Our model achieved an RMSE of 1.24 and Precision of 54.35%, outperforming the SVD recommendation model, which had an RMSE of 2.28 and Precision of 47.65%. Additionally, Figure 4.3 illustrated that our multi-modal model's performance improved when incorporating genre information compared to both the SVD recommendation model and our model that only utilized rating information.

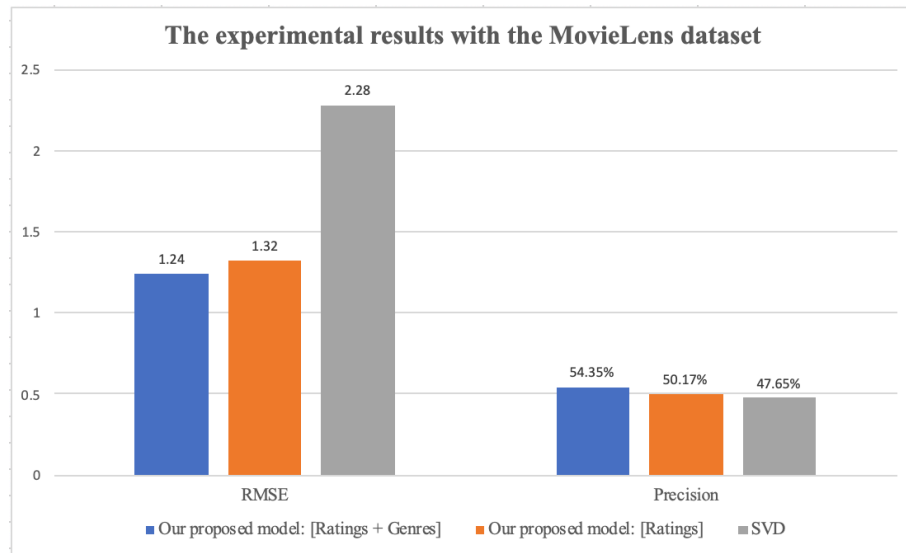


Figure 4.3 The comparison of the Precision and RMSE metrics among recommendation models on the MovieLens dataset.

The experimental results for the Amazon sub-datasets are also presented in Table 4.2. Our proposed model, which fused item descriptions and rating score

information, gave the best results with an RMSE of 1.01 and Precision of 82.86% for experiments conducted on the Amazon-Electronic dataset. This outcome was similar to our proposed model, which fused rating score, user review, item category, and description information, achieving an RMSE of 1.02 and Precision of 82.62%. In comparison to the SVD recommendation model, our proposed model yielded a 12.93% decrease in RMSE and a 19.14% increase in Precision. Additionally, the experiments indicated that combining category information and rating score information had an adverse effect on our recommender system's performance. Meanwhile, the graphical representation in Figure 4.4 illustrated that our proposed approach, which combined data from both the Utility Matrix and item descriptions, gave better results than the approach of using only information from the Utility Matrix.

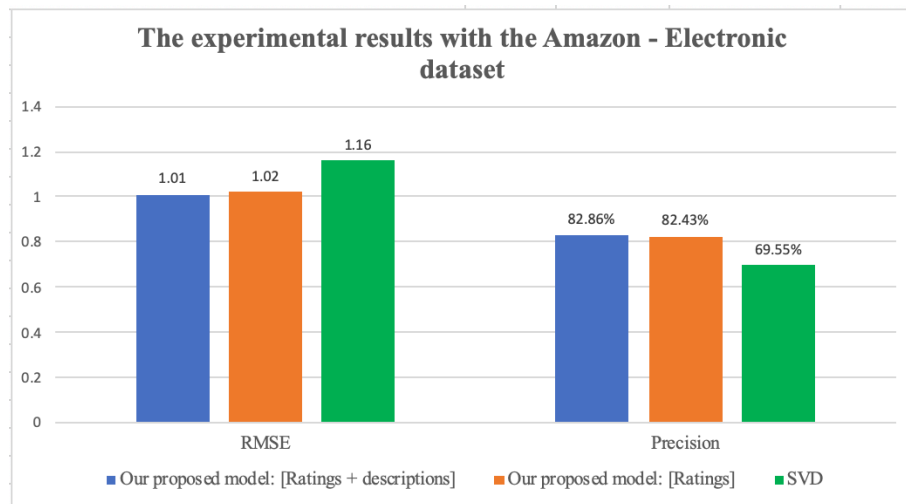


Figure 4.4 The comparison of the Precision and RMSE metrics among recommendation models on the Amazon - Electronic dataset.

Our proposed model, which combined item category and rating score information, yielded the best results with an RMSE of 0.88 and Precision of 86.53% for experiments conducted on the Amazon - Toys and Games dataset. In contrast to the SVD recommendation model, our model exhibited a significant 21.43% decrease in RMSE and a notable 26.49% increase in Precision. Additionally, the results in Figure 4.5 demonstrated that including user review information in our

recommendation model had a detrimental effect on the performance of the recommender system. We suspect that there may have been a mismatch between the rating scores and the user reviews.

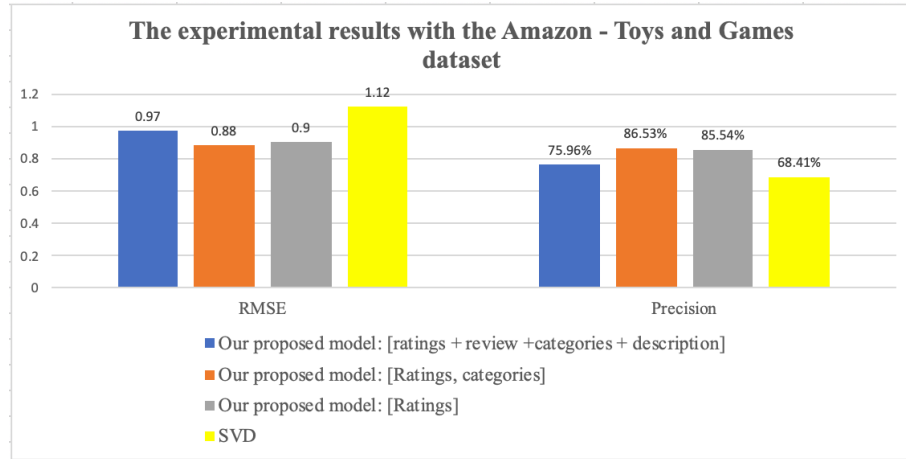


Figure 4.5 The comparison of the Precision and RMSE metrics among recommendation models on the Amazon - Toys and Games dataset.

4.5 The Chapter Summary

Our work proposes a recommendation model that enhances information for recommendation by transforming and merging modalities from various sources, such as the Utility Matrix, user reviews, item descriptions, and item categories. We start by utilizing MF to extract user and item latent features from the Utility Matrix. We also extract textual information from other sources using NLP techniques, including BERT and TF-IDF, while PCA and feature selection algorithms are used to extract useful features for all modalities. All features are then fed into a neural network to make rating predictions. Our experiments evaluated different feature sets corresponding to different modalities, and the results show that our proposed model outperformed SVD, a highly effective model for RSs. Specifically, our model achieved higher Precision and RMSE for both the Amazon and MovieLens datasets. The results also indicate that integrating information from multiple modalities is more effective than using only rating information for recommendation. However, modalities may contain noisy information and have different impacts on each data

type. Finally, the proposed model is efficient, and the multimodal approach holds promise for addressing the recommendation problem.

Table 4.2 The comparison of the Precision and RMSE metrics among recommendation models on the MovieLens and Amazon datasets.

The MovieLens dataset:						
Modality / Modalities	RMSE	Precision				
Our proposed model: User latent feature + item latent feature + item features (genres) [Ratings + genres]	1.24	54.35%				
Our proposed model: User latent feature + Item latent feature [Ratings]	1.32	50.17%				
SVD [Ratings]	2.28	47.65%				
The Amazon dataset:						
Modality / Modalities	RMSE	Precision	RMSE	Precision	RMSE	Precision
	Electronic		Toys and Games		Video Games	
user latent features + item latent features (we called feature R) + item features (category) + item features (description) + user review [ratings + review + categories + description]	1.02	82.62%	0.97	75.96%	1.19	64.34%
Feature R + item features (category) + item features (description) [Rating, category, description]	1.04	79.28%	0.91	84.28%	1.19	65.60%
Feature R + item features (category) [Rating, category]	1.05	78.26%	0.88	86.53%	1.19	65.60%
Feature R + item features (description) [Rating, description]	1.01	82.86%	0.91	84.85%	1.19	64.22%
Feature R [Ratings]	1.02	82.43%	0.90	85.54%	1.19	64.85%
SVD [Ratings]	1.16	69.55%	1.12	68.41%	1.25	64.27%

CHAPTER 5. A TRANSFORMER-BASED MULTIVIEW INTEGRATION MODEL FOR RECOMMENDER SYSTEMS

5.1 Introduction

RSs can be created through CF or CB methods. CF involves using ratings from similar users to forecast item ratings, while the content-based method anticipates ratings for akin items based on a user's previous ratings. Two renowned CF techniques include Neighborhood-based CF (Feng et al., 2020; Geetha et al., 2018), and MF (Hoyer, 2004; Z. Wang et al., 2021; Xue et al., 2017; E. Yang et al., 2021; H. Zhang et al., 2021). These strategies rely on information from the utility matrix. On the other hand, the CB approach in several studies (Geetha et al., 2018; Ghauth & Abdullah, 2010; Reddy et al., 2019b; Singla et al., 2020b; Wahyudi et al., 2020) is based on the concept that if an individual enjoys item i , they are likely to have an affinity for analogous products. Comparing item similarity can be achieved by referencing the Utility Matrix or item profiles from external sources. However, the CB approach has constraints in forecasting ratings for new items. To address this limitation, recent research has combined both CF and CB approaches using ensemble learning (Geetha et al., 2018; Ghazanfar & Prugel-Bennett, 2010; Mulay et al., 2022; Nayak et al., 2018; Okaka, 2018; Otegbade, 2016; Walek & Fajmon, 2023).

Traditionally, RSs have hinged upon a singular information source, like a user's historical purchasing or viewing data, to generate suggestions. However, there's a growing fascination with leveraging diverse information sources to enhance the precision and pertinence of recommendations. One proposed tactic is to adopt a multimodal approach, which entails integrating data from diverse sources, as highlighted in studies (Choudhury et al., 2021; Conceição et al., 2019; K. Liu et al., 2018; Ren et al., 2022; J. Wang et al., 2022). In recent research, neural networks

have also been employed (X. He et al., 2017; Nikzad–Khasmakhia et al., 2020; B. Yang et al., 2007) to fuse information from multiple sources for the purpose of predicting ratings. Hypergraphs present a more versatile and potent avenue for modeling intricate interactions and dependencies between users and items within RSs, as evidenced in earlier research (Fan et al., 2019; Pliakos & Kotropoulos, 2014; Xia et al., 2021).

In spite of the potential advantages associated with adopting a multimodal approach in RSs, there has been a noticeable scarcity of research focusing on the fusion of diverse information sources into a coherent model. This becomes particularly prominent when aiming to integrate the Utility Matrix with other information sources. A comprehensive review of relevant studies has revealed a trend where research employing the Utility Matrix tends to disregard the inclusion of supplementary information (Rendle et al., 2010; Rendle et al., 2020). This could possibly be attributed to the efficacy of collaborative MF models when dealing specifically with this type of information. Consequently, endeavors to incorporate other data sources, like user reviews or item descriptions, might lead to unwanted interference, potentially resulting in a decline in the accuracy of the system. Many other multimodal techniques that merge multiple sources often focus exclusively on textual data, neglecting the valuable insights contained within the Utility Matrix information, as evidenced in works such as (Qiu et al., 2021; F. Sun et al., 2019; B. Yang et al., 2007).

We firmly believe that the fusion of diverse data sources, encompassing the Utility Matrix, user profiles, and item descriptions, holds the potential to deliver deeply personalized recommendations that precisely mirror a user’s inclinations and preferences. Nonetheless, despite the apparent merits linked to a multimodal strategy within recommender systems, there exist noteworthy obstacles that demand attention. A primary challenge lies in devising a methodology for harmonizing

disparate information sources, all the while sidestepping the impact of noise or conflicting cues that could potentially undermine the accuracy of recommendations. This becomes especially crucial given that insights from various sources might either complement one another or redundantly overlap, necessitating a careful balancing.

The utilization of multimodal and multiview approaches presents a range of advantages that enhance their appeal in the realm of RSs. A particularly noteworthy benefit is their capacity to encompass diverse dimensions of user preferences and item attributes, leading to recommendations that are both more precise and diverse in nature. This effectiveness stems from the fact that different data sources can yield complementary perspectives that might elude a singular modality or view (J. Chen et al., 2017; L. Chen et al., 2022a; Tai et al., 2020).

Our research makes a substantial contribution to the realm of RSs through various avenues. Initially, we introduce a fresh model that merges both the Utility Matrix and textual resources. This innovative strategy entails employing feature extraction techniques from diverse information origins, while also implementing a conversion algorithm to segment the feature vectors of each user-item pair into sequences of token vectors. These token vectors subsequently serve as inputs for a classification model. Furthermore, our approach integrates a Transformer model, renowned for their prowess in addressing the intricate task of fusing heterogeneous information sources within RSs. This model possess the ability to autonomously select features, mitigating the impact of noise or conflicting cues, and exhibit potential to substantially heighten the efficacy of RSs in real-world scenarios. Lastly, we perform comprehensive experiments using the MovieLens and Amazon datasets to vividly showcase the efficacy of our novel model.

5.2 Related Works

Recent research underscores that CF recommendation models primarily hinge on user rating scores and textual review data for the computation and assessment of similarities between users or items. A model introduced by (Ghasemi & Momtazi, 2021) identifies akin users by considering their reviews and ratings, whereas (Terzi et al., 2014) modifies the user K-nearest neighbors (KNN) algorithm to gauge user similarity based on the likeness of their textual reviews. In several investigations, rating scores have been employed to compute user similarity, subsequently subjected to KNN algorithms for analysis, as exemplified by (Cui, 2017; Kamali et al., 2018; H.-M. Wang & Yu, 2015). Within CF-based RSs, two explored methodologies encompass user-based and item-based techniques. (Choudhury et al., 2021) have introduced a trust matrix measure that amalgamates user similarity and weighted trust propagation. Meanwhile, (L. Li et al., 2021) propose a hybrid strategy combining user-based CF with item-based CF and CB, while (Shi et al., 2020) advocate the incorporation of user embeddings to heighten the performance of SVD++-based CF. Addressing the cold-start issue, (Hasan & Roy, 2019) incorporate trust and genre data into item-item CF methods. Furthermore, (Duan et al., 2022) present a novel approach named Review-Based MF, which merges review-based CF with rating imputation. This solution tackles the problem of rating data sparsity by employing feature-level opinion mining from online review text. It constructs an item-topic rating matrix, populating the vacant values in the Utility Matrix, and then employs MF to generate personalized recommendations.

My research focuses on integrating multiple views from different sources in RSs. Research related to this topic can be categorized into three primary approaches: feature-level fusion, also known as early-fusion, decision-level fusion, which is also called late-fusion, and hybrid fusion. In addition, other studies on RSs using multimodal methods have also discussed techniques based on hypergraphs or tensors. Hypergraphs are a flexible and powerful approach for modeling complex user-item interactions in multimodal recommender systems. (Pliakos & Kotropoulos,

2014) proposed a method for image tagging and geo-location prediction using hypergraph learning with group sparsity constraints and social, image, and visual similarity information. (Xia et al., 2021) presented DHCN, a dual-channel hypergraph convolutional network that enhances session-based recommendation through self-supervised learning. (Fan et al., 2019) introduced GraphRec, a graph neural network framework that models interactions and opinions in the user-item graph to provide social recommendations.

Research related to early-fusion approach can be mentioned as (Ali et al., 2020), that proposed a smart healthcare system for heart disease prediction, using feature fusion and ensemble deep learning techniques. The method combines features from sensor data and electronic medical records, eliminates irrelevant and redundant features with the information gain technique, and computes a specific feature weight for each class using the conditional probability approach to improve system performance. (Thaipisutikul et al., 2022) introduced a feature-fusion framework called FIREC that uses deep learning to incorporate demographic-estimated features like age, gender, and race into facial representations to overcome information loss caused by masks. The system can rank potential identities of the person in a masked face image. (Cai et al., 2020) created a novel multimodal model to distinguish between depressed patients and normal controls by combining electroencephalogram (EEG) data from different sources. The EEG data was collected under neutral, negative, and positive audio stimulation. The authors used a feature-level fusion technique to merge the EEG data of different modalities and construct a depression recognition model. (Conceição et al., 2019) presented a novel multimodal framework for video recommendation based on deep learning, utilizing both visual and textual modalities. They first describe textual data using bag-of-words and TF-IDF models, fuse them with convolutional descriptors from visual data to construct a multimodal descriptor, and then use autoencoders to obtain a low-dimensional sparse representation for video documents, which is used to

produce a ranking of the top-N most relevant videos to the user. (X. Wang et al., 2021) developed the multimodal disentangled recommendation (MDR) model, which learns disentangled representations containing both complementary and common information from different modalities. This approach improves recommendation accuracy and enhances representation explainability. (Z. He et al., 2020) introduced a multi-view feature fusion recommendation algorithm named MVF, which utilizes an autoencoder to extract features of each view and constructs second-order and third-order interactive features. Then, the main interaction features are obtained by compressing the second-order and third-order interaction features using the singular value decomposition algorithm and Tucker tensor decomposition algorithm, respectively. Finally, the attention mechanism is used to fuse the main interaction features to get the representation of the item and user's preference model. (R. Sun et al., 2020) introduced Multi-modal Knowledge Graph Attention Network (MKGAT) as an approach to improving recommender systems by utilizing multi-modal knowledge. The approach incorporates a multi-modal graph attention mechanism to disseminate information across MMKGs, which leads to a merged embedding representation used for the recommendation process. (L. Chen et al., 2022b) introduced the MV-GAN model, which stands for Multi-View Graph Attention Network, for travel recommendation. This model improves user and product semantics by aggregating metapath-guided neighbors and fusing multiple views in a heterogeneous travel product recommendation graph, utilizing node-level and path-level attention networks and a view-level attention mechanism. (Dai et al., 2021) proposed the Attention-based Multi-View Feature fusion model (AMVF) to improve the availability of source domain information and recommend suitable items. The model learns latent transferable features and domain-specific features simultaneously by designing an attention-based feature fusion algorithm to learn different importance of various features, since various features make distinct contributions to recommendation. (X. Zhang, Luo, et al., 2020) introduced a novel

factorization model that integrates multi-view visual information with implicit feedback data for restaurant prediction and ranking. A deep convolution network is used to extract visual features from images, which are then incorporated into a collaborative filtering framework. To achieve better personalized recommendation, user-related weights are utilized to fuse multi-view visual features, indicating the personalized visual preference for restaurants, which is unique and independent for each user. (F. Liu et al., 2022) developed a recommendation system model called the dynamic multi-view fusion-based graph convolution network. The model uses multiple views and a continuous awareness mechanism to learn latent user interest topics and adjust the fusion weight of different propagation layers. Additionally, an attention mechanism and a principal component control mechanism are used to dynamically fuse features and predict the similarities between users and items. (Jing et al., 2022) developed an attentive DNN-based multi-view fusion recommendation algorithm, which uses a two-stage joint learning solution to integrate user and item attributes with user-item interaction data. The model utilizes CNNs and attention mechanisms to extract features from user and item attributes, and an extended DNN based on matrix factorization and multiple-layer perception to enhance feature extraction from user and item interaction information. (Ji et al., 2020) developed a hybrid model that integrates social relationships and multi-source data, incorporating both direct and indirect friends' influence in their recommendation algorithm. The proposed approach utilizes a joint representation learning model to fuse ratings, reviews, and social connections, integrating various data types from the data source perspective instead of combining recommendation outputs from different models.

Research related to late-fusion approach can be mentioned as (Truong & Lauw, 2019), that presented Multimodal Review Generation (MRG), a neural approach that combines a rating prediction component and a review text generation component. The authors suggested that incorporating shared user and item representations would enhance rating prediction using review text and fine-tune the sentiment of the

generated review text based on user and item of interest. (B. Li et al., 2022) investigated how to use image content information to recommend goods by analyzing the research status of CB image retrieval algorithms. They proposed a hierarchical commodity classification and retrieval system that involves a class decision layer for determining the category of commodity images and retrieving the corresponding category of commodity image features. The system also uses the corresponding relationship between different commodity visual features to recommend commodities and introduces a matching recommendation algorithm for commodities. (Shah et al., 2014) proposed a heuristic ranking approach for video recommendation based on heterogeneous late fusion, which considers three aspects: venue categories, visual scenes, and user listening history. They use SVM^{hmm} models constructed from geographic, visual, and audio features to obtain different types of video characteristics, and perform heuristic rankings to fuse the predicted confidence scores of multiple models. Their contributions include predicting scene moods from a real-world video dataset, customizing the video sound track recommendation functionality for mobile devices. (Pandeya & Lee, 2021) created a diverse music video emotion dataset and tested it on several unimodal and multimodal CNNs. They fine-tuned each unimodal CNN and compared their performance to determine the best model for integration into a multimodal structure. The resulting multimodal structure used cross-validation to prevent overfitting and achieved an overall prediction through the combination of all possible multimodal structures. In their study, (Bi et al., 2020) explored a late fusion technique using text and image modalities to classify e-commerce products on Rakuten. They developed deep neural networks for each modality and integrated them at the decision level. (K. Wang et al., 2017) suggested a deep learning-based method for decision-level information fusion. They divided the video modality into visual and audio modalities and processed them separately using improved VGG-16 and LSTM networks. Then, they combined the two modalities using an Extreme Learning Machine (ELM) for

decision-level information fusion. A decision-level fusion method is proposed by (Gumaei et al., 2022) to predict the COVID-19 patient health using well-calibrated ensemble classifiers, namely, random forest (RF), gradient boosting (GB), and extreme gradient boosting (XGB) methods. The method uses COVID-19 patient's health information, travel demographic, and geographical data and applies a soft voting technique to generate the final decision result. The proposed method aims to provide early monitoring and efficient treatment for COVID-19 patients by predicting the possible outcome of the COVID-19 case, recovered, or death.

Research related to Hybrid-fusion approach can be mentioned such as (Amer et al., 2018), in which a model based on Restricted Boltzmann Machines (RBMs) was used to learn shared representations from time-varying data across multiple modalities. The proposed Multimodal Discriminative Conditional Restricted Boltzmann Machines (MMDCRBMs) model combines CRBMs with a discriminative component for classification and is trained using labeled data to generate lower-level data for a specific label. (C. Wang et al., 2022) proposed a hybrid approach to identify gambling websites by extracting and fusing visual and semantic features of website screenshots. The approach involves fine-tuning a pre-trained ResNet34 model to extract visual features, extracting textual content through OCR, training a text classifier with Bi-LSTM to extract semantic features, fusing visual and semantic features with self-attention, and late-fusing the prediction results of image, text, and multimodal classifiers for the final prediction. A new approach to breast cancer survival prediction was presented by (Othman et al., 2023), using a hybrid deep learning (DL) model that combines CNN-LSTM and CNN-GRU to extract features automatically from the METABRIC dataset. The model distinguishes between long-term survivors and short-term survivors, and aims to reduce errors by pathologists. The authors recommend the use of this hybrid DL model in medical research for breast cancer survival prediction, and an ensemble model was proposed that achieves high accuracy. The ensemble model uses a voting

mechanism to make the final prediction. In their paper, (X. Zhang, Liu, et al., 2020) presented a new method that combines probabilistic MF with a hybrid approach. This approach comprises two sub-components: one sub-component uses auxiliary information to predict a user's rating scores based on their personal preferences, while the other sub-component models an item's textual appeal to various users using an attention-based convolutional neural network. The authors then introduced a unified framework to optimize these two sub-components with a global objective function. (Kumar et al., 2021) introduced a hybrid approach that utilizes deep learning to classify emotions conveyed by both images and their corresponding text into distinct emotional categories. The proposed method involves performing intermediate fusion on the image and text inputs, followed by late fusion on the output of the intermediate fusion and the original image and text inputs.

The research above focuses on the integration of multiple views from various sources in RSs and categorizes it into three primary approaches, including early fusion, late fusion, and hybrid fusion. The research also mentions the use of hypergraphs or tensors in RSs using multimodal methods. The early-fusion approach is discussed, and several studies related to it are mentioned, including a smart healthcare system for heart disease prediction, a feature-fusion framework that uses deep learning to overcome information loss caused by masks, a multimodal model to distinguish between depressed patients and normal controls, and a novel multimodal framework for video recommendation. Other approaches mentioned are MVF, MKGAT, and MV-GAN models, which utilize an autoencoder, multi-modal graph attention mechanism, and metapath-guided neighbors, respectively, to improve recommendation accuracy. The research also highlights the AMVF model, which learns latent transferable features and domain-specific features simultaneously and integrates multi-view visual information with implicit feedback data for restaurant prediction and ranking. Finally, the research discusses a RS model called the dynamic multi-view fusion-based graph convolution network, which uses multiple

views and a continuous awareness mechanism to learn latent user interest topics and adjust the fusion weight of different propagation layers.

Recent works have introduced neural network models as a means to harmonize information sources within RSs. NCF, a collaborative filtering framework presented by (X. He et al., 2017), harnesses neural networks to extend MF and employs a MLP to model nonlinearities inherent in the user-item interaction function. Notably, empirical observations substantiate the utilization of deeper neural network layers to enhance recommendation performance. The innovation known as BERTERS, put forth by (Nikzad–Khasmakhia et al., 2020), presents a multimodal classification approach that synergizes text and graph modalities specifically tailored for expert RSs. In this approach, BERT transforms text into vectors, while ExEm extracts features from the co-author network. These features are then fused with others to construct a definitive representation for the classifier. The work of (B. Yang et al., 2007) showcases an online video RS adept at multimodal fusion and relevance feedback. This system conceptualizes video recommendation as the search for the most pertinent videos grounded in multimodal relevance, entailing textual, visual, and aural attributes. By integrating users' click-through data, relevance feedback adjusts the weights both within and between modalities. An attention fusion function is harnessed to merge multimodal relevance effectively. Within the sequential recommendation domain, (F. Sun et al., 2019) introduces BERT4Rec, a model that leverages bidirectional self-attention to model user behavior sequences. Employing the Cloze objective for training, this model addresses sequential recommendations. Expanding the landscape, (Qiu et al., 2021) proposes a text-aware recommendation model grounded in a multi-attention neural network structure. Through modified LDA and paragraph vector learning, text vector representation is realized. This approach captures contextual cues via a Bidirectional LSTM (Bi-LSTM) layer and incorporates a CNN layer for emotional factor classification and prediction.

The incorporation of diverse information sources within RSs holds the promise of substantial benefits. However, the fusion of these sources remains a relatively underexplored domain, particularly when considering the integrating of the Utility Matrix with other data sources. It has come to our attention that studies centered around the Utility Matrix frequently disregard other information sources, potentially leading to interference and diminishing accuracy when attempting to merge sources like user reviews or item descriptions. To tackle this problem, we present a novel approach: a transformer-based multiview recommendation model adept at harmonizing information from a multitude of sources, including the Utility Matrix. This model is designed to furnish personalized recommendations that impeccably mirror user preferences. Nevertheless, the successful integration of such diverse sources is accompanied by substantial challenges, chief among them the task of circumventing noise or conflicting cues. Our study serves as an illustrative testament to the promise held by a multimodal strategy for enhancing RSs. Moreover, we propose the deployment of a transformer model to elevate their performance, offering a potential pathway to surmount the challenges posed by the effective integration of heterogeneous and diverse data sources.

5.3 The Proposed Model

5.3.1 Multi-view Transformer Model For Recommendation

Multiview fusion refers to the process of integrating data from various sources or perspectives, including images, text, audio, video, and other types of data, to create a comprehensive and accurate representation of a specific situation or event.

In the realm of machine learning and artificial intelligence, multiview fusion techniques are used to boost the performance of models that handle diverse and intricate data. By combining information from multiple perspectives, these models can achieve a more profound comprehension of underlying patterns and relationships in the data, resulting in enhanced predictions and decision-making.

Early fusion and late fusion are two methods used for multiview fusion. In early fusion, features or representations from different views are combined at the input level, transforming them into a shared feature space before being fed into the fusion model. This approach is ideal for views that are highly correlated and offer complementary information. On the other hand, late fusion involves combining the outputs of individual models trained on each modality separately. Each view is processed using a specific model, and the outputs are then combined using a fusion mechanism. Late fusion is employed for views that are not strongly correlated and provide distinct, independent information.

Our recommendation model, the Multiview Transformer, adopts an early fusion approach. The overview of our proposed model's architecture for the recommendation problem can be seen in Figure 5.1. The model is composed of three main components. The first component extracts and represents features from different information sources related to users and items. The second component involves segmenting the concatenated vector into n segments. Finally, the third component incorporates a transformer module for the prediction task.

Algorithm 9 Multiview Transformer Model For Recommendation

- 1: **Training Stage:**
 - 2: Input: The dataset contains multiview of a set of users and items.
 - 3: Output: A model for the rating
 - 4: **Step 1:** Feature Extraction
 - 5: **for** each modality **do**
 - 6: **if** user modality **then return** $V_u = v_{u1}, \dots, v_{uk}$ //A set of k feature vectors of a given user.
 - 7: **else if** Item modality **then return** $V_i = v_{i1}, \dots, v_{ij}$ //A set of j feature vectors of a given item.
 - 8: **end if**
 - 9: **end for**
 - 10: **Step 2:** Feature Conversion
Combining feature vectors from Step 1 and segment it into a sequence of token vectors as input to the classification model.
 - 11: **Step 3:** Classification: building the Transformer Model for the Prediction task (named as TMP)
 - Feeding n tokens from Step 2 into a transformer encoder model is used for the task of rating generation (i.e. prediction),
 - Training prediction model.
 - 12: **Inference stage:**
 - 13: Input: A given user-item pair (u, i) .
 - 14: Performing the feature extraction task:
 - $\leftarrow V_u = v_{u1}, \dots, v_{uk}$ //Feature vectors for user u
 - $\leftarrow V_i = v_{i1}, \dots, v_{ij}$ //Feature vectors for item i
 - 15: Combining V_u and V_i , then using the Feature Conversion algorithm to generate n segments.
 - 16: Feeding n tokens at Step 2 as input features for the TMP for generating the rating.
-

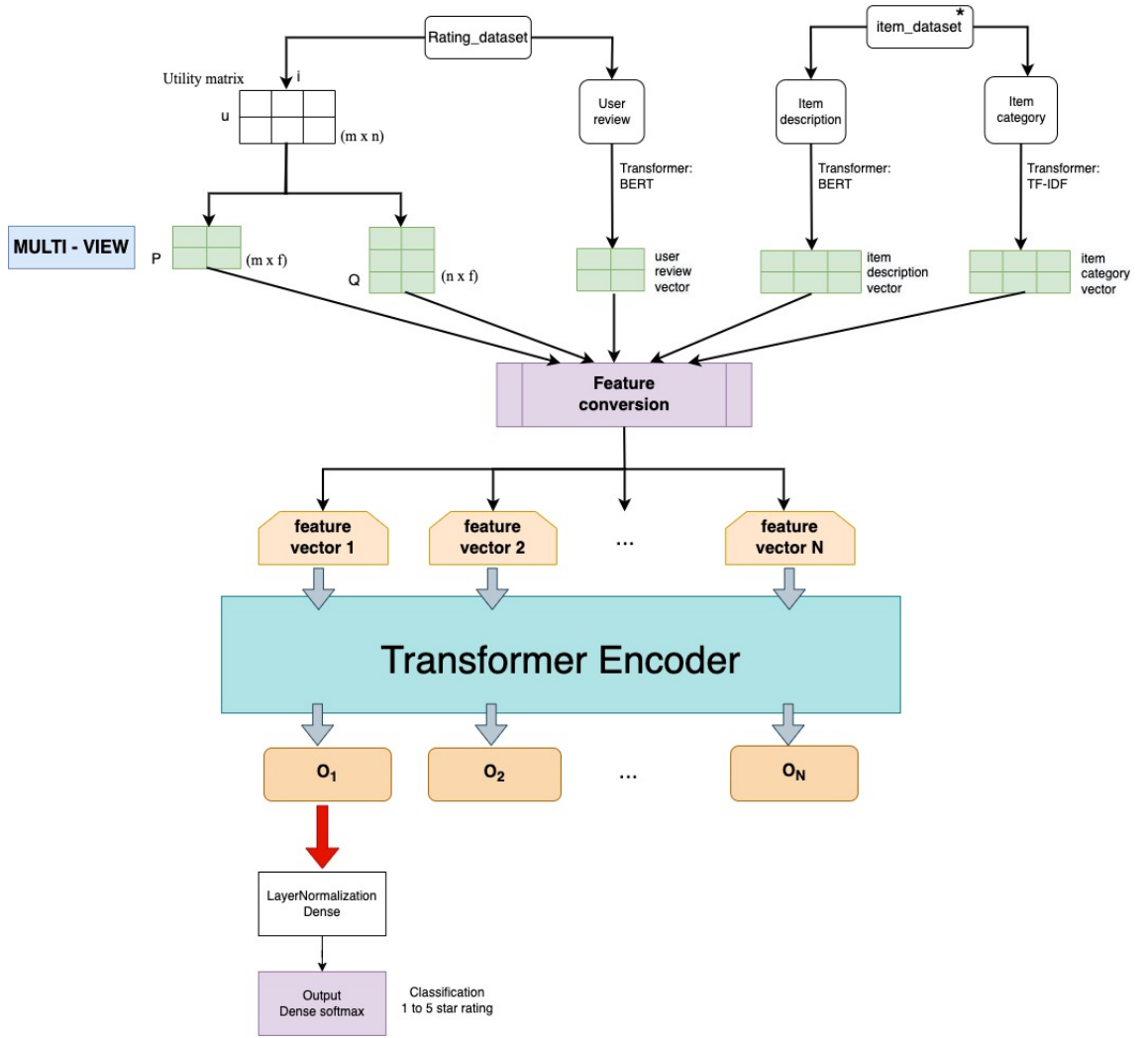


Figure 5.1 Multiview Fusion with Transformer Model for Recommender Systems.

5.3.2 The Feature Extraction Algorithm

The feature-extracting component plays a crucial role in our proposed model. To represent users and items, we generate various features by gathering information from four distinct sources: the Utility Matrix, user reviews, item descriptions, and item categories. We provide details about our approach below.

We utilize the matrix factorization technique, which we refer to as Rating-Feature, to extract latent features of users and items from the given Utility Matrix. This technique involves decomposing the Utility Matrix, which contains ratings for m users and n items, into two lower rank matrices. The decomposed

matrices represent users and items with f latent factors or features. For the purpose of factorization, we opt to determine the count of latent factors and employ the ALS algorithm. Our approach involves initializing the matrices for users and items with random values. Through iterative optimization, these matrices are adjusted to minimize the disparity between predicted and observed ratings. This iterative process alternates between maintaining the user matrix and refining the item matrix until a state of convergence is achieved. This culminates in the creation of two lower-dimensional matrices with dimensions of $m \times f$ and $n \times f$, corresponding to the user and item matrices, respectively.

1. A matrix of size $(m \times f)$ represents the user features, where each row pertains to a particular user and contains a vector of f features. We view this matrix as a representation of the user's characteristics, commonly known as the user view.
2. A matrix of size $(n \times f)$ represents the item features, where each item is represented by a vector of f features. Each row of this matrix corresponds to a specific item and is viewed as a representation of that item, commonly known as the item view.

To generate the Review-Feature, we use a pre-trained BERT model to create a vector representation of the text reviews written by users for items in the (user u , item i) pair. This vector representation serves as a view of user u .

Similarly, we generate the Description-Feature by utilizing a pre-trained BERT model to create a vector representation of the text description provided for item i . This vector representation is considered a view for item i .

To generate the Category-Feature, we create a vector representation by utilizing the list of categories to which item i belongs. We employ TF-IDF to measure the significance of words in the list of categories by taking into account their frequency in the list and the corpus. Since BERT is more suitable for extracting features from

complete sentences, we use TF-IDF instead of BERT to create the vector representation of these category features. This vector representation is also viewed as a representation of item i .

The following feature types, which will be extracted, are denoted as:

- The Rating-Feature refers to the evaluations of a user u and an item i , and is denoted as $\text{Rat}(u)$ and $\text{Rat}(i)$ respectively.
- The Review-Feature pertains to a user u and is indicated by $\text{Rev}(u)$.
- The Description-Feature pertains to an item i and is denoted by $\text{Des}(i)$.
- The Category-Feature pertains to an item i and is represented by $\text{Cat}(i)$.

5.3.3 The Feature Conversion Algorithm

Algorithm 10 The Feature Conversion Algorithm

- 1: The Feature Extraction module generates different types of features as corresponding feature vectors for each (user u , item i) pair:
 - $\text{FEM} \leftarrow fu_1, fu_2$ //Feature vectors for user u
 - $\text{FEM} \leftarrow fi_1, fi_2, fi_3$ //Feature vectors for item i
 - 2: Transform features:
 - $F1 = \text{Concat}(fu_1, fu_2, fi_1, fi_2, fi_3)$ //Combine these feature vectors and obtain the united vector
 - $F2 = \text{FFLayers}(F1)$ //Feed vector $F1$ into feedforward layers (denoted by FFLayers)
 - $\text{segment}(F2) \leftarrow (\text{token}_1, \text{token}_2, \dots, \text{token}_k)$ //A sequence of token vectors will be fed into the Transformer model for the task of ratings prediction.
-

After acquiring the feature representation as a sequence of token vectors through the aforementioned steps, we develop a Transformer model to classify the ratings.

5.4 Experiments and Results

5.4.1 The overview of datasets

Preprocessing datasets:

To evaluate the effectiveness of our proposed model, we used the MovieLens dataset and several sub-datasets from Amazon. To improve the accuracy of our analysis, we preprocessed the data by removing users who rarely rated items and items that received few ratings from the community. Additionally, in cases where a user assigned multiple ratings to the same item, we retained only the most recent rating.

Describing datasets:

Our experiments relied on the Amazon dataset, which can be accessed at <https://jmcauley.ucsd.edu/data/amazon/>. Specifically, we used sub-datasets such as Electronic, Video Games, and Toys and Games. To broaden the scope of our analysis, we integrated diverse information, such as rating scores, user reviews, item categories, and item descriptions.

To conduct our research, we utilized the MovieLens dataset, which can be obtained through <https://grouplens.org/datasets/movielens>. This dataset provided us with various data points, including user and movie identification, rating scores, and movie genres, which we leveraged in our analysis. Please refer to Table 5.1 for a summary of the MovieLens and Amazon datasets.

- The MovieLens dataset includes 77,763 ratings of 27,041 users and 8,203 movies.
- The Electronic dataset includes 80,472 ratings of 1,042 users and 21,200 items.
- The Video Games dataset includes 98,769 ratings of 2,291 users and 24,708 items.
- The Toys and Games dataset includes 68,102 ratings of 5,462 users and 3,028

items.

Table 5.1 The overview of the datasets.

Dataset	Ratings	Users	Items
MovieLens	77763	27041	8203
Amazon: Toys and Games	68102	5462	3028
Amazon: Video and Games	98769	2291	24708
Amazon: Electronic	80472	1042	21200

5.4.2 Evaluation Metrics

To evaluate the proposed model, we use metrics including MAE, RMSE, and Precision.

5.4.3 Experimental setups for datasets

For the experimental phase, the dataset was partitioned into smaller subsets intended for training, validation, and testing purposes. The specifics of this division are outlined in Table 5.2. To elaborate further, the individual counts of ratings for the training, validation, and testing sets in each dataset are as follows:

- In the MovieLens dataset, the training set comprises 62,211 ratings, while the validation set contains 3,110 ratings, and the testing set encompasses 12,442 ratings.
- For the Toys and Games dataset, the training set consists of 54,482 ratings, the validation set holds 2,724 ratings, and the testing set encompasses 10,896 ratings.
- Within the Electronic dataset, the training set encompasses 64,378 ratings, the validation set consists of 3,218 ratings, and the testing set comprises 12,876 ratings.
- Lastly, in the Video and Games dataset, the training set contains 79,016 ratings, the validation set comprises 3,950 ratings, and the testing set holds 15,803 ratings.

Table 5.2 Setting up the data for the experiments.

Dataset	Train	Validation	Test
MovieLens	62211	3110	12442
Amazon - Toys and Games	54482	2724	10896
Amazon - Electronic	64378	3218	12876
Amazon - Video and Games	79016	3950	15803

5.4.4 Experimental setups for Models

Our research aims to show that using the Transformer model to combine various perspectives is more effective than relying on a single view, such as the Utility Matrix, and superior to using the SVD model for the Utility Matrix. To substantiate this hypothesis, we must conduct empirical testing using a consistent configuration. Consequently, we opted for the test configuration without prioritizing the selection of the optimal configuration. The parameter values were derived from previous research and were tested with several different values to identify the best configuration from that set.

Regarding the attribute vectors' magnitude, we have established the following values:

- The attribute vectors in our dataset are generated using BERT for user reviews, resulting in a magnitude of 768 dimensions that matches the output vector size of the BERT-base model.
- We experimented with various magnitudes of the latent feature vector (20, 50, and 100) to represent users and items using the factorization matrix approach. Based on the validation set's performance evaluation, we determined that a magnitude of 50 was the optimal choice.
- We employed the complete set of tokens in our dataset to create attribute representation using the TF-IDF measure. As a result, the size of the resulting representation vectors will match the number of keywords used.

The user latent features, item latent features, item genre/category features, user review features, and item description features are denoted as f_{u_1} , f_{i_1} , f_{i_2} , f_{u_2} , and f_{i_3} , respectively. To configure our Multiview Transformer recommendation model, we specified dimensional parameters for each input feature vector, as presented in Table 5.3.

- The MovieLens dataset has f_{u_1} , f_{i_1} , and f_{i_2} dimensions of 50, 50, and 20, respectively.
- The Amazon - Toys and Games dataset has f_{u_1} , f_{i_1} , f_{i_2} , f_{u_2} , and f_{i_3} dimensions of 50, 50, 321, 768, and 768, respectively.
- The Amazon - Electronics dataset has f_{u_1} , f_{i_1} , f_{i_2} , f_{u_2} , and f_{i_3} dimensions of 50, 50, 944, 768, and 768, respectively.
- The Amazon - Video and Games dataset has f_{u_1} , f_{i_1} , f_{i_2} , f_{u_2} , and f_{i_3} dimensions of 50, 50, 16978, 768, and 768, respectively.

To examine the impact of utilizing different views of the data in the Multiview selection, we conducted various experiments, including:

- We conducted an experiment on data represented by two views, which were generated from the Utility Matrix and represented users and items.
- Another experiment was performed on data represented by all available views. For the MovieLens dataset, there were three views in total, including two from the Utility Matrix and one from Genre. For the Amazon dataset, there were five views in total, including two from the Utility Matrix and three from user reviews, item descriptions, and item categories.

We have developed the following models for the purpose of predicting ratings:

- The training method for the Transformer model: We have transformed into a classification problem. Each rating ranging from 1 to 5 is mapped to a

corresponding classification label within the range of 0 to 4. The model takes in preprocessed inputs and subsequently predicts classifier outputs labeled from 0 to 4.

- After trying various configurations of the Transformer model in our proposed approach, with different numbers of layers (2, 4, or 6) and hidden state values (20 and 50), we assessed their performance on the validation set and identified the optimal configuration to be (Layers = 4, Hidden state = 20, Heads = 4).
- To demonstrate that our proposed Transformer model outperforms existing models, we designed two baseline models: a Feedforward Neural Network (MLP) and GRU.
- Furthermore, we have evaluated a powerful baseline model, SVD, on the Utility Matrix, which is widely recognized as one of the most effective models for the RS problem.

Our model underwent training over 50 epochs, and we conducted the experiment three times for each model. Given the minimal variance observed among the experimental iterations, we identified the optimal outcome from the trials and incorporated it into Table 5.4.

Within the context of the graph-based recommendation model, we devised a graph structure comprising two distinct node types: users and items.

- In the Amazon dataset, we incorporated category and description attributes as attributes for the item nodes. Additionally, the user nodes were enriched with the review attribute.
- For the MovieLens dataset, we added the genre attribute to the item nodes.
- The graph was designed to establish connections between user and item nodes, encapsulating their interactions. To quantify the intensity of these interactions,

we assigned ratings as the weight property of the edges. This mechanism allowed us to depict the degree of preference demonstrated by each user toward every item, a pivotal facet in our RS.

Table 5.3 Setting up the parameters for the multiview transformer RS

The dataset	Dimensions of user latent features	Dimensions of item latent features	Dimensions of item genre / category features	Dimensions of user review features	Dimensions of item description features
MovieLens	50	50	20		
Toys and Games	50	50	321	768	768
Electronic	50	50	944	768	768
Video and Games	50	50	16978	768	768

5.4.5 Results

We compared the performance of our proposed Multiview transformer recommendation model with the SVD, MLP, and GRU recommendation models on the MovieLens and Amazon datasets to evaluate its effectiveness. Our experiments supported two hypotheses:

- Integrating information from both the Utility Matrix and textual features using the Transformer model yields better outcomes than relying solely on features generated from the Utility Matrix.
- The Transformer model exhibits superior performance compared to other models, namely GRU and MLP, in terms of delivering better results.

Experimental results for four datasets, including The MovieLens dataset and three Amazon sub-datasets (Toys and Games, Video and Games, and Electronic), are presented in Table 5.4. The results support our first hypothesis, which suggests that our proposed model outperforms the SVD model, which only uses the Utility Matrix, in all three metrics (MAE, RMSE, and Precision) across all four datasets.

Additionally, our second hypothesis, which suggests that the Transformer model performs better than the GRU and MLP models, is also supported. The Transformer model exhibits the best Precision results, and outperforms the other models in the MAE and RMSE metrics for most datasets, with the exception of the MovieLens dataset where the GRU model produces the best results in both MAE and RMSE metrics. Similarly, for the Amazon - Electronic dataset, the MLP model yields the best RMSE.

Specifically, when evaluating our proposed transformer recommendation model on the MovieLens dataset, which integrated three views including user latent features, item latent features, and item genre features, we obtained MAE, RMSE, and Precision metrics of 0.964, 1.216, and 67.16%, respectively. Our transformer recommendation model exhibited the best Precision compared to the MLP, GRU, and SVD recommendation models (as shown in Figure 5.2). Moreover, our transformer recommendation model provided slightly better results when integrating three views than when integrating only two views.

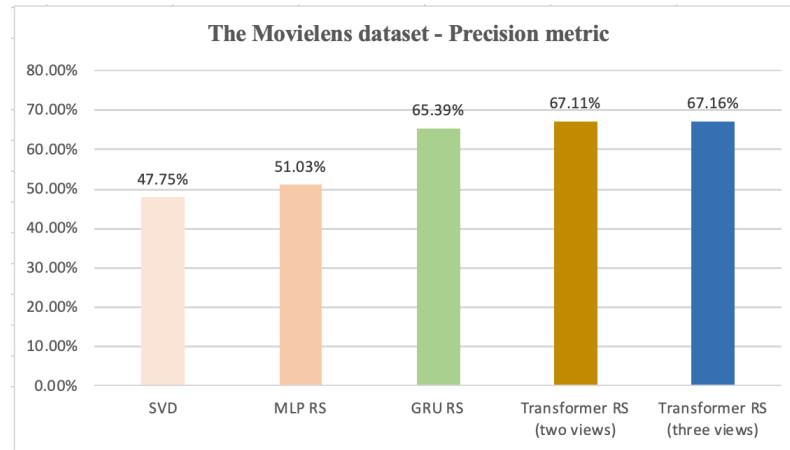


Figure 5.2 The comparison of the Precision metric among recommendation models on the MovieLens dataset.

During the experiments conducted on the Amazon - Toys and Games dataset, we integrated five views into our proposed transformer recommendation model. These views included user latent features, item latent features, item category

Table 5.4 The comparison of the MAE, RMSE, and Precision metrics among recommendation models on the MovieLens and Amazon datasets.

Models	MAE	RMSE	Precision
MovieLens:			
MLP RS [three views]	1.261	1.677	51.03%
GRU RS [three views]	0.852	1.062	65.39%
Transformer-based RS [two views]	0.973	1.230	67.11%
Transformer-based RS [three views]	0.964	1.216	67.16%
SVD RS	1.562	2.293	47.75%
Amazon: Toys and Games:			
MLP RS [five views]	0.527	0.776	86.65%
GRU RS [five views]	0.729	0.889	86.31%
Transformer-based RS [two views]	0.570	1.09	86.85%
Transformer-based RS [five views]	0.445	0.743	92.07%
SVD RS	0.844	1.120	68.06%
Amazon: Video and Games:			
MLP RS [five views]	0.729	0.999	73.74%
GRU RS [five views]	0.941	1.174	76.89%
Transformer-based RS [two views]	0.842	2.062	77.09%
Transformer-based RS [five views]	0.689	1.471	83.31%
SVD RS	1.572	1.254	63.99%
Amazon: Electronic:			
MLP RS [five views]	0.628	0.914	83.22%
GRU RS [five views]	0.78	1.007	85.80%
Transformer-based RS [two views]	0.606	1.362	85.94%
Transformer-based RS [five views]	0.554	1.195	87.83%
SVD RS	0.85	1.159	69.59%

features, item description features, and features extracted from user reviews. The model yielded superior results in terms of MAE, RMSE, and Precision metrics, with values of 0.445, 0.743, and 92.07%, respectively, as compared to MLP, GRU, and SVD recommendation models (as shown in Figure 5.3). Moreover, our proposed transformer recommendation model performed better when integrating five views as compared to integrating only two views that contained user latent features and item latent features.

In the experiments conducted on the Amazon - Video and Games dataset, our

proposed transformer recommendation model incorporated five views, namely user latent features, item latent features, item category features, item description features, and features extracted from user reviews. The model yielded MAE, RMSE, and Precision metrics of 0.689, 1.471, and 83.31%, respectively, which were better than those of recommendation models based on MLP, GRU, and SVD (as shown in Figure 5.4). Furthermore, our proposed transformer recommendation model exhibited superior results in terms of MAE and Precision than the other models. Additionally, when compared to the integration of only two views containing user latent features and item latent features, our proposed transformer recommendation model delivered better results when integrating all five views.

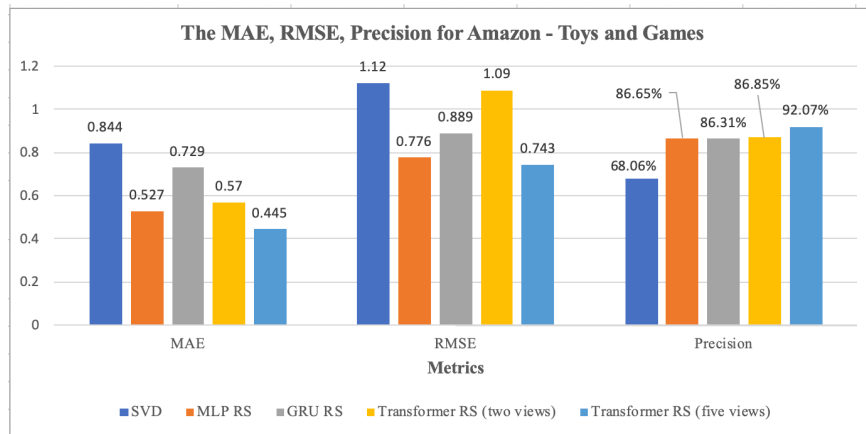


Figure 5.3 The comparison of the MAE, RMSE, and Precision metrics among recommendation models on the Amazon - Toys and Games dataset.

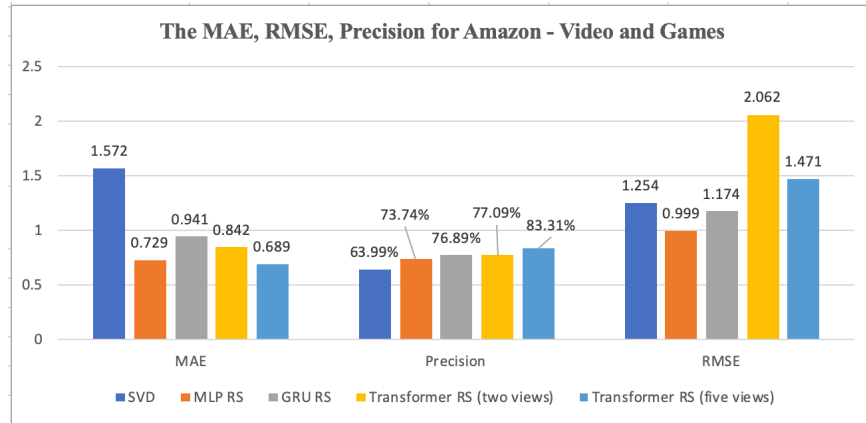


Figure 5.4 The comparison of the MAE, RMSE, and Precision metrics among recommendation models on the Amazon - Video and Games dataset.

Experimental results were also obtained for the Amazon - Electronics dataset, which demonstrated the effectiveness of our proposed transformer recommendation model integrating five views with MAE, RMSE, and Precision metrics of 0.554, 1.195, and 87.83%, respectively. Our proposed model outperformed recommendation models based on MLP, GRU, and SVD in terms of MAE and Precision (as illustrated in Figure 5.5). Moreover, the model performed better when integrating five views rather than only two views containing user latent features and item latent features.

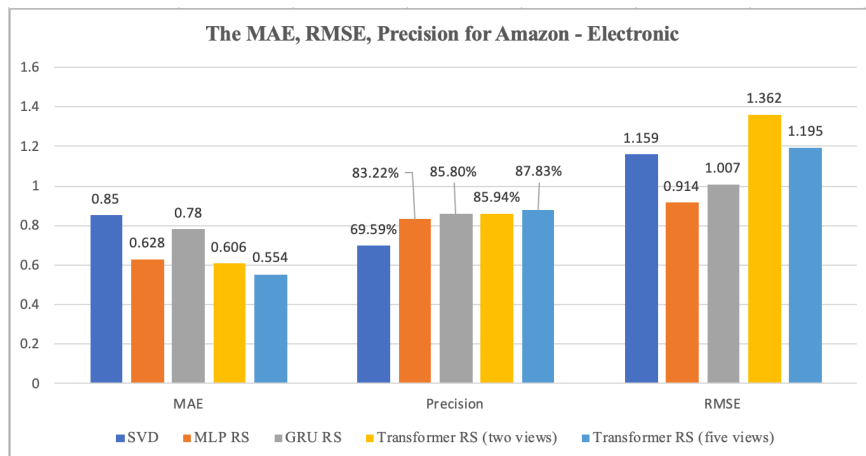


Figure 5.5 The comparison of the MAE, RMSE, and Precision metrics among recommendation models on the Amazon - Electronic dataset.

Apart from comparing our proposed model to recommendation models based on SVD and DNN (MLP and GRU), we conducted experiments using the graph-based recommendation model on the same train and test sets obtained from the MovieLens and Amazon sub-datasets. Our experiments provide empirical evidence that our proposed recommendation model outperforms the graph-based recommendation model. Specifically, the outcomes presented in Table 5.5 demonstrate that our proposed model achieves higher recommendation accuracy. The Precision metric shows an improvement ranging from 11.14% to 108%, the MAE metric indicates a decrease ranging from 31.19% to 65.37%, and the RMSE metric showcases a decrease ranging from 15.56% to 59.24%.

Table 5.5 The comparison of the MAE, RMSE, and Precision metrics between our proposed model and a graph-based RS on the MovieLens and Amazon datasets.

Dataset	Methods	Numbers of Views	MAE	RMSE	Precision
MovieLens	Graph	3	1.401	1.908	60.43%
	Our proposed model: Transformer	3	0.964	1.216	67.16%
Amazon: Toys and Games	Graph	5	1.285	1.823	60.30%
	Our proposed model: Transformer	5	0.445	0.743	92.07%
Amazon: Video and Games	Graph	5	1.305	1.742	56.01%
	Our proposed model: Transformer	5	0.689	1.471	83.31%
Amazon: Electronic	Graph	5	1.542	1.863	42.25%
	Our proposed model: Transformer	5	0.554	1.195	87.83%

We have also conducted a comparative analysis of our proposed model's results and those reported in recent studies on multi-modal methods for RSs (The detail is shown in Table 5.6). We used the same recommendation accuracy metrics as our proposed model to compare. Since replicating all experiments from other studies exactly is unfeasible, we comparatively evaluated the percentage of efficiency improvement of the research models with the SVD recommendation model. Our

proposed recommendation model improved the recommendation accuracy compared to the SVD recommendation model by up to 56.17% on the MAE metric and up to 31.26% on the RMSE metric when experimented on different datasets. For instance, (G. Li et al., 2019) designed a hybrid RS that uses images to tackle the "data sparseness" issue. Their model produced two recommendation lists (LA and LH) based on cross-modal correlations and user preferences, respectively. However, the experimental results showed a decrease in recommendation accuracy by 15.69% on the MAE metric compared to the SVD model, albeit with an increase in the RMSE metric to 2.24%. On the other hand, (Santamaria-Granados et al., 2021) suggested TERS to generate tourist recommendations based on the user's emotional state, using HR measurements and deep learning algorithms. They reported an increase in recommendation accuracy by 18.95% on the MAE metric and 29.11% on the RMSE metric compared to the SVD model.

Table 5.6 The comparison of the MAE and RMSE between our proposed model and some other research about Multi-modal RSs.

Models	MAE (Rate reduction compared to SVD-based RSs)	RMSE (Rate reduction compared to SVD-based RSs)
Our proposed model	up to 56.17%	up to 31.36%
(G. Li et al., 2019)	15.69%	2.24%
(Santamaria-Granados et al., 2021)	18.95%	29.11%

Furthermore, the research model GraphRec by (Fan et al., 2019) achieved the best experimental results on different datasets with an MAE metric of 0.7387 and an RMSE metric of 0.9794. In contrast, our proposed recommendation model achieved the best experimental results with an MAE metric of 0.445 and an RMSE metric of 0.743.

We also conduct additional experiments with the multiview transformer-based integration model for recommender systems, using selected information sources with

k best features. Subsequently, we compare these experimental results with those of the transformer-based multiview integration model on unselected information sources with k best features, as well as with the proposed model from Chapter 4, which fuses information sources with selected k best features using a neural network. The experimental results in Table 5.7 show that the transformer-based multiview integration model for recommender systems, which utilizes information sources without selecting only the k best features, yields nearly the best results when tested on datasets. This can be explained by its ability to learn effective features from the training data, enabling it to capture complex relationships and dependencies within the combined information from the Utility Matrix and textual sources, resulting in improved recommendation accuracy.

Table 5.7 The experimental results of the transformer-based multiview integration model for recommender systems are compared in three scenarios: using unselected information sources of k best features, using selected information sources of k best features, and with a proposed model from Chapter 4, which fuses information sources selected k best features using a neural network.

Model	MAE	RMSE	Precision
Movielens:			
K best features with DNN	1.092	1.24	54.35%
K best features with Transformer	0.979	1.228	66.96%
Fusion with Transformer	0.964	1.216	67.16%
Amazon: Toys and Games:			
K best features with DNN	0.716	0.88	86.53%
K best features with Transformer	0.536	0.994	88.76%
Fusion with Transformer	0.445	0.743	92.07%
Amazon: Video and Games:			
K best features with DNN	0.946	1.19	65.60%
K best features with Transformer	0.758	1.294	81.17%
Fusion with Transformer	0.689	1.471	83.31%
Amazon: Electronic:			
K best features with DNN	0.764	1.02	82.62%
K best features with Transformer	0.606	1.169	85.99%
Fusion with Transformer	0.554	1.195	87.83%

5.5 The Chapter Summary

This study introduces a novel approach called multiview fusion for recommender systems, addressing the challenge of integrating multiple information sources such as the Utility Matrix and textual data. Unlike previous studies, our method successfully tackles this problem by leveraging the Transformer Model and feature extraction techniques for each information source.

Our experimental findings indicate that the proposed model outperforms both the SVD method and the baseline model (MLP) that solely rely on user and item representation from the Utility Matrix. Specifically, our proposed model achieves higher recommendation accuracy compared to the baseline model (MLP) in terms of Precision, with improvements ranging from 6.03% to 35.48% for the Amazon and MovieLens datasets. Furthermore, it demonstrates enhanced accuracy in terms of MAE, with reductions ranging from 10.79% to 31.03% for the Amazon datasets. In comparison to the widely recognized SVD model, our proposed approach shows an increase in Precision ranging from 26.21% to 40.65% and an improvement in MAE ranging from 34.82% to 56.17% for the Amazon datasets.

Furthermore, the experimental results obtained from the MovieLens and Amazon datasets provide evidence that our proposed model, which integrates information from both the Utility Matrix and textual sources, outperforms using only the information from the Utility Matrix in terms of recommendation accuracy. However, it is important to note that the performance may vary depending on the characteristics of the data, and it cannot be assumed that incorporating multiple views will always enhance performance. Additionally, the quality of additional data can impact the overall model performance. We have conducted an analysis that demonstrates the superiority of our proposed model over the SVD approach in all four datasets, with three out of the four datasets showing improved performance.

Moreover, our model exhibits better performance compared to the graph-based recommendation model. It achieves a significant increase of up to 108% in the

Precision metric, a decrease of up to 65.37% in the MAE metric, and a decrease of up to 59.24% in the RMSE metric.

In future work, we plan to expand our approach by incorporating new information sources and further enhancing the efficiency of the Transformer-based model for integrating multiple views.

CHAPTER 6. CONCLUSIONS

6.1 Achievements

This Ph.D. dissertation's objective is to integrate diverse and heterogeneous sources within recommender systems to enhance recommendation accuracy. Our first contribution to the field of multimodal recommender systems involves adopting a late fusion approach, wherein outputs from individual recommendation models are combined for decision-making in predicting user-item ratings. Our second and third contributions pertain to the early fusion approach, which involves integrating features from various information sources. These information sources undergo distinct processing techniques to achieve a unified representation. Subsequently, advanced machine learning models are employed to predict user preferences in these multimodal recommendation systems. The third contribution builds upon the second, leading to the advancement of recommender systems enhancement.

In the first study, we developed a hybrid recommender system that combines predicted ratings from three individual models using voted-based and weighted-based methods. The weighted-based approach utilizes a neural network (MLP), while the voted-based approach includes maximum, minimum, and average strategies. Different combination strategies were employed to merge the predicted ratings, and the resulting combined ratings were used as input for a neural network in the hybrid system. We also incorporated the KNN algorithm into the individual recommendation models, enhancing their performance. Experimental results show that the weighted-based hybrid model outperforms the best-performing three individual recommendation models on the MovieLens and Amazon datasets.

The second study focuses on creating a model that utilizes multiple sources of information about users and items through a multimodality approach. The model extracts latent features of users and items using MF techniques and combines them with other features extracted by other techniques (such as TF-IDF, BERT, etc.) from

various sources such as user reviews, item descriptions, and item categories. Principal Component Analysis (PCA) and Feature Selection techniques are also utilized to increase accuracy. Then we have used a neural network (MLP) for the rating prediction task. The study shows that the proposed model outperforms SVD in terms of recommendation accuracy, and the results demonstrate the effectiveness of the proposed model.

The third study is a continuation of the second study, aiming to further enhance recommendation accuracy. The proposed method utilizes the Transformer Model, which efficiently integrates information from Utility Matrix and textual sources. The transformer model leverages its ability to learn effective features from the training data. Its capacity to capture complex relationships and dependencies within the combined information from the utility matrix and textual sources enables it to generate more accurate predictions. The study shows that the proposed model outperforms MLP for the MAE metric. A reduction range of 10.79% to 31.03% for Amazon sub-datasets and a decrease in the MAE metric by 34.82% to 56.17% for Amazon sub-datasets compared to SVD. These results exhibit the efficacy of the proposed method in improving recommendation accuracy.

In general, new contributions in my work to the research field of multiview integration for recommender systems include:

1. Developing a hybrid recommender system that combines outcomings from three individual recommendation models through voted-based or weighted-based.
2. Developing a multi-modal fusion model for integrating the Utility Matrix sources and textual sources, that's not commonly explored in multimodal RSs.
3. Exploiting a Transformer model for the multimodal fusion model in recommender systems.

6.2 Discussion And Future Research Directions

The results presented above demonstrate the effectiveness of utilizing multiple sources of information in recommender systems through multiview and multimodal approaches. These approaches involve extracting information from multiple sources, such as user reviews, item descriptions, and item categories, and combining them to provide more accurate recommendations.

To compare the improvement of the recommendation accuracy of the proposed models in the thesis, we have summarized the experimental results in Table 6.1. We utilized two metrics, MAE and Precision, for this assessment. MAE is suitable for a general evaluation of system accuracy, providing the average error in predictions without being overly concerned with large disparities between predictions and real values. Precision measures the percentage of correctly predicted user preferences for products across all predictions, aligning with the same evaluation goal as MAE. A smaller MAE and a larger Precision indicate higher recommendation accuracy.

To enable the general evaluation of a model i using these two metrics, we converted the experimental results to MAE and Precision indexes as follows:

$$MAE_{general} = (MAE_i - MAE_{min})$$

$$Precision_{general} = -(Precision_i - Precision_{max})$$

As the smaller the MAE and the larger the Precision, we take a negative value for $(Precision_i - Precision_{max})$ in the value conversion formula.

To evaluate the model i , we used the following formula: $general_evaluation = (MAE_{general} + Precision_{general})/2$. The smaller the $general_evaluation$ value of the model, the better its recommended accuracy.

The overall evaluation results presented in Table 6.1 show that Transformer-based Multiview integration model for recommender systems gives the best results on both Electronic and Toys and Games datasets. The results of this evaluation demonstrate that using the Transformer-based Multiview integration

Table 6.1 A general comparison of the proposed models in the study based on MAE and Precision metrics.

Model	MAE	Precision	MAE rate	Precision rate	General Evaluation	Rank
Electronic:						
The hybrid model for RSs (Chapter 3)	0.656	83.36%	0.102	0.045	0.073	3
K best features with DNN (Chapter 4)	0.764	82.62%	0.210	0.052	0.131	4
K best features with Transformer (data in chapter 4 + the model in chapter 5)	0.606	85.99%	0.052	0.018	0.035	2
Fusion with Transformer (Chapter 5)	0.554	87.83%	0	0	0	1
Toys and Games:						
The hybrid model for RSs (Chapter 3)	0.599	81.40%	0.154	0.107	0.130	3
K best features with DNN (Chapter 4)	0.716	86.53%	0.271	0.055	0.163	4
K best features with Transformer (data in chapter 4 + the model in chapter 5)	0.536	88.76%	0.091	0.033	0.062	2
Fusion with Transformer (Chapter 5)	0.445	92.07%	0	0	0	1

model for recommender systems significantly enhances recommendation results. Therefore, organizations/businesses can apply this proposed model to their e-business systems to help increase sales.

Our proposed Transformer-based Multiview model outperformed others because of the efficacy of the scaled dot-product attention mechanism of the Transformer model, that empowers the model to discern the significance of individual tokens within a sequence. Notably, this sequence is segmented through the utilization of the algorithm denoted as Algorithm 10. This attention mechanism, in turn, grants the model the capacity to capture interdependencies among features more adeptly than conventional architectures like MLPs, RNNs, or CNNs.

Another potential research direction is to investigate the effectiveness of

incorporating user context in multiview or multimodal recommender systems. User context can include factors such as user location, time of day, and user behavior, and incorporating this information can lead to more personalized and accurate recommendations.

Furthermore, it is important to evaluate the effectiveness of these approaches in different domains and with different types of data. For example, multiview or multimodal approaches may be more effective for certain types of products or services, or with certain types of user data. Therefore, future research should focus on evaluating the effectiveness of these approaches in different domains and with different types of data.

Finally, there is also a need to develop evaluation metrics that can capture the performance of multiview or multimodal recommender systems more accurately. Traditional evaluation metrics such as accuracy and precision may not be sufficient for evaluating the performance of these systems, as they may not capture the complexity of the recommendations provided.

In conclusion, the results presented above demonstrate the effectiveness of multiview or multimodal approaches in improving the accuracy of recommender systems. However, there is still much room for further research in this field, including incorporating user context, evaluating the effectiveness in different domains and with different types of data, and developing more accurate evaluation metrics.

LIST OF PUBLICATIONS

Throughout the implementation phase of the thesis, the Ph.D. student, assuming the role of the first author, successfully published the following works. To ensure the significance and quality of these publications, the student meticulously assessed their rankings using SCImago for journals and Conference Ranks for conferences. Consequently, the works listed below represent a culmination of the student's research efforts and were ordered based on their respective rankings.

[P1] **Ho, T. L.** Le, A. C., Vu, D. H., "A Hybrid Model for Recommendation Systems," *In Advances in Intelligent Information Hiding and Multimedia Signal Processing: Proceeding of the 16th International Conference on IIHMSP in conjunction with the 13th international conference on FITAT*, November 5-7, 2020, Ho Chi Minh City, Vietnam, Volume 2 (pp. 316-323), Springer Singapore, doi:10.1007/978-981-33-6757-9_40. **SCOPUS-indexed**

[P2] **Thi-Linh Ho**, Anh-Cuong Le, and Dinh-Hong Vu. Enhancing Recommender Systems by Fusing Diverse Information Sources through Data Transformation and Feature Selection. *KSII Transactions on Internet and Information Systems*. 2023, 17, 5, 1413-1432. DOI: 10.3837/tiis.2023.05.006. **SCIE, Q3**.

[P3] **Ho, T.-L.**; Le, A.-C.; Vu, D.-H. Multiview Fusion Using Transformer Model for Recommender Systems: Integrating the Utility Matrix and Textual Sources. *Appl. Sci*. 2023, 13, 10, 6324. <http://doi.org/10.3390/app13106324>. **SCIE, Q2**.

[P4] **T. -L. Ho**, and A. -C. Le, "Multi-view Transformation in Recommender Systems," *2021 International Conference on System Science and Engineering (ICSSE)*, Ho Chi Minh City, Vietnam, 2021, pp. 88-91, doi: 10.1109/ICSSE52999.2021.9538423. **SCOPUS-indexed**.

Scientific research topic

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